Efficient Modelling of Lateral Heterogeneity on Depth Stratification via the Impedance Marching

From the elastodynamic matrix–vector wave equation to a laterally coupled Dirichlet-to-Neumann march: derivation, validation, and what it actually buys

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Abstract

The elastodynamic matrix–vector wave equation $\partial_3 \mathbf{q} = \mathscr{A}\mathbf{q} + \mathbf{d}$ expresses the wave field as a first-order recursion in depth. This note asks whether the cube moment method can be applied to it directly, to obtain a single-site T-matrix without passing through the Navier equation. The answer is yes, and the construction is not new: the thesis already carries it in $2\frac{1}{2}$-D. What is new here is the 3-D Cartesian operator, and a measurement of what the route is worth.

Three results. *First*, the operator matrix $\mathscr{A}$ is derived from the stiffness tensor and reproduces the thesis system matrix (Akdef) under one constant similarity transform; the paper's ν_1, ν_2 are the thesis's ζ, χ. A fourth, arrived at late and by the wrong road, is that the thesis's own spectral representation of $\mathbf{A}$ — analytic eigenvectors, a piecewise branch, and an inverse obtained from the symplectic structure rather than by inversion — removes three numerical hazards outright that the operator route had to discover and repair one at a time. *Second*, the test space is *derived* rather than chosen: $\mathbf{J}_6$ is antisymmetric and $\Delta\mathscr{A}^{\mathsf{T}}\mathbf{J}_6 = -\mathbf{J}_6\Delta\mathscr{A}$, so $\mathbf{J}_6\Delta\mathscr{A}$ is symmetric and the only natural projection is test $= \mathbf{J}_6 \times$ trial. This dissolves an apparent mismatch of ten conditions against nine unknowns. *Third*, and least expected, the resulting coupling agrees with the ordinary Navier weak form at Born order but differs at $O(\Delta c^2)$ entirely within the traction channel — and that difference splits into a genuine O_h-invariant resummation *and* a spurious preference for the x_3 axis, in comparable amounts. A single-site score against an independent arbiter in a homogeneous background would mix the two and settle neither. The directional preference ceases to be an artefact once the background is layered, which is where the route was wanted.

The T-matrix itself is assembled in §9 and the comparison is reported channel by channel. The sharpest statement is not a norm: e_{12} and e_{13} are equivalent under O_h and so must amplify identically, and in the first-order T they split by 8.4×10^{-4} while the ordinary route splits them by exactly zero. That split is an arbiter-free *lower bound* on the first-order error, $\geq \delta/2$; it bounds nothing about the ordinary route, which can be symmetrically wrong, so no ordering of the two follows from it.

The ordering is settled elsewhere, in the one geometry where both formulations apply and an external reference exists. A periodic patch of space-filling cubes *is* a homogeneous layer, so a reflectivity recursion gives its reflection coefficient exactly; running both routes through the same lattice sum and the same solve, with only T_0 substituted, the first-order T is the more accurate of the two, by a margin that grows with contrast and that *grows further as the shared lattice error is refined away* — which is what distinguishes a real advantage from an accidental cancellation (§14).

That first-order system is then used for what the title names. Because the kernel is one-sided in depth, the integral equation splits and the problem becomes an initial-value march in the impedance $\mathbf{Y}$, defined by $\boldsymbol{\tau}_3 = \mathbf{Y}\mathbf{u}$ — the *Dirichlet-to-Neumann map* of everything below the current depth, which §16.9 argues is the single idea the rest of this note keeps meeting under other names. Marching $\mathbf{Y}$ rather than a reflection matrix removes the mode decomposition entirely, and with it the branch loss that limits the mode-basis form; carrying the lateral grid *inside* the marched variable then handles lateral heterogeneity without an outer iteration. Against an exact Mie sphere the march is fourth order in depth, and the residual that depth refinement leaves is linear in contrast and converges under lateral refinement to about $0.29\,\varepsilon$ at a spacing of 2.5 diameters, which second-order Born of the periodic array predicts independently, in real and imaginary part: it is the array's own double scattering. §24 sets out the substantial prior art — the construction is that of Haines, Hulme & Yu (2004) — and §24.4 adopts and measures three of their computational devices.

Every claim below is backed by an executable check; the scripts report 25/25, 44/44, 14/14, 17/17, 9/9, 7/7, 7/7 and 11/11, and a separate gate binds the numerical propagator to the derived $\mathscr{A}$. Section 21 states plainly what is *not* yet established.

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1 Conventions and objects

1.1 Indices, time, and units

The positive x_3 axis points downward; x_1, x_2 are lateral. Greek indices α, β run over the lateral pair $\{1, 2\}$ only, Latin indices over $\{1, 2, 3\}$. The time convention is $e^{-i\omega t}$ throughout, so $\partial_t \to -i\omega$ and the particle velocity is $\mathbf{v} = -i\omega\mathbf{u}$. The lateral transform is $e^{+i(k_1 x_1 + k_2 x_2)}$, hence $\partial_\alpha \to ik_\alpha$.

Numerical work uses seismic units: km, $\mathrm{km\,s^{-1}}$, $\mathrm{g\,cm^{-3}}$, GPa. These are consistent, since $\mathrm{GPa} = \mathrm{g\,cm^{-3}} \cdot (\mathrm{km\,s^{-1}})^2$.

Two orderings appear in the sources and must not be conflated. The paper orders components $(1, 2, 3)$; the thesis orders them $(z, x, y) = (x_3, x_1, x_2)$. The map between them is written explicitly in §3.2.

1.2 The field vector

The matrix–vector wave equation is

$$\partial_3 \mathbf{q} = \mathscr{A}\,\mathbf{q} + \mathbf{d}, \qquad \mathbf{q} = \begin{pmatrix} \mathbf{q}_1 \\ \mathbf{q}_2 \end{pmatrix}, \quad \mathscr{A} = \begin{pmatrix} \mathscr{A}_{11} & \mathscr{A}_{12} \\ \mathscr{A}_{21} & \mathscr{A}_{22} \end{pmatrix}, \tag{1}$$

with the elastodynamic choice

$$\mathbf{q}_1 = -\boldsymbol{\tau}_3, \qquad \mathbf{q}_2 = \mathbf{v}, \qquad (\boldsymbol{\tau}_j)_i = \tau_{ij}, \tag{2}$$

so that $\mathbf{q} = (-\tau_{13},\, -\tau_{23},\, -\tau_{33},\, v_1,\, v_2,\, v_3)^{\mathsf{T}}$. This is not an arbitrary pairing: $-\boldsymbol{\tau}_3$ and $\mathbf{v}$ are conjugate in the vertical power flux, $j = \frac{1}{4}(-\boldsymbol{\tau}_3^{\dagger}\mathbf{v} - \mathbf{v}^{\dagger}\boldsymbol{\tau}_3)$, and that is what makes the metric of §1.3 the natural one.

Transpose, and why it is marked this way. $(\cdot)^{\mathsf{T}}$ is the transpose throughout and $(\cdot)^{\dagger}$ the conjugate transpose. Neither is ever a power. The upright sans-serif marker is used deliberately: this note carries $\boldsymbol{\tau}_3$ as half of its own state vector, so a transpose written $(\cdot)^{\top}$ sits a symbol next to τ that is hard to tell from it, and one written $(\cdot)^{t}$ collides with the time of the $e^{-i\omega t}$ convention. Earlier drafts used both, and inconsistently.

1.3 The metric $\mathbf{J}_6$

The symplectic metric is

$$\mathbf{J}_6 = \begin{pmatrix} \mathbf{0} & \mathbf{I}_3 \\ -\mathbf{I}_3 & \mathbf{0} \end{pmatrix}, \qquad \mathbf{J}_6^{\mathsf{T}} = -\mathbf{J}_6, \qquad \mathbf{J}_6^2 = -\mathbf{I}_6. \tag{3}$$

A naming warning. This project, following the thesis, calls this matrix $\mathbf{J}_6$; it is already defined as such in the package. The paper calls the *same* matrix $\mathbf{N}$, and uses $\mathbf{J}$ for $\mathrm{diag}(\mathbf{I}_3, -\mathbf{I}_3)$, a different object. Translate on the way in. Every occurrence of $\mathbf{N}$ in the paper's equations (16)–(30) corresponds to $\mathbf{J}_6$ here.

The operator matrix is *quasi-Hamiltonian* with respect to it,

$$\mathscr{A}^{\mathsf{T}}\mathbf{J}_6 = -\mathbf{J}_6\mathscr{A}, \tag{4}$$

which is the paper's eq. (27) and the thesis's `ATdef`. Everything in §6 follows from (4) together with the antisymmetry of $\mathbf{J}_6$, and from nothing else.

2 The first-order system for an isotropic medium

2.1 From the stiffness tensor to the stiffness matrices

For an isotropic solid,

$$c_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \qquad \rho_{ij} = \rho\,\delta_{ij}, \tag{5}$$

with $\lambda = \lambda(\mathbf{x}, \omega)$, $\mu = \mu(\mathbf{x}, \omega)$, $\rho = \rho(\mathbf{x}, \omega)$ and $K_c = \lambda + 2\mu$. The stiffness matrices are $(\mathbf{C}_{jl})_{ik} = c_{ijkl}$. In particular

$$\mathbf{C}_{33} = \mathrm{diag}(\mu, \mu, K_c), \qquad \mathbf{C}_{33}^{-1} = \mathrm{diag}(\mu^{-1}, \mu^{-1}, K_c^{-1}), \tag{6}$$

and $\mathbf{C}_{jl}^{\mathsf{T}} = \mathbf{C}_{lj}$, which follows from $c_{ijkl} = c_{klij}$. All nine blocks are reproduced from (5) and checked entry by entry.

2.2 Eliminating the in-plane tractions

The components $\boldsymbol{\tau}_1, \boldsymbol{\tau}_2$ do not appear in (2) and must be removed. Doing so introduces the reduced stiffness

$$\mathbf{U}_{\alpha l} = \mathbf{C}_{\alpha l} - \mathbf{C}_{\alpha 3}\mathbf{C}_{33}^{-1}\mathbf{C}_{3l}, \tag{7}$$

with the two structural properties $\mathbf{U}_{\alpha 3} = \mathbf{0}$ and $\mathbf{U}_{\alpha\beta}^{\mathsf{T}} = \mathbf{U}_{\beta\alpha}$. Both are checked rather than assumed, because it is they — not the algebra that produced them — that make the symmetry relations of §3.3 hold.

The isotropic $\mathbf{U}_{\alpha\beta}$ is where two constants are *born*:

$$\nu_1 \equiv (\mathbf{U}_{11})_{11} = \frac{4\mu(\lambda+\mu)}{\lambda+2\mu}, \qquad \nu_2 \equiv (\mathbf{U}_{12})_{12} = \frac{2\mu\lambda}{\lambda+2\mu}, \qquad \nu_1 - \nu_2 = 2\mu. \tag{8}$$

They are read off $\mathbf{U}$ and only then compared with the published definitions; they are not independent quantities. Every $\mathbf{U}_{\alpha\beta}$ has a null third row and column, which is the statement that τ_{33} has been fully eliminated.

2.3 The operator matrices

The general construction gives

$$\mathscr{A}_{11} = -\partial_\alpha\, \mathbf{C}_{\alpha 3}\mathbf{C}_{33}^{-1}, \quad \mathscr{A}_{12} = i\omega\rho - \tfrac{1}{i\omega}\,\partial_\alpha \mathbf{U}_{\alpha\beta}\partial_\beta, \quad \mathscr{A}_{21} = i\omega\,\mathbf{C}_{33}^{-1}, \quad \mathscr{A}_{22} = -\mathbf{C}_{33}^{-1}\mathbf{C}_{3\beta}\,\partial_\beta, \tag{9}$$

and for the isotropic medium these evaluate to

$$\mathscr{A}_{11} = \begin{pmatrix} 0 & 0 & -\partial_1 \frac{\lambda}{\lambda+2\mu} \\ 0 & 0 & -\partial_2 \frac{\lambda}{\lambda+2\mu} \\ -\partial_1 & -\partial_2 & 0 \end{pmatrix}, \qquad \mathscr{A}_{22} = \begin{pmatrix} 0 & 0 & -\partial_1 \\ 0 & 0 & -\partial_2 \\ -\frac{\lambda}{\lambda+2\mu}\partial_1 & -\frac{\lambda}{\lambda+2\mu}\partial_2 & 0 \end{pmatrix}, \tag{10}$$

$$\mathscr{A}_{12} = \begin{pmatrix} i\omega\rho - \frac{1}{i\omega}(\partial_1\nu_1\partial_1 + \partial_2\mu\partial_2) & -\frac{1}{i\omega}(\partial_2\mu\partial_1 + \partial_1\nu_2\partial_2) & 0 \\ -\frac{1}{i\omega}(\partial_2\nu_2\partial_1 + \partial_1\mu\partial_2) & i\omega\rho - \frac{1}{i\omega}(\partial_1\mu\partial_1 + \partial_2\nu_1\partial_2) & 0 \\ 0 & 0 & i\omega\rho \end{pmatrix}, \tag{11}$$

with $\mathscr{A}_{21} = i\omega\,\mathrm{diag}(\mu^{-1}, \mu^{-1}, K_c^{-1})$.

The ordering is the content. Compare the $(1,3)$ entry of $\mathscr{A}_{11}$ with the $(3,1)$ entry of $\mathscr{A}_{22}$. In the first the derivative stands to the *left* of the modulus and differentiates the product; in the second it stands to the *right* and differentiates the field alone. For a laterally invariant medium the distinction collapses and is invisible. For a laterally varying one it is the physics. This is why the checks in §3.1 compare *operators* — by their action on generic fields — rather than expressions, and why they are accompanied by a negative control that corrupts precisely this ordering.

3 $\mathscr{A}$ against the thesis, and against the published operator form

One object is used here and not derived in this note: the thesis system matrix [29]. The published matrix–vector operator form [28] is *not* a second source. It contains nothing beyond the thesis — the two are the same object under one constant similarity transform, as (14) shows entry by entry — and it is used here for the one thing an independent publication of the same result is good for: corroboration. Two derivations made separately and agreeing is evidence; it is not a contribution, and nothing in this note rests on it.

> **A note on where this starts.** An earlier version of this note presented the published operator matrices as the foundation and the thesis system matrix as an arbiter for them. That is the wrong way round. The thesis carries $\mathbf{A}(k_x, k_y)$ in §2.1, equation (Akdef), and the purpose of this work is to extend and test the thesis, not to check the thesis against a later paper. What the operator form genuinely contributes is narrower: the thesis has the system only in the laterally *invariant* case, where $\partial_\alpha \to ik_\alpha$ has already been taken, and a cube requires the coefficients to vary laterally. So the operator form is a *generalisation* of (Akdef), and it is that generalisation — not the system — that is new here. The consequences of getting this backwards were not only presentational; they are collected in §11.

3.1 Arbiter A: the published isotropic operator matrices

The blocks (10)–(11) are transcribed separately and compared against those built from (9). The comparison is made by acting on a vector of generic symbolic functions, so it tests the derivative ordering as well as the coefficients. All four match.

A check that cannot fail is worth nothing, so a corrupted $\mathscr{A}_{22}$ — with the derivative wrongly placed outside the modulus — is also tested, and is rejected.

3.2 The anchor: the thesis system matrix (Akdef)

The thesis carries a 6×6 Fourier-domain system matrix $\mathbf{A}(k_x, k_y)$ for a laterally invariant medium, in a *different* field vector

$$\mathbf{b} = (u_3,\ u_1,\ u_2,\ \tau_{33},\ \tau_{31},\ \tau_{32})^{\mathsf{T}}, \tag{12}$$

i.e. displacement–stress rather than velocity–traction, ordered (z, x, y) rather than $(1, 2, 3)$, and normalised so that $\rho\omega^2$ appears where the paper has $i\omega\rho$. It uses the shorthand

$$\gamma = \frac{\lambda}{\lambda + 2\mu}, \quad a = \frac{1}{\lambda + 2\mu}, \quad b = \frac{1}{\mu}, \quad \zeta = \frac{4\mu(\lambda + \mu)}{\lambda + 2\mu}, \quad \chi = \frac{2\mu\lambda}{\lambda + 2\mu}. \tag{13}$$

The thesis attaches a warning to these that must travel with them: the shorthand $\gamma, a, b, \zeta, \chi$ is *local to* (Akdef), and is *not* the γ, ζ, χ of the List of Symbols, which denote the Chapter 5 material-contrast ratios. Since this note uses ζ and χ throughout as the in-plane stiffnesses that survive the elimination of ε_{33}, the collision is worth stating once rather than inheriting silently. With that said, $\zeta - \chi \equiv 2\mu$, which is the thesis's own footnote and the cleanest check on the pair.

Comparing (8) with these: $\zeta \equiv \nu_1$ and $\chi \equiv \nu_2$. The thesis's in-plane stiffnesses and the paper's auxiliary moduli are the same two quantities, born in the same elimination (7). This is verified, not asserted.

Setting λ, μ, ρ constant and acting on $e^{i(k_1 x_1 + k_2 x_2)}$ reduces $\mathscr{A}$ to a matrix. With the constant map $\mathbf{b} = \mathbf{E}\mathbf{q}$ built from $\mathbf{u} = \mathbf{v}/(-i\omega)$, $\boldsymbol{\tau}_3 = -\mathbf{q}_{1:3}$ and the permutation $(1,2,3) \to (3,1,2)$, the identity

$$\mathbf{A}_{\text{thesis}} = \mathbf{E}\,\mathbf{A}_{\text{paper}}\,\mathbf{E}^{-1} \tag{14}$$

holds entry by entry. Since the two matrices were derived independently and published separately, this is evidence rather than bookkeeping.

3.3 Structural checks

Three further properties are verified on the reduced 6×6:

1. the operator symmetry relations $\mathscr{A}_{11}^{\mathsf{T}} = -\mathscr{A}_{22}$, $\mathscr{A}_{12}^{\mathsf{T}} = \mathscr{A}_{12}$, $\mathscr{A}_{21}^{\mathsf{T}} = \mathscr{A}_{21}$, where the transpose is the formal adjoint under integration by parts ($\partial_\alpha^{\mathsf{T}} = -\partial_\alpha$);
2. the quasi-Hamiltonian relation (4), which in the wavenumber domain reads $\mathbf{A}^{\mathsf{T}}(-\mathbf{k})\mathbf{J}_6 = -\mathbf{J}_6\mathbf{A}(\mathbf{k})$;
3. the characteristic polynomial factorises as $(s^2 + k_{z,P}^2)(s^2 + k_{z,S}^2)^2$, recovering $\alpha^2 = (\lambda + 2\mu)/\rho$ and $\beta^2 = \mu/\rho$ from a construction in which no wave speed was ever written down.

Item 3 is the one that binds the algebra to physics: nothing upstream knows the wave speeds.

4 The contrast operator

4.1 Definition and the Lippmann–Schwinger equation

Inside the scatterer the medium is $(\lambda + \Delta\lambda,\ \mu + \Delta\mu,\ \rho + \Delta\rho)$. Writing $\mathscr{A}^{\text{tot}} = \mathscr{A} + \Delta\mathscr{A}$, the total field solves the background system driven by its own contrast,

$$(\partial_3 - \mathscr{A})\,\mathbf{q} = \Delta\mathscr{A}\,\mathbf{q} + \mathbf{d}, \tag{15}$$

which is the Lippmann–Schwinger equation [18] of the first-order system.

4.2 The foundation check

Before (15) means anything, $\mathbf{q}$ must actually satisfy (1). Building $\mathbf{q}$ from a *generic* displacement field in a *laterally varying* medium and forming $\partial_3\mathbf{q} - \mathscr{A}\mathbf{q}$ gives two halves with different strengths:

- rows 4–6 vanish **identically**, for any $\mathbf{u}$ whatever. They are the constitutive relation rearranged; no equation of motion is used. Failure here would mean the elimination (7) is wrong.

- rows 1–3 equal **exactly** the Navier body force $f_i = -\rho\omega^2 u_i - \partial_j \tau_{ij}$. Anything left over would be a defect in $\mathscr{A}_{11}$ or $\mathscr{A}_{12}$.

Both hold. This is a considerably stronger statement than the plane-wave and eigenvalue checks of §3.3, because $\mathbf{u}$ is generic and the moduli vary laterally, so every ordering term in (10) is exercised.

4.3 $\Delta\mathscr{A}$ is not linear in the contrast

The Navier potential is exactly linear in $(\Delta\lambda, \Delta\mu, \Delta\rho)$. $\Delta\mathscr{A}$ is not, because the elimination inverted $\mathbf{C}_{33}$:

$$(\Delta\mathscr{A}_{21})_{11} = i\omega\left(\frac{1}{\mu+\Delta\mu} - \frac{1}{\mu}\right) = -\frac{i\omega\,\Delta\mu}{\mu(\mu+\Delta\mu)}, \tag{16}$$

whose ratio to its linearisation $-i\omega\Delta\mu/\mu^2$ is exactly $\mu/(\mu+\Delta\mu)$. At the project's validated moderate contrast ($\alpha = 5$, $\beta = 3$, $\rho = 2.5$, so $\lambda = 17.5$, $\mu = 22.5$ GPa; $\Delta\lambda = 2$, $\Delta\mu = 1$ GPa):

quantity	value
$\Delta\mu/\mu$	0.04444
exact / linear for $\Delta b = \Delta(1/\mu)$	0.95745
exact / linear for $\Delta a = \Delta(1/K_c)$	0.93985

The first-order form has already summed part of the local response exactly. §7 measures what that is worth, and finds the answer is not the obvious one.

5 The augmentation

5.1 Why a derivative must be moved

Inside the scatterer the contrasts are constant, so every coefficient of $\Delta\mathscr{A}$ carries the indicator $\mathbf{1}_V$. A derivative standing to the *left* of such a coefficient differentiates the indicator and produces a surface delta on the cube faces. Integration by parts over all space moves that derivative onto the test function instead, where it meets something smooth:

$$\int \phi_i \left(-\partial_\alpha (m\,\mathbf{1}_V\,\psi_j)\right) d\mathbf{x} = + \int (\partial_\alpha \phi_i)\, m\,\mathbf{1}_V\,\psi_j\, d\mathbf{x}, \tag{17}$$

and the face term never arises. This is the step the thesis describes as transferring the derivatives onto the source and receiver coordinates of the propagator, and it is what converts a singular integro-differential equation into a nonsingular, compact integral equation.

5.2 The augmented dimension

After the transfer every term has the form $(\partial^a \phi)^{\mathsf{T}} \cdot \text{const} \cdot (\partial^b \psi)$ with $a, b \in \{0, 1\}$, so the form is *multiplicative* on the enlarged vector $\mathscr{L}\psi = (\psi,\ \partial_1\psi,\ \partial_2\psi)$ — nominally $6 + 6 + 6 = 18$ slots. The true dimension is smaller and is counted, not assumed.

Result. Exactly $10 = 6 + 4$ slots are used. The four extra are $\partial_1 v_1,\ \partial_2 v_1,\ \partial_1 v_2,\ \partial_2 v_2$: the lateral gradients of the two *horizontal* velocity components, and nothing else.

Setting $\partial_2 \to ik_y$, as a $2\frac{1}{2}$-D formulation does, makes two of those four multiplicative, leaving $6 + 2 = 8$. That is precisely the thesis's count for its augmented coupling matrix, reproduced here from an independent 3-D Cartesian derivation. The agreement is a corroboration of the thesis construction, not a restatement of it.

The transfer identity is verified three ways: the augmented matrix is constant (no derivative operators survive); the augmented form reproduces the transferred bilinear; and the untransferred and transferred bilinears differ by an explicitly exhibited lateral divergence $\partial_\alpha P_\alpha$.

6 The test space, derived

6.1 The argument

At this point there are ten augmented slots and, for an affine trial space, nine parameters. The temptation is to choose nine conditions. That temptation must be resisted: a subset chosen by scoring against the arbiter it is later judged by is a fitted parameter wearing a derivation's clothes.

There is no choice to make. $\mathbf{J}_6$ is *anti*symmetric, and (4) applies to $\Delta\mathscr{A}$ as well as to $\mathscr{A}$ (it is linear in the operator), so

$$(\mathbf{J}_6\Delta\mathscr{A})^\mathsf{T} = \Delta\mathscr{A}^\mathsf{T}\mathbf{J}_6^\mathsf{T} = \Delta\mathscr{A}^\mathsf{T}(-\mathbf{J}_6) = \mathbf{J}_6\Delta\mathscr{A}. \tag{18}$$

$\mathbf{J}_6\Delta\mathscr{A}$ is a **symmetric** matrix, so

$$b[\phi,\psi] = \phi^\mathsf{T}\,\mathbf{J}_6\Delta\mathscr{A}\,\psi \tag{19}$$

is a symmetric bilinear form. A symmetric form admits exactly one natural projection:

$$\boxed{\text{test space} \;=\; \mathbf{J}_6 \times \text{trial space.}} \tag{20}$$

Verified, together with the negative control: drop $\mathbf{J}_6$ and the symmetry is lost, both at operator level and in the projected system.

6.2 The ten-versus-nine mismatch dissolves

The ten slots were never test conditions. They measure how many derivative channels the *form* touches — its representation — not how many independent functionals it supplies. Each trial function ψ_k gives exactly one test functional $(\mathbf{J}_6\psi_k)^\mathsf{T}$, so the number of conditions equals the trial dimension by construction.

6.3 The effective coupling matrix

Collecting the last three subsections gives the object that goes inside the integral. The form $b[\phi,\psi] = \phi^\mathsf{T}\mathbf{J}_6\Delta\mathscr{A}\,\psi$ became *multiplicative* on the augmented vector $\mathscr{L}\psi$ once the derivatives

were transferred, so it has a matrix there; call it $\Delta\mathbf{C}$,

$$b[\phi,\psi] = (\mathscr{L}\phi)^{\mathsf{T}}\,\Delta\mathbf{C}\,(\mathscr{L}\psi) \quad \text{pointwise}, \tag{21}$$

the *augmented contrast matrix*. It is not a stiffness contrast and not $\Delta\mathscr{A}$: it is $\mathbf{J}_6\Delta\mathscr{A}$ rewritten on the enlarged vector, so it is symmetric by (18), it carries no derivatives of its own — they are all in $\mathscr{L}$ — and by the count above only $10 = 6 + 4$ of the nominal 18 slots of $\mathscr{L}\psi$ are occupied. It is distinct from the Navier coupling $\Delta\mathbf{C}_{\mathrm{Nav}}$ of §7.3, which is the object the two routes are later compared *against*, not a name for this one.

For the affine trial space — nine parameters, three rigid translations and six uniform strains — the projected coupling is then

$$(\Delta\mathbf{C}_{\mathrm{eff}})_{kl} = \int_V (\mathscr{L}\psi_k)^{\mathsf{T}}\,\Delta\mathbf{C}\,(\mathscr{L}\psi_l)\,dV \tag{22}$$

is the 3-D Cartesian analogue of the thesis's effective coupling matrix. It is 9×9, symmetric, of full rank 9, vanishes with the contrast, and — because the trial functions are polynomials — requires **no Green's function at all**: the cube integral is exact.

The rigid-translation block is diagonal,

$$(\Delta\mathbf{C}_{\mathrm{eff}})_{1:3,1:3} = i\omega^3\,\Delta\rho\,V\,\mathbf{I}_3, \qquad V = (2a)^3. \tag{23}$$

The power of ω is a basis statement, not a physical one. $\Delta\mathscr{A}$'s density entry is $i\omega\Delta\rho$ acting on the velocity slot; the trial carries $\mathbf{v} = -i\omega\mathbf{u}$ there; and $\mathbf{J}_6$ pulls the test function's velocity slot into the traction rows, supplying a second $-i\omega$. Hence $i\omega\cdot(-i\omega)\cdot(-i\omega) = i\omega^3$. The displacement-basis answer $i\omega\Delta\rho V$ is wrong by ω^2. This is the displacement-versus-flux confusion in a new place, and the check asserts the correct factor *and* asserts the incorrect one is incorrect, so it cannot drift back.

7 What the first-order coupling actually buys

7.1 The trial spaces coincide

It is tempting to attribute any difference from the Navier formulation to the nonlinearity of §4.3, or to a different ansatz. Neither is right. Inside the scatterer the moduli are constant, so $\boldsymbol{\tau}_3 = \mathbf{C}^{\mathrm{tot}}_{33}\,\boldsymbol{\varepsilon}_{i3}$ is a constant *invertible* map. A uniform strain gives a uniform $\boldsymbol{\tau}_3$ and an affine $\mathbf{v}$, and conversely: "$\boldsymbol{\varepsilon}$ uniform" and "$\boldsymbol{\tau}_3$ uniform plus in-plane $\boldsymbol{\varepsilon}$ uniform" describe the same nine-parameter family.

The two formulations are therefore **different projections of the same residual onto the same trial space**. The difference lives in the test conditions.

7.2 The density channel cannot differ

With $\Delta\lambda = \Delta\mu = 0$, $\Delta\mathscr{A}$ is exactly $i\omega\Delta\rho$ in rows 1–3, columns 4–6, and reaches *no* part of the stress-glut slot: a density contrast is a pure body force. The propagator's body-force velocity

block is $-i\omega\mathbf{G}$, so both routes return

$$A_u = \frac{1}{1-\omega^2\Delta\rho\,\Gamma_0}, \qquad \Gamma_0 = \int_V \mathbf{G}\,dV, \tag{24}$$

identically. The modulus channels are the only place the two can part.

7.3 The units the two couplings are written in

The statement just made is about the *amplification factor*, which is a ratio, and ratios are where this comparison is safe. The couplings themselves are not in the same units, and it is worth saying so plainly because every other check in this note is also a ratio, and none of them can see it.

The $\mathbf{J_6}$ pairing tests against the field q, whose velocity slots carry $v = -i\omega u$. The Navier weak form, and this package's local T-matrix, test against the *displacement*. So

$$\Delta\mathbf{C}_{\text{eff}} = i\omega \times \big[\text{the same object in displacement units}\big], \qquad \Delta\mathbf{C}_{\text{eff}}\big|_{\text{rigid}} = i\omega^3\Delta\rho V \ \text{ against } \ \omega^2\Delta\rho V. \tag{25}$$

This is a change of units and not of physics. It cancels out of every amplification factor, every channel ratio and every Born limit expressed as a ratio; it does *not* cancel the moment $\Delta\mathbf{C}_{\text{eff}}$ is combined with an object built in displacement units. The $i\omega$ appearing in (73) is this same factor seen from the other side.

The check that establishes it is necessarily absolute rather than relative: after dividing by $i\omega$, the residual $\|\Delta\mathbf{C}_{\text{eff}}/i\omega - \Delta\mathbf{C}_{\text{Nav}}\|/\|\Delta\mathbf{C}_{\text{Nav}}\|$ falls as 4.19×10^{-2}, 4.19×10^{-3}, 4.19×10^{-4}, 4.19×10^{-5} under decade reductions of the contrast — the *first* power, so the residual is the genuine $O(\Delta c^2)$ difference of §7 against an $O(\Delta c)$ coupling, and not a second, smaller units error hiding beneath the first.

7.4 Born agreement, and an $O(\Delta c^2)$ difference

Reading a proportionality constant off the rigid block alone gives $i\omega$, and testing it on the remaining eighty entries **refutes** the hypothesis $\Delta\mathbf{C}_{\text{eff}} = i\omega\,\Delta\mathbf{C}_{\text{Navier}}$. The two couplings agree exactly at $O(\Delta c^0)$ and $O(\Delta c^1)$ — identical Born limits, so nothing is broken — and differ from second order.

Where. Eleven of eighty-one entries differ. With the parameter order

$$(c_1, c_2, c_3,\ e_{11}, e_{22}, e_{33},\ e_{12}, e_{13}, e_{23})$$

the nonzero block is $\{e_{11}, e_{22}, e_{33}\}^2$ together with (e_{13}, e_{13}) and (e_{23}, e_{23}).

That is exactly the set of strains $\boldsymbol{\tau}_3$ sees:

$$\tau_{13} = 2\mu\, e_{13}, \qquad \tau_{23} = 2\mu\, e_{23}, \qquad \tau_{33} = \lambda(e_{11}+e_{22}+e_{33}) + 2\mu\, e_{33}. \tag{26}$$

The in-plane shear e_{12} is **untouched**, as it must be: τ_{12} was eliminated, not resummed. This was

a falsifiable prediction about which entries may be nonzero, and it held. A representative entry is

$$\left(\Delta\mathbf{C}_{\text{eff}} - i\omega\,\Delta\mathbf{C}_{\text{Navier}}\right)_{e_{11}e_{11}} = -\frac{i\omega V\,\Delta\lambda^2}{\lambda + 2\mu}, \tag{27}$$

pure second order, divided by K_c: the remainder of the geometric series for $1/(K_c + \Delta K_c)$.

7.5 Half of it is a broken symmetry

An isotropic cube in an isotropic whole space has an O_h-symmetric T-matrix; no direction is special. But the first-order system is a recursion in *depth*: it eliminated τ_1, τ_2 and kept τ_3, so x_3 is preferred by construction. The tell is already above — (e_{13}, e_{13}) and (e_{23}, e_{23}) differ while (e_{12}, e_{12}) does not, and those three are O_h-equivalent.

Applying the cyclic element $x \to y \to z \to x$ (positive controls: $P^3 = \mathbf{I}$, and the Navier coupling *is* invariant):

contrast (GPa)	invariant	breaking	ratio
pure shear, $\Delta\mu = 1$	2.62×10^{-2}	1.94×10^{-2}	1.3520
moderate, (2, 1, 0.1)	5.17×10^{-4}	2.93×10^{-4}	1.7669
strong, (20, 10, 0.5)	1.05×10^{-2}	5.92×10^{-3}	1.7669

Table 1. Decomposition of $\Delta\mathbf{C}_{\text{eff}} - i\omega\Delta\mathbf{C}_{\text{Navier}}$, relative to $\|\Delta\mathbf{C}_{\text{eff}}\|$, at $\lambda = 17.5$, $\mu = 22.5$ GPa. The two mixed rows share a contrast *direction* ($\Delta\lambda{:}\Delta\mu = 2{:}1$) at different magnitudes and return the identical ratio, so the split is a function of contrast direction alone — as a leading-order $O(\Delta c^2)$ split must be.

So the difference is neither pure artefact nor pure physics. It carries both a genuine O_h-invariant resummation the Navier form lacks, and a spurious preference for x_3.

A caveat on this test. The group used is the cyclic subgroup of order 3, not all 48 elements of O_h. Invariance under it is necessary for O_h invariance, not sufficient.

7.6 What follows, and what does not

Two readings are available and the measurement above does not separate them.

Against the route: an approximation that breaks a symmetry the exact answer possesses has introduced an error that no propagator can remove.

For the route: the correction is not scattered. It lands exactly on the traction channel, treating $\mathbf{C}_{33}$ exactly instead of leaving it to the Neumann series, and leaves the in-plane channel alone. Vertical P incidence excites e_{33}; vertical S incidence excites e_{13} or e_{23}; both are corrected. The in-plane shear, which the side-scattered field carries, is unchanged. If the response is dominated by reflection and transmission, the asymmetry is "z treated better" rather than "z treated worse".

> **Consequence for any comparison.** Do not score the raw coupling against an independent arbiter in a homogeneous background. Three effects are in play at comparable magnitude — the Galerkin-versus-collocation projection, the O_h-invariant

resummation, and the artefact — and a single number separates none of them. Any comparison must be channel-resolved.

Whether the resummation is an *improvement* is not established here. The local inversion $1/(K_c + \Delta K_c)$ is exact; whether that makes the closed T-matrix better depends on how the ansatz error and the series truncation interact, and those do not simply add.

8 The propagator, and the bridge to it

8.1 It already exists

Everything downstream needs the Green's matrix of (1). Write it $\mathbf{\Gamma}$, defined by

$$\big[\partial_z - \mathbf{A}(\mathbf{k})\big]\,\mathbf{\Gamma}(\mathbf{k}; z, z') = \mathbf{I}_6\,\delta(z - z'), \tag{28}$$

together with the radiation condition at $z \to \pm\infty$. It is a 6×6 matrix in the state-vector basis of (2), not a Green's *tensor*: its jth column is the state vector produced at z by a unit jump in the jth component of $\mathbf{q}$ imposed at z', so it carries the displacement-to-displacement, traction-to-displacement and traction-to-traction responses together. Three consequences are used repeatedly and are worth naming here rather than where each is first needed. (28) makes $\mathbf{\Gamma}$ satisfy the *homogeneous* system in z away from the source plane, which is (29) below; it fixes the jump across $z = z'$ to be exactly $\mathbf{I}_6$, which §15 measures and which §16.1 shows is what makes the integral equation and the differential equation the same statement; and, with the radiation condition, it makes $\mathbf{\Gamma}$ one-sided — built from downgoing modes below the source and upgoing modes above — which is what (74) tests and what §15 uses to name the difference between a layered and a whole-space background.

One warning about the symbol. $\mathbf{\Gamma}$ here is unrelated to the scalar $\Gamma_0 = \int_V \mathbf{G}\,dV$ of (24), which is a volume moment of the Kelvin tensor and not this object evaluated anywhere. The two are kept apart by weight and by subscript only, so the bold face is load-bearing.

It does not need to be built: the stratified 6×6 Green's function and its validated wrapper already return it, in the basis $(u_z, u_x, u_y, T_{zz}, T_{xz}, T_{yz})$ — which is exactly the basis the thesis writes $\mathbf{A}$ in. That the two are the same object had, however, been *assumed*.

8.2 The bridge identity

Away from the source plane the Green's matrix solves the homogeneous system in its receiver coordinate, so for two receiver depths in one uniform layer

$$\mathbf{\Gamma}(z_2, z') = \exp\!\big(\mathbf{A}\,(z_2 - z_1)\big)\,\mathbf{\Gamma}(z_1, z'), \tag{29}$$

with no free parameters. The identity is *local*: it holds however complicated the medium between the source and z_1 is, because the ODE knows only the layer the receiver sits in. The gate therefore places the source above a fast slab and the receivers below it, so transmission, reflection, conversion and interbed multiples are all present in $\mathbf{\Gamma}$ and (29) must still hold exactly.

8.3 Reporting residuals against the arithmetic floor

A bare relative residual is not comparable across incidence angles. Beyond the critical slowness the propagator carries both a growing and a decaying evanescent branch, so $\exp(\mathbf{A}\,dz)$ has dynamic range $\sim e^{|k|dz}$ — about 1.6×10^{7} at the widest angle used. Forming $\exp(\mathbf{A}\,dz)\,\mathbf{\Gamma}(z_1)$ then amplifies the rounding already present by $\|\exp(\mathbf{A}\,dz)\|$, and no correct implementation can beat

$$\text{floor} = \epsilon\,\|\exp(\mathbf{A}\,dz)\|\,\frac{\|\mathbf{\Gamma}(z_1)\|}{\|\mathbf{\Gamma}(z_2)\|}. \tag{30}$$

Residuals below are quoted in units of that floor, so $O(1)$ means the identity holds to the last bit available at that conditioning.

k_x	k_y	$\lvert k\rvert/\omega$	raw	corrected
2.00	1.30	0.0759	7.68×10^{13}	1.26×10^{-2}
0.40	0.05	0.0128	2.95×10^{14}	2.43×10^{-2}
14.00	9.00	0.5298	4.31×10^{13}	9.37×10^{-3}
control, $\zeta\leftrightarrow\chi$ swapped			7.68×10^{13}	1.36×10^{12}

Table 2. Residual of (29) in units of the arithmetic floor. $\omega = 31.4159$ rad s^{-1}, $dz = 1$ km. The third row is evanescent for both branches.

> **The corrected object is the physical state vector of $\partial_3\mathbf{q} = \mathscr{A}\mathbf{q}$**; the raw one misses by fourteen orders of magnitude.

The negative control swaps ζ and χ — two in-plane stiffnesses that differ by exactly 2μ, a plausible transcription slip that no symmetry check would catch. It is rejected by fourteen orders.

9 The single-site T-matrix

This is the object the construction was for. On the nine-parameter affine trial space,

$$\boxed{\mathbf{T} = \Delta\mathbf{C}_{\text{eff}}\big(\mathbf{I}_9 - \mathscr{G}\,\Delta\mathbf{C}_{\text{eff}}\big)^{-1},} \tag{31}$$

with $\Delta\mathbf{C}_{\text{eff}}$ the projected coupling (22) and $\mathscr{G}$ of (33) the self propagator moment of the cube.

9.1 What is new and what is borrowed

New is $\Delta\mathbf{C}_{\text{eff}}$. Borrowed is $\mathscr{G}$: the first-order route changes the *coupling*, not the Green's function. The two moments it needs are

$$\mathbf{G}_{ij} = \int_V G^0_{ij}\,dV, \qquad M_{in,pk} = \int_V \partial'_p\partial'_k G^0_{in}\,dV. \tag{32}$$

These are *derived* in §9.2 below, not taken from a table. In the parameter basis

$$\mathscr{G} = \frac{1}{V}\begin{pmatrix} \Gamma_0\,\mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & -\mathbf{S}\,\mathbf{W}^{-1} \end{pmatrix}, \qquad \mathbf{W} = \operatorname{diag}(1,1,1,2,2,2), \tag{33}$$

where $\mathbf{S}$ is the second moment M of (32) carried into the 6×6 Voigt basis and symmetrised on its outer index pair,

$$S_{mn,jk} = \tfrac{1}{2}\big(M_{mj,kn} + M_{nj,km}\big), \qquad M_{in,pk} = \int_V \partial'_p \partial'_k G^0_{in}\, dV, \tag{34}$$

so that $\mathbf{S}$ maps a uniform strain imposed on the cube to the response the cube's *own* radiated field produces back inside it, averaged over the cube. That self-response is what is called the **depolarisation** here and at the two places below where the term recurs: the name is the standard one from inclusion theory, where for an ellipsoid the corresponding object is Eshelby's tensor [4]. It is the object the $r = 0$ delta of §9.2 belongs to, and the reason that delta cannot be dropped — omitting it reverses the sign of $\mathbf{S}$, which is the failure §9.2 and §10.2.1 both warn about. The symmetrisation is not cosmetic and is not supplied by any symmetry M already has. M is symmetric in $p \leftrightarrow k$ because the derivatives commute, and in $i \leftrightarrow n$ because G^0 is; but m and n in (34) sit in *different slots* — m is a Green's index in the first term and a derivative index in the second — so nothing makes $\mathbf{S}$ symmetric on its outer pair except writing it that way. $\mathbf{W}$ undoes the engineering doubling of the shear slots.

$\mathbf{S}$ here is not the $\mathbf{E}$ of (14), nor the scaling $\boldsymbol{\Sigma}$ of (59), nor the convention factors $\mathbf{S}_u, \mathbf{S}_d$ of (117); earlier drafts wrote all four with the same letter and defined only two of them.

Note that (33) is asserted, not derived — which is exactly why §9.5 exists.

9.2 The self moments, derived

The integrand of M goes as $1/r^3$, which is not absolutely integrable in three dimensions. The customary remedy is to quote a tabulated Eshelby tensor [4, 22] and add a delta-function term at $r = 0$ by hand; omit the delta and the depolarisation comes out with the *wrong sign*. Nothing below is tabulated and nothing is added by hand.

9.2.1 The static Green's tensor

In the static limit the elastodynamic tensor reduces to Kelvin's,

$$G^0_{in}(\mathbf{r}) = \frac{a_0\,\delta_{in}}{r} + \frac{b_0\,x_i x_n}{r^3}, \qquad a_0 = \frac{1}{8\pi}\Big(\frac{1}{\mu} + \frac{1}{\lambda+2\mu}\Big), \quad b_0 = \frac{1}{8\pi}\Big(\frac{1}{\mu} - \frac{1}{\lambda+2\mu}\Big), \tag{35}$$

the coefficients being fixed by requiring that (35) satisfy the Navier operator away from the origin. An equivalent form makes the distributional content visible, since $\partial_i\partial_n r = \delta_{in}/r - x_i x_n/r^3$:

$$G^0_{in} = \frac{1}{4\pi\mu}\Big[\frac{\delta_{in}}{r} + h_c\,\partial_i\partial_n r\Big], \qquad h_c = -\frac{\lambda+\mu}{2(\lambda+2\mu)}, \tag{36}$$

and $a_0 = (1+h_c)/4\pi\mu$, $b_0 = -h_c/4\pi\mu$ identically.

9.2.2 The moment as a distribution, and why no delta is inserted

Write the moment as a pairing against the test function $w\,\mathbf{1}_V$ rather than as an integral of a function. With $T = \partial_{\text{rest}} r^m$ and $\partial_q \mathbf{1}_V = -n_q\,\delta_{\partial V}$,

$$\langle \partial_q T,\ w\,\mathbf{1}_V\rangle = \oint_{\partial V} n_q\, w\, T\, dA \;-\; \langle T,\ (\partial_q w)\,\mathbf{1}_V\rangle. \tag{37}$$

That is the entire construction. Two features make it exact rather than merely plausible:

- T is only ever evaluated *on the boundary*, at distance $\Delta/2$ from the origin, where every derivative of r^m is smooth and bounded. The singular pointwise structure of $\partial^D r^m$ at the origin is never formed, so there is no principal value to take and no delta to insert.
- the recursion strictly lowers both the derivative count and the weight degree, terminating at $\int_V x_w\, r^m\, dV$, an ordinary convergent integral for $m \geq -1$.

The delta term is therefore neither added nor droppable — it is already inside the surface integral. The sharp test is the Laplacian sum rule $\sum_p E[-1;\{p,p\}] = -4\pi$, which a spherical excision returns as 0.

9.2.3 The face integrals, and one of them by pure geometry

On the face $x_3 = \Delta/2$, scaling the in-face coordinates by $\Delta/2$ reduces every surface term to one of three integrals over $[-1,1]^2$:

$$\begin{aligned} j_1 &= \iint \frac{du\,dv}{(1+u^2+v^2)^{3/2}} && = \frac{2\pi}{3}, \\ j_2 &= \iint \frac{u^2\,du\,dv}{(1+u^2+v^2)^{5/2}} && = -\frac{2(\sqrt{3}-\pi)}{9}, \\ k_1 &= \iint \frac{du\,dv}{(1+u^2+v^2)^{5/2}} && = \frac{2(2\sqrt{3}+\pi)}{9}. \end{aligned} \tag{38}$$

All three are evaluated in closed form. The first needs no integration at all: j_1 *is* the solid angle subtended by one face at the cube's centre, and the six faces subtend 4π, so $j_1 = 4\pi/6 = 2\pi/3$. That identity is a check on the scaling and orientation of (38), obtained independently of any quadrature.

The zeroth moment needs one further integral, which also has a closed form:

$$\int_{[-1,1]^3} \frac{dV}{r} = -2\big(\pi + \log 64 - 12\log(1+\sqrt{3})\big) = \tfrac{4}{3}\log\big(70226+40545\sqrt{3}\big) - 2\pi = 9.520309455918214\ldots \tag{39}$$

The two closed forms are independent evaluations of the same constant and agree to the precision carried; the second is the one this package uses elsewhere.

9.2.4 The closed forms

Assembling (35) through the recursion (37) with the face integrals (38) gives, with

$$M_{in,pk} = A\,\delta_{in}\delta_{pk} + B\,(\delta_{ip}\delta_{nk} + \delta_{ik}\delta_{np}) + C\,E_{inpk}, \qquad E_{inpk} = \begin{cases} 1 & i = n = p = k \\ 0 & \text{otherwise,} \end{cases} \tag{40}$$

the third structure being the one a cubic — as opposed to isotropic — geometry admits,

$$B = \frac{\sqrt{3}\,(\lambda + \mu)}{6\pi\mu(\lambda + 2\mu)}, \tag{41}$$

$$A = B - \frac{1}{3\mu}, \tag{42}$$

$$C = -\Big(5 - \frac{2\pi}{\sqrt{3}}\Big)B, \tag{43}$$

$$G_{ij} = \delta_{ij}\Big(a_0 + \frac{b_0}{3}\Big)\Big(\frac{\Delta}{2}\Big)^2 \int_{[-1,1]^3} \frac{dV}{r}. \tag{44}$$

A, B and C are scale-free; G carries Δ^2. Note that A, B and C are *not* three independent constants — the whole rank-four moment is carried by B together with the elementary $1/3\mu$, which is a structural statement the tabulated presentation obscures. G_{ij} must be a multiple of δ_{ij}, because a rank-two cubic invariant has no other form: the cube cannot distinguish its own axes at that order.

9.2.5 The check that certifies the delta content

From (36) with $\nabla^2(1/r) = -4\pi\delta^3$ and $\nabla^2(\partial_i\partial_n r) = 2\partial_i\partial_n(1/r)$, whose cube integral is $-\frac{4\pi}{3}\delta_{in}$,

$$\sum_p M_{in,pp} = -\frac{\delta_{in}}{\mu}\Big(1 + \frac{2h_c}{3}\Big) = -\,\delta_{in}\,\frac{2\lambda + 5\mu}{3\mu(\lambda + 2\mu)}, \tag{45}$$

derived without reference to the recursion at all. The assembled moment satisfies (45) identically, and the export refuses to write its output otherwise. A spherically excised moment returns zero on the left-hand side, so (45) is exactly the statement that the $r = 0$ content is present and correctly weighted.

9.3 The tabulated route agrees

This package also carries the conventional tabulated Eshelby route, in which j_1, j_2 and k_1 are quoted rather than integrated. Evaluated alongside (41)–(44) at $\lambda = 17.5$, $\mu = 22.5$ GPa, $\Delta = 1$:

moment	integrated	tabulated	relative
A	$-1.220110745874 \times 10^{-11}$	$-1.220110745874 \times 10^{-11}$	0
B	$2.613707356074 \times 10^{-12}$	$2.613707356074 \times 10^{-12}$	3.1×10^{-16}
C	$-3.587055298887 \times 10^{-12}$	$-3.587055298887 \times 10^{-12}$	2.3×10^{-16}
G	$6.622004020056 \times 10^{-12}$	$6.622004020056 \times 10^{-12}$	0

Table 3. The table reproduces the integration to machine precision. The direction of that statement matters: the integration *verifies* the table; the assembly reads the integrated values.

The dynamic (smooth radiation) part carries no table — it is already exact polynomial integration — and is taken from the existing route with its tabulated static piece removed and the integrated one substituted.

9.4 Two implementations of the coupling

$\Delta\mathbf{C}_{\text{eff}}$ is rebuilt in Python directly from $\Delta\mathscr{A}$, independently of the symbolic derivation, and compared against the values that derivation exports. Worst relative difference over three contrasts: 1.4×10^{-15} for $\Delta\mathbf{C}_{\text{eff}}$ and exactly zero for the Navier coupling. Two implementations agreeing is the evidence standard; one agreeing with itself is not.

9.5 Calibration: every normalisation fixed against existing code

Run the *same* assembly with the ordinary Navier coupling in place of $\Delta\mathbf{C}_{\text{eff}}$. It must reproduce this package's validated four amplification factors. That pins the volume factors, the sign of $\mathscr{G}$ and the weight $\mathbf{W}$ — the *bookkeeping* of (33) — with no freedom left to tune. It does not certify the moments themselves; that is what the integration of the previous subsection is for. Against $\alpha = 5000$, $\beta = 3000$, $\rho = 2500$ (SI), $\omega = 60$, $a = 0.5$:

case	factor	assembled vs existing (rel.)
pure shear ($\Delta\mu = 1$ GPa)	A_u, A_θ, A_e^{diag}, A_e^{off}	0, 2.2×10^{-16}, 0, 0
moderate (2, 1, 100)	same four	0, 2.3×10^{-16}, 1.1×10^{-16}, 0

Table 4. The assembly reproduces the existing amplification factors to machine precision. Only after this is the coupling swapped.

One term had to be isolated to get there. The weak form's $\int_V \Delta\rho\, \omega^2\, \boldsymbol{\phi} \cdot \mathbf{u}\, dV$ contributes to the *strain*–strain block as well as the rigid one, because an affine trial displacement is not constant: $\int_V x_a x_b\, dV = \delta_{ab} V a^2/3$. This is a finite-size dynamic coupling between the density contrast and the strain channels, and the factorised amplification factors do not carry it — there A_u and the strain amplifications are independent. It shifts A_θ by 1.5×10^{-7} at the moderate contrast and vanishes identically with $\Delta\rho$. It is a small correction that the factorisation drops, not a defect in either object, and it is reported rather than absorbed.

9.6 The first-order T, channel by channel

A single norm would average a correction that lives in a few channels against others that do not move. The comparison is therefore resolved by channel.

case	channel	relative difference from the Navier route
pure shear	θ (bulk)	1.13×10^{-4}
	e_{diag}	1.1×10^{-16}
	e_{12} (in-plane shear)	0
	e_{13} (out-of-plane shear)	$\mathbf{8.36 \times 10^{-4}}$
	u (density)	0
moderate	θ (bulk)	1.74×10^{-3}
	e_{diag}	1.1×10^{-16}
	e_{12}	0
	e_{13}	$\mathbf{8.36 \times 10^{-4}}$
	u	0

Table 5. Amplification $(\mathbf{I} - \mathcal{G}\Delta\mathbf{C})^{-1}$ along each channel. The density channel is identical in both routes, as §7 requires; the in-plane shear is untouched; the bulk and out-of-plane shear channels move.

9.7 The broken symmetry, now visible in T itself

e_{12} and e_{13} are equivalent under O_h. For an isotropic cube in an isotropic background their amplifications *must* be equal. They are, in the Navier route, to exactly zero. In the first-order route they split:

	Navier	first-order
$\lvert A(e_{12}) - A(e_{13})\rvert \,/\, \lvert A(e_{12})\rvert$	0	8.36×10^{-4}

Table 6. The O_h violation of §7.5, measured on the T-matrix rather than on the coupling. Identical at both contrasts, because both carry the same $\Delta\mu$ and the shear channels see only that.

This is the sharpest form of the result. It is not a norm over a matrix: it is two numbers that a symmetry argument says must agree, and do not.

What the split does and does not license. It is tempting to read this as the first-order T failing an external test while the ordinary route passes. That reading claims more than the check can deliver, and the two halves must be kept apart.

What it gives. An **arbiter-free lower bound on the error of the first-order** T. If $A(e_{12})$ and $A(e_{13})$ must be equal and differ by δ, then whichever lies closer to the truth, the larger of the two errors is at least $\delta/2 \approx 4.2 \times 10^{-4}$. An absolute error estimate obtained from a symmetry alone, with nothing external to compare against, is worth having — and it is the absolute companion this project requires alongside any ratio test.

What it does not give. Any bound whatever on the ordinary route's error. A symmetric approximation can be symmetrically wrong, missing the truth in *both* channels by more than 8.4×10^{-4} while satisfying the symmetry exactly. Symmetry compliance is not accuracy. Setting a bound on one formulation against no bound on the other is not a comparison, and no ordering of the two follows from this table.

What is nonetheless attributable. The breaking is not manufactured by the projection mismatch of §9: $\mathscr{G}$ is itself O_h-symmetric, and §7.5 measured the coupling's asymmetry symbolically with no propagator present at all. The asymmetry is a property of the coupling, and it survives into $\mathbf{T}$.

The channel set has to include e_{13}. The coupling leaves e_{12} untouched by construction, so a comparison spanning only θ, e_{diag} and e_{12} reports that the shear channels agree — true, and uninformative. A channel-resolved comparison is only as good as the channels chosen, and the O_h-equivalent pair is the one that carries the information.

9.8 What this does and does not settle

The Born limits agree: at a contrast of 10^{-6} the two couplings differ by 4.2×10^{-8}, i.e. at $O(\Delta c)$ as they must.

> **The caveat that no number above reveals.** The propagator moment used is the *single* (collocation) one, while the $\mathbf{J}_6$ pairing makes the scheme Galerkin. The consistent pairing needs the *double* moment. This is the same class of mismatch the package has already measured as a genuine error elsewhere, so the table above should be read as the first-order coupling evaluated through a collocation propagator, not as the fully consistent first-order T. No winner is declared, and none should be read in.

10 The propagator moment from the first-order system

§9 took $\mathscr{G}$ from the ordinary Kelvin tensor, so the T-matrix assembled there is first-order in its *coupling* only. This section replaces that half. The plan is `docs/plans/2026-09-18-first-order-propagator-moment.md`; what follows is written as each step lands, and says plainly where the front is.

The object is the sandwiched form

$$\left[\Delta\mathbf{C}_{\mathrm{eff}}\,\mathscr{G}\,\Delta\mathbf{C}_{\mathrm{eff}}\right]_{kl} = \left\langle \mathbf{J}_6\psi_k \;\middle|\; \Delta\mathscr{A}\,\boldsymbol{\Gamma}\,\Delta\mathscr{A} \;\middle|\; \psi_l \right\rangle. \tag{46}$$

Plain Galerkin on $\mathbf{q}$ is not available: its mass matrix $\langle \mathbf{J}_6\psi_k, \psi_l\rangle$ is antisymmetric, and an antisymmetric matrix in *odd* dimension — here nine — is singular.

10.1 Step 1: Γ, built from $\mathscr{A}$ alone

Solving (28) by the spectral representation, the radiating Green's matrix is

$$\boldsymbol{\Gamma}(z,z') = \begin{cases} +\mathbf{P}_d\, e^{\mathbf{A}(z-z')}, & z > z', \\ -\mathbf{P}_u\, e^{\mathbf{A}(z-z')}, & z < z', \end{cases} \tag{47}$$

with $\mathbf{P}_d, \mathbf{P}_u$ the spectral projectors onto the down- and up-going eigenspaces, so that $\boldsymbol{\Gamma}$ decays away from the source in both directions and jumps by $\mathbf{P}_d + \mathbf{P}_u = \mathbf{I}$ across it.

Projectors, not eigenvectors. The thesis writes the eigen-matrix column by column with normalisation factors $\epsilon_P, \epsilon_S, \epsilon_H$ left symbolic. Those factors are fixed by a flux normalisation that degenerates at normal incidence, where the horizontal direction no longer distinguishes the *SV* and *SH* polarisations — exactly the $k_\parallel \to 0$ difficulty. The spectral projector

$$\mathbf{P}_d = \mathbf{V}_d\big(\mathbf{W}_d^{\mathsf{T}}\mathbf{V}_d\big)^{-1}\mathbf{W}_d^{\mathsf{T}} \tag{48}$$

is invariant under any rescaling of the columns of $\mathbf{V}_d$, so building $\mathbf{\Gamma}$ from projectors removes the ambiguity instead of carrying it. That invariance is verified directly, by rescaling every eigenvector by a random complex factor and confirming $\mathbf{P}_d$ is unchanged.

The matrix exponential must never be formed. Equation (47) is a statement, not an algorithm. Evaluating it as written — projecting *after* exponentiating — builds the growing branch and then cancels it. At a separation of 640 km the growing entries reach $e^{+28} \approx 10^{12}$, so the cancellation leaves 10^{-4} of absolute noise on a true answer of 3×10^{-8}. This is the Thomson–Haskell instability [24, 10], the reason invariant imbedding exists, reappearing inside our own construction; it announces itself as a decay rate that turns *negative* at large separation. Applying the exponential to the projected eigenvalues never forms the growing branch at all,

$$\mathbf{\Gamma}(z-z') = \pm \sum_{i\,\in\,\mathrm{down/up}} e^{s_i(z-z')}\,\mathbf{v}_i\mathbf{w}_i^{\mathsf{T}}, \qquad \mathbf{w}_i\cdot\mathbf{v}_j = \delta_{ij}, \tag{49}$$

and is what is used. The moment integral needs only $|z-z'| \le 2a$, where the instability cannot bite; a layered background with thick layers is where it would.

ω is complex from the outset. The vertical wavenumbers $k_{z,c} = \sqrt{k_c^2 - k^2}$ vanish on the circles $|k| = k_P$ and $|k| = k_S$, which the lateral integral must cross, and k_z sits in a denominator. At real ω a quadrature straddles an integrable $1/\sqrt{\cdot}$ singularity and does not converge; the directional-sweep work measured $1.9\times10^{-1} \to 1.1\times10^{-1}$ over an eightfold refinement, non-monotone. With damping 0.02 the smallest $|s|$ *on* the P circle is 0.240 rather than zero.

Checks. Thirteen, all passing: the spectrum is $\{\pm ik_{z,P}\}$ and $\{\pm ik_{z,S}\}$ twice; $\mathbf{P}_d + \mathbf{P}_u = \mathbf{I}$, both idempotent, both commuting with $\mathbf{A}$; $\mathbf{P}_d$ invariant under rescaling; $\partial_z\mathbf{\Gamma} = \mathbf{A}\mathbf{\Gamma}$ on both sides; the jump exactly $\mathbf{I}$; and the decay rate converging to the slowest eigenvalue, $0.034990 \to 0.033901 \to 0.032857 \to 0.032436$ against 0.032393 — a rate, not a bar, because at any one finite separation the faster mode still contributes.

10.2 Step 2: the vertical integral, in closed form

The vertical integral is exact, but it is *not* an integral over the square. The two halves of Γ carry different matrices, so the double integral splits over the two triangles and they cannot be recombined: treating the kernel as a function of $|z-z'|$ alone would symmetrise the propagator and destroy the up/down distinction the first-order system exists to carry.

With affine trial fields the weights are 1 and z, so four triangular integrals are needed,

$$T_{mn}(q) = \int_{-a}^{a} dz \int_{-a}^{z} dz'\; z^m z'^n\, e^{-q(z-z')}, \tag{50}$$

the lower triangle being T_{nm} by relabelling. All four are elementary:

$$T_{00} = \frac{2aq - 1 + e^{-2aq}}{q^2}, \tag{51}$$

$$T_{10} = -T_{01} = \frac{aq - 1 + (1+aq)\,e^{-2aq}}{q^3}, \tag{52}$$

$$T_{11} = \frac{3 - 3(1+aq)^2 e^{-2aq} + a^2q^2(2aq-3)}{3q^4}. \tag{53}$$

Summing the two triangles with a common matrix recovers the familiar square integral $2(2aq-1+e^{-2aq})/q^2$, and the odd weight cancels between them by parity — both checked, as is the large-q behaviour $T_{00} \sim 2a/q$, each triangle carrying half of the square's $4a/q$.

The decisive check of this step is not the closed forms themselves but the *assembly*: the vertical integration of Γ performed through (51)–(53), matrices and all, reproduces a brute-force double quadrature of Γ to 2.5×10^{-9} relative.

10.2.1 The same integrals as a series, and why they are needed

(51)–(53) are exact and, evaluated as written, lose most of their digits as $q \to 0$. Expanding (53) gives

$$3 - 3(1+x)^2 e^{-2x} + x^2(2x-3) = \tfrac{4}{5}x^5 - \tfrac{2}{3}x^6 + O(x^7), \qquad x = aq, \tag{54}$$

so a numerator of order x^5 is assembled from terms of order unity: at $x = 0.025$ about eight digits are gone. That is not a corner of the parameter space — it is the whole propagating window $k < \omega/\beta$, which is precisely where the radiation lives.

The remedy is the series. Substituting $u = z - z'$ in (50) and expanding the exponential,

$$T_{mn}(q) = \sum_{j\ge 0} \frac{(-q)^j}{j!} \int_0^{2a} u^j J_{mn}(u)\, du, \qquad J_{mn}(u) = \int_{-a}^{a-u} (z'+u)^m z'^n\, dz', \tag{55}$$

and every u-integral is elementary. With $b = 2a$,

$$J_{00} = b - u, \quad J_{10} = -J_{01} = \tfrac{b}{2}u - \tfrac{u^2}{2}, \quad J_{11} = \tfrac{b^3}{12} - \tfrac{b^2}{4}u + \tfrac{u^3}{6}, \tag{56}$$

giving coefficients of $(-qb)^j/j!$ equal to

$$\frac{b^2}{(j+1)(j+2)}, \qquad \frac{b^3}{2(j+2)(j+3)}, \qquad b^4\Big[\frac{1}{12(j+1)} - \frac{1}{4(j+2)} + \frac{1}{6(j+4)}\Big] \tag{57}$$

for $(0,0)$, $(1,0)$ and $(1,1)$ respectively. Two structural checks: at $q = 0$ the first gives $2a^2$ and the second $2a^3/3$, which are $\iint_{z>z'} 1$ and $\iint_{z>z'} z$; the third gives *zero*, and indeed $\iint_{z>z'} zz' = 0$ by parity. Its $j = 1$ term is $\frac{4}{15}a^5 q$, which is exactly what (53) tends to, $\frac{4}{5}a^5q^5/3q^4$. The series is entire, so it is valid for the imaginary q that propagating modes carry; it is used below $|2aq| = 3$,

where it is also the accurate branch.

10.3 Step 3: the lateral integral

These are **Sommerfeld-type integrals** — integrals over horizontal wavenumber, with the vertical dependence already reduced to the propagator, of the kind that layered-media Green's functions have always been evaluated by. Naming them so is not decoration: it says at once which family of numerical difficulty this is, and that the difficulty is a known one with a known remedy rather than something peculiar to the first-order formulation. What is peculiar to it is only *why* the integrand fails to decay, which the power count below settles.

Count powers as $|k| \to \infty$ for the *untransferred* form, in which $\partial_\alpha \to ik_\alpha$: the box form factor gives k^{-2}, (51) gives $q^{-1} \sim |k|^{-1}$, the two contrast operators give k^2, and the measure gives k. The strain channel does not converge. (The transferred form of §10.3.6 does; that is one of the things the augmentation buys, and it is taken up there.)

The divergence is not an artefact to be trimmed. A delta function in real space is a constant in k-space, so the non-decaying tail of the integrand *is* the Eshelby term of §9.2. A bare cutoff discards precisely the piece whose omission reverses the sign of the depolarisation. The remedy is to split the integrand into a convergent remainder and an asymptote that is integrated in closed form:

$$I(\mathbf{k}) = \big[I(\mathbf{k}) - I_\infty(\mathbf{k})\big] + I_\infty(\mathbf{k}), \tag{58}$$

with the bracket integrated numerically — convergent, and its convergence measured as a rate — and $\int I_\infty \, d^2k$ evaluated in closed form over the whole plane, where the delta content resides. The asymptote is taken of the z-integrated kernel (51)–(53), with the box form factors kept exact: they are what makes the remainder converge, so expanding them would defeat the subtraction.

Where the credit sits. Two different things are borrowed here and neither is devised in this note.

The *method* — subtract the large-wavenumber asymptote of a slowly convergent integrand, integrate the remainder numerically and the asymptote in closed form — is the standard remedy for slowly convergent Sommerfeld-type integrals, and is standard in exactly this setting, wavenumber integration for layered elastic media, where it is known as **static term subtraction** [11, 12, 19]. Hisada's case is the hardest one and is also ours: when source and receiver approach the same depth the integrand ceases to decay, and for equal depths the stress integrand *grows*. The self moment of a cube has z and z' in the same body, so it sits at that limit by construction. The same technique is called **singularity extraction** in the computational-electromagnetics literature on layered-media Green's functions [21], where the subtracted piece is the quasi-static term.

The *physics* that says which asymptote to subtract is Eshelby's [4, 22]: the non-decaying tail is the delta at $r = 0$, and dropping it reverses the sign of the depolarisation. This note takes that from there rather than rediscovering it.

What is new is only the setting — the horizontal wavenumber integrals the first-order formulation mandates, in which the tail arises from the two contrast operators rather than from a quoted inclusion tensor. "Split-and-subtract" is used below as a shorthand for that arrangement and is a label, not a citation.

10.3.1 The scaling that makes the expansion uniform

$\mathbf{A}$ is not uniformly $O(k)$. Rows 1–3 carry k, but rows 5–6 carry k^2, because the basis mixes displacement with traction and the two have different dimensions; expanding $\mathbf{A}$ in $1/k$ as it stands mixes orders between blocks. The traction of an evanescent field scales as $\mu k u$, so the natural balance is $\mathbf{q} \to (\mathbf{u}, \boldsymbol{\tau}_3/k)$. Under $\boldsymbol{\Sigma} = \mathrm{diag}(1,1,1,k,k,k)$,

$$\widetilde{\mathbf{A}} = \boldsymbol{\Sigma}^{-1}\mathbf{A}\,\boldsymbol{\Sigma} \tag{59}$$

is uniformly $O(k)$: the upper-right block gains a factor k, the lower-left loses one, and the diagonal blocks are untouched. Measured, $\|\mathbf{A}\|/k$ grows as $5.3\times10^3 \to 5.3\times10^5$ over two decades in k while $\|\widetilde{\mathbf{A}}\|/k$ settles at $53.0100 \to 53.0101$. Being a similarity transform it changes no eigenvalue, and $|s|/k \to 1$ for all six.

The scaling is also what makes the decomposition numerically usable. On the unscaled matrix the down- and up-going eigenvectors become nearly parallel as k grows — the evanescent limit degenerates — the eigen-decomposition is reported badly conditioned, and the kernel entries collapse into rounding error, from which no asymptotic power can be read. The decomposition is therefore done on $\widetilde{\mathbf{A}}$ and mapped back, $\mathbf{K} = \boldsymbol{\Sigma}\,\widetilde{\mathbf{K}}\,\boldsymbol{\Sigma}^{-1}$, checked against the verified kernel of step 2 at moderate k.

10.3.2 Which entries fail to decay

The order in $1/k$ that each entry of the z-integrated kernel carries is read off numerically, over decades in k, rather than derived: a measured power is harder to fool oneself with than one obtained from a 6×6 eigen-perturbation. A dot marks an entry that vanishes, so no power can be read.

	u_z	u_x	u_y	τ_{zz}	τ_{xz}	τ_{yz}
u_z	·	−1	−1	−2	·	·
u_x	−1	·	·	·	−2	−2
u_y	−1	·	·	·	−2	−2
τ_{zz}	**0**	·	·	·	−1	−1
τ_{xz}	·	**0**	**0**	−1	·	·
τ_{yz}	·	**0**	**0**	−1	·	·

Table 7. Power of k carried by each entry of the z-integrated kernel as $|k| \to \infty$. Exactly one block fails to decay.

> **The entire large-wavenumber problem sits in one block:** the traction rows against the displacement columns. Everything else decays at least as $1/k$ and needs no subtraction.

That block is also the one the physics predicts. In the evanescent limit a short-wavelength displacement produces a traction $\tau \sim \mu k u$, while the vertical integral (51) supplies $1/k$; the ratio is k-independent. A traction response to a displacement that does not depend on k *is* a local stiffness

— which is the Eshelby delta of §9.2, seen from the first-order side. The two representations identify the same object, and I_∞ has to reproduce exactly this block.

10.3.3 The asymptote in closed form

The limit of the non-decaying block is not merely numerical. Writing $\hat{\mathbf{k}}$ for the lateral direction,

$$C\big[\tau_{zz} \leftarrow u_z\big] = -\nu_1 = -\frac{4\mu(\lambda+\mu)}{\lambda+2\mu}, \tag{60}$$

$$C\big[\tau_{\alpha z} \leftarrow u_\beta\big] = \mu(\hat{k}_\alpha \hat{k}_\beta - \delta_{\alpha\beta}). \tag{61}$$

> **ν_1 is not a new constant.** It is the same ν_1 that appears in $\mathscr{A}_{12}$ — born in the elimination (7), and shown in §3.2 to be the thesis's ζ. The subtraction is expressible entirely in the operator's own constants. A wrongly identified asymptote would have no reason to land on ν_1 and μ.

Both are established as *rates*. The relative residuals fall $1.50\times10^{-3} \to 1.50\times10^{-4} \to 1.50\times10^{-5}$ across decades in k, ratios 10.0006 and 10.011 — clean $O(1/k)$, which is the $O(k_c/k)$ correction to the evanescent limit. Agreement at a single k would not distinguish a correct closed form from a coincidence.

The in-plane form is the sharper test. Two constants are fitted on *two* angles and then predict *six*: $P = -22.4977$ and $Q = +22.4948$, both equal to $\mu = 22.5$, with predictions tracking measurements to five digits across $\theta \in [0, 2.4]$. The isotropic entry (60) is constant to seven digits over the same range.

What this settles about the integration. The asymptote *is* direction-dependent — the relative variation over θ is 0.39 — so I_∞ is not a pure delta and the regulator cannot be avoided. But the structure is the most favourable available: $\delta_{\alpha\beta}$ and $\hat{k}_\alpha\hat{k}_\beta$ are the only two rank-two invariants a direction can build, and they transform to a local delta and a principal-value dipole kernel respectively, both standard. The $\tau_{zz} \leftarrow u_z$ entry is isotropic and so is a pure delta of weight $-\nu_1$.

10.3.4 The lateral integral, and where it must be evaluated

The box form factor is separable — the lateral transform of the cube's indicator being

$$(2a)^2 \operatorname{sinc}(k_1 a)\, \operatorname{sinc}(k_2 a).$$

With (60)–(61) and one lateral derivative from each $\Delta\mathscr{A}$, the integrand carries $k_c k_d$ against C, giving two families,

$$\int d^2k\, W\, k_c k_d \qquad \text{and} \qquad \int d^2k\, W\, k_a k_b k_c k_d / k^2,$$

of which the first is separable and the second is not. The elementary one-dimensional integrals are

$$\int_{-\infty}^{\infty} \operatorname{sinc}^2(ka)\, dk = \frac{\pi}{a}, \qquad \int_{-\infty}^{\infty} k^2 \operatorname{sinc}^2(ka)\, e^{-\epsilon|k|} dk = \frac{4}{4a^2\epsilon + \epsilon^3} = \frac{1}{a^2\epsilon} - \frac{\epsilon}{4a^4}. \tag{62}$$

The second is a pure pole with no finite part. So $\int I_\infty\, d^2k$ is not a number, and the split-and-subtract cannot be closed in k-space alone: adding back a pure pole restores exactly the divergence it was introduced to remove.

The method is not wrong; the tail is being evaluated in the wrong representation. Forming $|\widehat{S}|^2$ first and then integrating term by term destroys the structure that makes the constant part finite. By Parseval,

$$\iint f(\mathbf{x})\, K(\mathbf{x}-\mathbf{x}')\, g(\mathbf{x}') = \int \frac{d^2k}{(2\pi)^2}\, \widehat{f}(-\mathbf{k})\, \widehat{K}(\mathbf{k})\, \widehat{g}(\mathbf{k}), \tag{63}$$

so a $\widehat{K}$ tending to a constant C contributes $C\int_V f\, g\, d\mathbf{x}$ — a *local* term, finite, needing no regulator at all. This is verified directly in one dimension, both sides returning $1/2a$.

The decomposition therefore splits by representation rather than by magnitude:

piece	real-space object	evaluation
isotropic, $-\nu_1$ and $P\delta_{\alpha\beta}$	delta	local, elementary
traceless $\hat{k}_\alpha\hat{k}_\beta - \frac{1}{2}\delta_{\alpha\beta}$	2-D principal-value dipole	conditionally convergent
remainder $I - I_\infty$	—	numerically, in k

Table 8. Split-and-subtract survives, but $\int I_\infty$ is evaluated in real space. Only the remainder stays in k.

The traceless part is of degree -2 with vanishing angular average — exactly the conditional convergence the real-space engine of §9.2 handles by peeling derivatives onto the cube faces. The two representations meet again here: the same distributional structure, in two dimensions rather than three.

10.3.5 The principal-value piece vanishes

In two dimensions $\nabla^2 \log\rho = 2\pi\delta^2$, and $\partial_\alpha\partial_\beta \log\rho = (\delta_{\alpha\beta} - 2\hat{n}_\alpha\hat{n}_\beta)/\rho^2$ away from the origin; taking the trace fixes the delta weight, so that

$$\partial_\alpha\partial_\beta \log\rho = \mathrm{PV}\left[\frac{\delta_{\alpha\beta} - 2\hat{n}_\alpha\hat{n}_\beta}{\rho^2}\right] + \pi\, \delta_{\alpha\beta}\, \delta^2(\mathbf{x}). \tag{64}$$

By translation invariance the double integral over the square collapses onto the autocorrelation of its indicator, $w(\mathbf{u}) = (2a - |u_1|)(2a - |u_2|)$, exactly — removing two of the four dimensions — and both derivatives move onto w, which is legitimate because w has compact support.

w is piecewise bilinear, so $\partial_1\partial_1 w$ is a sum of line deltas. It has three terms, not one: w vanishes at $|u_1| = 2a$ but $\partial_1 w$ does *not* — it jumps from $\mp(2a - |u_2|)$ to zero — so there are edge deltas at $u_1 = \pm 2a$ alongside the interior one at $u_1 = 0$. With them,

$$D_{11} = \underbrace{3}_{\text{interior}} + \underbrace{(\pi - 3)}_{\text{edges}} = \pi = \pi(2a)^2\Big|_{a=1/2}, \qquad D_{12} = 0, \tag{65}$$

and the trace identity $\mathrm{tr}\, D = \langle \nabla^2 \log\rho, w\rangle = 2\pi w(0) = 2\pi(2a)^2$ holds exactly. Dropping the edge deltas gives 6 in place of 2π, so that identity is what detects them.

Hence $\langle \mathrm{PV}_{\alpha\beta}, w\rangle = D_{\alpha\beta} - \pi\delta_{\alpha\beta} w(0) = 0$: the traceless dipole contributes nothing.

It could not have contributed. $\langle \mathrm{PV}_{\alpha\beta}, w\rangle$ is a symmetric two-tensor invariant under the square's D_4 symmetry, and the only such tensor is a multiple of $\delta_{\alpha\beta}$ — the traceless part has nowhere to live. The conditional convergence is real, but for this cross-section it integrates away.

The whole of $\int I_\infty$ is therefore the local delta term. The piece that looked hardest — the conditionally convergent, shape-dependent dipole — is identically zero here, and what remains is elementary.

10.3.6 Why the transferred form converges

In the untransferred form $\partial_\alpha \to ik_\alpha$, each contrast operator contributes a factor of k, and the pole of (62) follows. After the by-parts transfer of (17) the derivatives act on the *polynomials, in real space*: ∂_α of an affine trial function is a constant, whose lateral transform is again the box form factor. No factor of k survives, and the lateral integral converges.

The augmentation therefore does two things at once: it keeps the indicator from being differentiated, and it makes the lateral integral convergent. Both are the same integration by parts.

Convergence is nevertheless slow. With $\widehat{K} \to C$ the truncation error falls only as $1/\Lambda$, which is what the subtraction is for.

10.3.7 The tail, in closed form

The tail is elementary. For the in-plane entry $C = \mu(\hat{k}_1^2 - 1) = -\mu\, k_2^2/k^2$, so

$$\int d^2k\, W\, C = -\mu \int d^2k\, \operatorname{sinc}^2(k_1 a) \operatorname{sinc}^2(k_2 a)\, \frac{k_2^2}{k^2} = -\frac{\mu}{2}\Big(\frac{\pi}{a}\Big)^2, \tag{66}$$

because $k_1^2/k^2 + k_2^2/k^2 = 1$ and the weight is symmetric under $k_1 \leftrightarrow k_2$, so each piece is exactly half of $\int d^2k\, W = (\pi/a)^2$. No quadrature is required; direct integration returns 19.7376 against $(\pi/a)^2/2 = 19.7392$, the difference being the quadrature's own error on a slowly converging integrand.

10.3.8 The remainder, and the test that closes the construction

Truncating the lateral integral at $|k| < \Lambda$ for the $\tau_{xz} \leftarrow u_x$ entry, with and without the subtraction:

The split is only worth making if both halves give the same answer. Truncating at $|k| < \Lambda$ for the $\tau_{xz} \leftarrow u_x$ entry:

Λ	unsubtracted	remainder	remainder + tail
80	−318.2214	118.83504	−325.29716
160	−321.6764	118.82727	−325.30493
320	−323.4539	118.82513	−325.30707
640	−324.3400	118.82466	−325.30754

Table 9. The subtracted route is converged to six figures by $\Lambda = 80$; the unsubtracted one is still 0.97 short of the limit at $\Lambda = 640$.

The remainder's successive differences fall $7.8 \times 10^{-3} \to 2.1 \times 10^{-3} \to 4.7 \times 10^{-4}$, a ratio near 0.27, against the unsubtracted 0.50 — the latter being the $1/\Lambda$ decay the constant part imposes. Extrapolating both:

$$\underbrace{-325.226}_{\text{unsubtracted limit}} \quad \text{versus} \quad \underbrace{-325.308}_{\text{remainder+closed-form tail}}, \qquad \text{relative difference } 2.5 \times 10^{-4}, \tag{67}$$

and that residue is dominated by the single-term extrapolation of the slowly converging unsubtracted series, not by the construction. At $\Lambda = 640$ the unsubtracted error is 0.97 and the subtracted one 5×10^{-4}: a factor of about two thousand.

Split-and-subtract is verified. The two routes to the same number agree, the tail is closed in form by (66), and the remainder converges three orders of magnitude faster than the direct integral.

This is the method, and §13.4 is where it pays. The finding above — that $\int I_\infty$ is a pure pole with no finite part, so the split *cannot* be closed in k-space and must be taken to real space — reads here as a difficulty encountered and worked around. It is better than that: it is the general statement, and the moments A and C are obtained by applying it verbatim. There the asymptotic integrand is again a function of direction alone, its $n = 0$ part is again a delta, and its $n \neq 0$ part is again a principal-value dipole over the square. The only change is which angular harmonics survive, which the symmetry of the trial function decides.

Worth stating plainly, since the cost was paid twice: four separate attempts were made to close the A and C tail in k-space — cosine-integral tails, graded angular meshes for the logarithmic singularities that then appear, a square-versus-disc geometry correction — before this paragraph was reread. Every one of them was a way of doing what it says cannot be done.

What steps 1–3 establish. Γ from spectral projectors; the triangular z integrals and their assembly; the scaling, the power table, and the asymptote $-\nu_1$, $\mu(\hat{k}\hat{k} - \delta)$ verified as an $O(1/k)$ rate; $\int I_\infty$ decomposed by representation, its principal-value part vanishing by D_4 symmetry and its remaining part closed by (66); and the closure test (67). All of that is machinery for the lateral integral. It does not yet say what the integrand is, and the next subsection is about a question that has to be answered before it can.

10.4 Step 4: how the augmentation composes with the Schwinger form

The augmentation of §6 turns the differential ΔA into a multiplicative ΔC_{eff} by moving each outer lateral derivative onto the test function. That is what keeps $\partial_\alpha \mathbf{1}_V$ — the surface layer on the faces of the cube — from ever being formed. The Schwinger object

$$\big[\Delta C\, \mathscr{G}\, \Delta C\big]_{kl} = \big\langle J_6 \psi_k \,\big|\, \Delta A\, \Gamma\, \Delta A \,\big|\, \psi_l \big\rangle \tag{68}$$

carries *two* contrast operators, and the right-hand one has no test function next to it: what Γ needs is $\Delta A \psi_l$ as an actual source, which is $\partial_\alpha(\Delta c\, \mathbf{1}_V\, \partial_\beta \psi_l)$ — precisely the object the transfer exists to avoid. Read that way the two constructions look incompatible.

They are not, and the reason is that the Born term and the Schwinger term differ in a way that matters. In the Born term the indicator of the contrast and the test function sit at the *same* point, so $\partial_\alpha(\mathbf{1}_V c)$ would have to be multiplied by another factor supported on $\bar{V}$; a surface layer times an indicator is not defined, and the transfer is therefore not a convenience but the only well-posed form. In the Schwinger term the two indicators are separated by Γ. The source at $\mathbf{x}'$ is paired, through a kernel, with a test object at $\mathbf{x}$; nothing coincides, the surface layer is harmless — and in the lateral Fourier domain it need not be formed at all, because

$$\mathscr{F}\big[\partial_\alpha(\mathbf{1}_V c)\big](\mathbf{k}) = i k_\alpha\, \mathscr{F}\big[\mathbf{1}_V c\big](\mathbf{k}) \tag{69}$$

is exact. The derivative is carried by the *transform*; the indicator stays inside the form factor, undifferentiated.

The bookkeeping that follows is uniform. Every derivative in A is lateral — the depth derivative is the left-hand side of $\partial_3 q = Aq + d$ — so in the lateral Fourier domain each is a factor ik_α at the single wavenumber the integral runs over. One entry of (68) has four derivative slots, and they are placed as

leg	slot	where it goes	
left	outer	onto the test polynomial	(the augmentation)
left	inner	onto the propagated field	$\to\ ik_\beta$
right	outer	onto the source transform	$\to\ ik_\alpha$, by (69)
right	inner	onto the trial polynomial	(the augmentation)

The two outer slots are the ones in question. The left one goes exactly where ΔC_{eff} already puts it; the right one goes onto the kernel. No surface term is evaluated on a face, and the augmentation is unchanged.

(69) is checked numerically rather than asserted. Paired with a smooth bump f it states $\int f\, \partial_1 \mathbf{1}_S = -\int_S \partial_1 f$, whose right side is an ordinary integral over the square; evaluating the left side in the Fourier domain as $(2\pi)^{-2} \int \hat{f}(-\mathbf{k})\, ik_1 F(\mathbf{k})$ reproduces it to 1.0×10^{-13}, with a zero imaginary residue and the opposite sign rejected by a factor of two.

The propagator, checked against the Kelvin tensor

Before any of this can be assembled, one normalisation has to be settled that the bridge identity of §9 does not reach: the bridge compares Γ at two depths and so tests propagation, not the strength of the source jump. Inverting Γ laterally at a fixed vertical separation and reading the velocity–force block,

$$\frac{1}{-i\omega}\,(2\pi)^{-2}\int d^2k\;\Gamma_{3+i,\,j}(\mathbf{k};\Delta z)\;\;\longrightarrow\;\;G_{ij}(\Delta z), \tag{70}$$

the analytic Kelvin tensor comes back to 7.3×10^{-4} and still converging, with off-diagonal leakage 2.9×10^{-4}; the residue is consistent with the $O((k_S r)^2)$ the static comparison drops. This fixes both conventions at once — that a force f_i enters row i of $\partial_3 q = Aq + s$, and that $q_{3+i} = v_i = -i\omega u_i$ comes out. The quadrature must be polar: the kernel depends on $|\mathbf{k}|$ and on direction separately and so is not smooth at the origin, where a Cartesian tensor rule converges only as $1/n$ (measured: $4.0\times10^{-2}\to1.2\times10^{-2}$ over a ninefold refinement, against $5.4\times10^{-3}\to7.3\times10^{-4}$ in polar). This is the opposite of the lateral moment itself, whose form factors are separable sinc products that polar cannot resolve.

The density channel, assembled

With $\Delta\lambda = \Delta\mu = 0$ only the multiplicative $i\omega\Delta\rho$ term of ΔA survives, both legs reduce to constants, and (68) can be predicted in closed form:

$$\big\langle J_6\psi_0 \;\big|\; \Delta A\,\Gamma\,\Delta A \;\big|\; \psi_0\big\rangle = i\omega^5\Delta\rho^2\,M, \qquad M = \int_V\!\!\int_V G_{11} = \Big(a_0 + \tfrac{b_0}{3}\Big)D_0, \tag{71}$$

with a_0, b_0 the Kelvin coefficients of §9.2 and D_0 the purely geometric double-volume moment $\int_V\!\int_V 1/r$. D_0 is evaluated by writing the double integral as a single one against the autocorrelation weight $\prod_i(2a - |u_i|)$ — exact, and three dimensions cheaper — and removing the $1/r$ outright with a Duffy substitution $\mathbf{u} = t(\xi,\eta,1)$, whose Jacobian t^2 cancels it:

$$D_0 = 1.8823126444, \qquad \int_V 1/r = 2.3800773640. \tag{72}$$

The lateral assembly with *both* contrast operators in place reproduces (71) to 6.9×10^{-5}, the two routes sharing nothing but the medium: one is a spectral construction on A integrated over lateral wavenumber, the other a real-space moment of the Kelvin tensor.

What the nine-parameter $\mathscr{G}$ is, and is not

The same comparison locates an object that had been taken on trust. The $\mathscr{G}$ used in §9 was *calibrated* against this package's T_9 operator rather than derived, and it is not the moment that (68) calls for. The gap factorises exactly:

$$\frac{[\Delta C\,\mathscr{G}\,\Delta C]_{00}}{i\omega^5\Delta\rho^2 M} = i\omega\,\frac{\int_V 1/r \;+\; i\,(\text{radiation})}{D_0}, \qquad \text{residual } 3.6\times10^{-5}, \tag{73}$$

where the radiation term is the closed form $V\omega(\alpha^{-3}+2\beta^{-3})/12\pi\rho$ that the moment export is itself gated on. Two statements, both of which matter:

- The geometric factor $\int_V 1/r\big/D_0 = 1.2644$ is the ratio of a *point*-volume moment to a *volume*-volume moment. The calibrated $\mathcal{G}$ is a collocation object; (68) is a Galerkin one. That is consistent with this project's own finding that the single site is best posed by collocation, but the two cannot be mixed without the factor.
- The remaining $i\omega$ is the normalisation of the q-field trial space: $\Delta C\,\mathcal{G}\,\Delta C$ carries two J_6 test factors while (68) carries one, so a $\mathcal{G}$ that composes correctly must absorb the trial-function normalisation. How it should do so is not settled here; see §21.

The derivative channels, and three defects the test exposed

The density channel settles the basis, the kernel, the z integrals and the quadrature, but its contrast operator is *multiplicative*: it does not exercise (69) at all. The channels that do are those carrying $\Delta\lambda$ and $\Delta\mu$, and for them a sharp statement is available with no new arbiter. $\mathbf{J}_6\Delta\mathscr{A}$ is symmetric — that is what derived the test space in §6 — and $\mathbf{\Gamma}$ is reciprocal,

$$\mathbf{\Gamma}(-\mathbf{k};z',z)^{\mathsf{T}} = \mathbf{J}_6\,\mathbf{\Gamma}(\mathbf{k};z,z')\,\mathbf{J}_6, \tag{74}$$

which follows from $\mathscr{A}(-\mathbf{k})^{\mathsf{T}}\mathbf{J}_6 = -\mathbf{J}_6\mathscr{A}(\mathbf{k})$ and $\mathbf{J}_6^{-1} = -\mathbf{J}_6$. Hence the assembled matrix must be symmetric. It is not symmetric by construction: the assembly builds its two legs by opposite rules, transferring the left outer derivative onto the test polynomial and carrying the right one on the kernel as ik. Symmetry is therefore a test of exactly the bookkeeping in question, and on a quadrature grid symmetric under $\mathbf{k}\to-\mathbf{k}$ it is an *algebraic* property, holding to round-off even where the grid is far too coarse to give the value accurately.

Applied, it found three defects, each in code that had already passed its own checks:

1. **The up/down split was decided by round-off inside the propagating window.** Classifying by $\Re(\mathrm{ev}) < 0$ is right for evanescent modes and meaningless for propagating ones: below $k = \omega/\beta$ the vertical wavenumber is real, $\Re(\mathrm{ev})$ vanishes to arithmetic noise, and its sign was being decided independently at $+\mathbf{k}$ and $-\mathbf{k}$. The reciprocity residual (74) at $|k| = 1.1\times10^{-3}$ was 1.245 — a complete failure, in the region that carries all of the radiation. The limit $\omega\to\omega(1+i\epsilon)$ settles it without needing ϵ: a propagating mode has $\mathrm{ev} = ik_z$ and the perturbation gives k_z a positive imaginary part, so $\Re(\mathrm{ev})\to0^-$ exactly when $\Im(\mathrm{ev}) > 0$. Residual after: 3.7×10^{-13}.
2. **The triangular z integral cancels catastrophically at small aq.** The numerator of T^{11} is $O((aq)^5)$ while its individual terms are $O(1)$, so by $aq\approx0.025$ it has lost about eight digits. That is not a corner case — it is again the propagating window. Expanding T^{mn} as a power series in q below $|2aq| = 3$, with the coefficients of §10.2.1, took that node from 1.3×10^{-6} to 2.1×10^{-13}.
3. **The scaling used to condition the eigenproblem is in the wrong direction.** $\mathrm{diag}(1,1,1,k,k,k)$ was introduced because the up- and down-going eigenvectors become parallel as k grows. Measured against *no scaling at all* it makes the eigenvector condition

number *worse* at every k above the propagating window — by two orders at $k = 60$. Parlett–Reinsch balancing [23] beats both by about twelve orders there (8.7×10^{7} against 1.6×10^{20}), because it is free to treat the third component differently from the first two, which no scalar rule can.

With all three repaired the assembled 9×9 matrix is symmetric to 4.7×10^{-12}, while reversing the sign of the right leg's ik — the one step (69) licenses — breaks it at 7.2×10^{-2}. That is a separation of ten orders on precisely the disputed bookkeeping, and the derivative terms carry 80% of the matrix, so the test is not vacuous.

The radiation reaction, from the first-order system

Repairing the branch selection changed the physics, not just the arithmetic. The previous classification suppressed the outgoing condition; with it restored, the propagator carries radiation damping, and both real-space arbiters above had to acquire the term they were missing. The sharpest form of the check is pointwise:

$$\mathrm{I}G_{11}(\Delta z)\Big|_{\text{first-order}} = 5.2134\times10^{-14} \qquad \text{against} \qquad \frac{\omega}{12\pi\rho}\Big(\frac{1}{\alpha^3}+\frac{2}{\beta^3}\Big) = 5.2250\times10^{-14}, \tag{75}$$

agreeing to 2.2×10^{-3}. This quantity has *no static counterpart at all*: it is not a correction to the Kelvin tensor but a term the static theory does not contain, and it arrives here out of the spectral projectors of $\mathscr{A}$ with nothing put in by hand.

Resolving it needs the quadrature to respect the two branch radii $|k| = \omega/\alpha$ and ω/β, where the vertical wavenumber vanishes. Those are *circles*: no Cartesian panel can follow one, and an unresolved square-root costs several per cent. The lateral integral is therefore split — Cartesian outside the propagating window, where the integrand is a separable product of sinc-type form factors that polar cannot resolve; polar inside it, where the branch radii are panel edges and $ka < 0.02$ makes the form factors constant to 10^{-4}. The central block of the Cartesian grid is exactly the inner square, so the two tile the box with no overlap and no gap. With that in place the density channel converges cleanly — 9.858, 9.834, 9.828×10^{-4} over a fourfold refinement — where before it drifted.

Status of Task 1. The design question is settled; the assembly computes a definite object; the density channel is verified against an independent real-space arbiter to 9.8×10^{-4}; the derivative channels are verified by a symmetry with a ten-order separation from its control; and the radiation reaction comes out right. What remains is to read the shear moment B off the assembled matrix and compare it with $\sqrt{3}(\lambda+\mu)/6\pi\mu(\lambda+2\mu)$, and to settle the $i\omega$ of (73).

11 The thesis spectral representation, and what it removes

Three of the four numerical defects of §10.4 are not properties of the physics. They are consequences of asking a numerical eigensolver to do work that §2.1 of the thesis has already done in closed form, and they do not arise at all if $\mathbf{\Gamma}$ is built the way the thesis builds it.

11.1 What the thesis gives

Alongside (Akdef) the thesis carries the complete spectral representation

$$\mathbf{A}(k_x, k_y) = \mathbf{D}_z \, \mathbf{\Lambda} \, \mathbf{D}_z^{-1}, \qquad \mathbf{\Lambda} = i \, \mathrm{diag}\big[k_{z,P}, k_{z,S}, k_{z,H}, -k_{z,P}, -k_{z,S}, -k_{z,H}\big], \tag{76}$$

with the columns of $\mathbf{D}_z$ written out analytically — $|\pm, P\rangle$, $|\pm, S\rangle$, $|\pm, H\rangle$, the + columns downgoing — and three further things that matter more than they look:

1. **The branch is defined piecewise.** $k_{z,c} = \sqrt{\hat{K}_c^2 - k_x^2}$ where $|k_x| < |\hat{K}_c|$ and $i\sqrt{k_x^2 - \hat{K}_c^2}$ otherwise: the positive root in both regimes, with the up/down assignment carried by the *column ordering* of (76) rather than by the sign of any computed quantity.

2. **The shear basis is chosen non-degenerate.** The thesis notes $k_{z,S} = k_{z,H}$ and keeps the two labelled apart regardless, remarking that its quasi-*SV* and quasi-*SH* vectors — unlike the conventional seismological decomposition — remain independent at $k_x = k_y = 0$.

3. **The inverse requires no inversion.** From the quasi-Hamiltonian relation,

$$\mathbf{D}_z^{-1}(k_x, k_y) = -i \, \mathbf{J}_6 \, \mathbf{D}_z^{\mathsf{T}}(-k_x, -k_y) \, \mathbf{J}_6. \tag{77}$$

All of it is verified here rather than taken on trust, and the checks are coupled: (76) cannot reproduce (Akdef) unless the eigenvectors, the ordering, the branch and (77) are all correct together. It does, to 6.2×10^{-14}. The normalisation factors ϵ_c, which the thesis specifies only as "chosen so that $\mathbf{D}_z^{-1}$ takes a simple form", are *read off* rather than guessed: with $\epsilon = 1$ the product (77)$\cdot\mathbf{D}_z$ is diagonal — to 1.1×10^{-15}, itself a test of the eigenvectors — and ϵ_c^2 is the reciprocal of each diagonal entry.

The one place $\mathbf{D}_z$ is genuinely singular, and why it is not a difficulty. Reading ϵ_c off the diagonal fails where a diagonal entry vanishes, and that is not round-off: it is the branch point $k_{z,c} = 0$, at $|k| = \hat{K}_P$ and $|k| = \hat{K}_S$ exactly, where that wave type's up- and down-going columns coalesce and $\mathbf{D}_z$ has rank five. It must be tested on $k_{z,c}$ itself and not on the spread of the diagonal — the entries of $\mathbf{D}_z$ span many orders at moderate k, so a smallest-entry threshold measures dynamic range and fires where nothing is wrong.

The degeneracy exists only in the lossless idealisation. Since $k_{z,c}^2 = \hat{K}_c^2 - k^2$ and $\hat{K}_c = \omega/c$, it cannot vanish at real k once ω carries an imaginary part, so the attenuation that §10 already adopts for the quadrature — "ω is complex from the outset" — removes the coalescence outright rather than dodging it. At $Q = 200$ the closest approach of $|k_{z,S}|$ across its own circle is 1.4×10^{-3} against a scale of 2×10^{-2}; $\mathbf{D}_z$ inverts *on* the circle to 1.2×10^{-15}, where the lossless evaluation is singular; and (76) is rebuilt there to 3.7×10^{-16}. The device is the one the package already uses: putting the loss in a complex slowness rather than a complex ω is the same statement, since $k_{z,c}$ depends on the two only through ω/c, and the two agree to 6.3×10^{-6} at $Q = 200$ — the $O(Q^{-2})$ difference. Nudging a node off the point remains the fallback for a deliberately lossless calculation, and placing a panel edge there, as the lateral gates do, is the right treatment of the integrable $1/\sqrt{\cdot}$ that survives in that limit.

11.2 What it removes

found numerically, working from the operator form	already settled in §2.1
$\mathfrak{K}(\lambda)$ carries no information in the propagating window; its sign was decided independently at $\pm\mathbf{k}$, giving a reciprocity residual of 1.245	the branch of $k_{z,c}$, defined piecewise, with the ordering of (76) carrying up/down
the doubly degenerate shear eigenspace returns an arbitrary SV/SH mixture; an intermittent factor of four in a mode-converted coefficient	quasi-SV and quasi-SH chosen non-degenerate, and said to be so
eigenvector conditioning of 1.6×10^{20}; a scalar scaling that proved to be in the wrong direction; then Parlett–Reinsch balancing	no eigenvectors are computed
$\mathbf{D}_z^{-1}$ by numerical inversion	(77)

Table 10. The fourth defect, the cancellation in T^{11} of §10.2.1, is genuine: it concerns the z integral and not the spectral representation.

Rebuilt on (76), $\mathbf{\Gamma}$ satisfies the reciprocity relation (74),

$$\mathbf{\Gamma}(-\mathbf{k}; z', z)^{\mathsf{T}} = \mathbf{J}_6\, \mathbf{\Gamma}(\mathbf{k}; z, z')\, \mathbf{J}_6, \tag{78}$$

to a residual of 2.2×10^{-11} *relative to* $\max|\mathbf{\Gamma}(\mathbf{k})|$, with no classification performed anywhere. Two things make that a real test rather than an arithmetic one. The source and receiver are *exchanged*, not just $\mathbf{k}$ reversed, and exchanging them negates $z - z'$ — so the right-hand side is built from the downgoing columns of (76) and the left-hand side from the upgoing ones, and no one-sided object can satisfy it. And the sampling runs to $|\mathbf{k}| = 3.22\,\mathrm{rad\,m^{-1}}$ against $\omega/\beta = 0.02$, a factor of 160, so it is an evanescent-regime claim and not merely a propagating one. On the same construction $\mathbf{\Gamma}$ reproduces the Kelvin tensor to 1.0×10^{-3} and converging, and carries the radiation reaction to 4.4×10^{-3} — *with no ϵ prescription*, since the outgoing condition is already in the branch of $k_{z,c}$. In the thesis basis the extraction is also plainer: a force f_j enters row $3 + j$ with a minus sign and the displacement comes out in rows 0–2, so $G_{ij} = -\mathbf{\Gamma}_{i,3+j}$ with no velocity factor at all.

12 A layer in a whole space: the external arbiter

Every comparison so far in this note sets two formulations against each other. None of them says which is right. A single layer embedded in a whole space removes that limitation, because its reflection coefficient is known independently — from a validated reflectivity recursion [15], and from a closed form short enough to derive here — and because it exercises the operator in a way the cube cannot.

12.1 Why a layer is the right test

For a laterally invariant contrast the operator difference is simply

$$\Delta\mathscr{A}(z) = \left[\mathscr{A}_{\text{total}}(\mathbf{k}) - \mathscr{A}_{\text{background}}(\mathbf{k})\right] \mathbf{1}_{[0,h]}(z), \tag{79}$$

a piecewise-constant matrix. **There is no augmentation, no by-parts transfer, no surface term and no differentiated indicator**, because $\mathscr{A}$ carries no ∂_3: a jump in the medium is simply a jump in the coefficients of an ODE, and is harmless. That is the structural advantage of the first-order form, and it is precisely the property the whole cube development of §6–§10.4 had to work around. So this tests $\mathscr{A}$, $\mathbf{\Gamma}$ and the Lippmann–Schwinger composition with none of the cube machinery in the way.

It also tests two things the cube checks could not. It is *nonlinear* in the contrast, so it reaches the resummation rather than the Born term alone; and $\Delta\mathscr{A}$ carries $\mathbf{k}$ explicitly through $\mathscr{A}_{12} \sim \nu_1 k^2/i\omega$, so scanning slowness tests a k dependence that the density channel of §10.4, being multiplicative, left untouched.

12.2 The closed form, by continuity

At normal incidence the answer follows from matching. With $e^{-i\omega t}$ and u the vertical displacement, each medium carries $u = De^{ikz} + Ue^{-ikz}$ with $k = \omega/c$, and the traction is $\tau = \rho c^2 \partial_z u = i\rho c\,\omega\,(De^{ikz} - Ue^{-ikz})$. Continuity of u and τ at $z = 0$ and $z = h$, with no up-going wave below, gives

$$R = \frac{r_{12} + r_{23}\, e^{2i\omega h/\alpha_2}}{1 + r_{12} r_{23}\, e^{2i\omega h/\alpha_2}}, \qquad r_{12} = \frac{Z_1 - Z_2}{Z_1 + Z_2} = -r_{23}, \qquad Z = \rho c. \tag{80}$$

The point of deriving it rather than quoting it is that the *sign* is then fixed by the derivation. Throughout this section

$$R \;=\; \frac{u_3 \text{ of the up-going field at } z = 0^-}{u_3 \text{ of the incident field at } z = 0^-}, \tag{81}$$

on both sides of every comparison, which leaves no convention free.

12.3 The first-order route

The laterally invariant Lippmann–Schwinger equation

$$q(z) = q_{\text{inc}}(z) + \int_0^h \mathbf{\Gamma}(\mathbf{k}; z, z')\, \Delta\mathscr{A}\, q(z')\, dz' \tag{82}$$

is solved by piecewise-constant collocation, the cell integrals taken in closed form from the spectral representation of $\mathbf{\Gamma}$ and *split at* $z' = z$: the jump across the diagonal is the source delta, and quadrature across it would smear exactly the object that defines the propagator.

One structural simplification is used, and it is exact. With $\mathbf{k}$ along x_1 the system is block diagonal: $\{-\tau_{23}, v_2\}$ is *SH* and does not couple to $\{-\tau_{13}, -\tau_{33}, v_1, v_3\}$. Working in the 4×4 *P-SV* block is therefore no approximation — and it removes the double degeneracy of q_S, which is

a genuine hazard rather than a nicety: a degenerate eigenspace has no preferred basis, so `eig` returns an arbitrary mixture of SV and SH. Before the block restriction the SS coefficient was correct at three slownesses and wrong *by a factor of four* at the fourth, which reads as a critical angle and is linear algebra.

12.4 What it gives

contrast	(80)	first order	relative
+2%	$-0.019937 - 0.019792i$	$-0.019937 - 0.019792i$	2.7×10^{-5}
+10%	$-0.127846 - 0.087946i$	$-0.127849 - 0.087948i$	2.4×10^{-5}
+40%	$-0.578990 - 0.067730i$	$-0.578999 - 0.067736i$	1.7×10^{-5}

Table 11. Normal incidence, $h = 200\,\mathrm{m}$, $\omega h/\alpha = 2.4$. The +40% row is the one that matters: it is far from Born, so it tests the *resummation*. Refinement in the cell count gives ratios 4.00, 4.00, 4.00 — clean second order.

Against the reflectivity recursion the comparison runs to $\sin i = 0.75$:

$$\frac{R_{PP}^{\text{first order}}}{R_{PP}^{\text{Kennett}}} = -1.00004,\ -1.00004,\ -1.00003,\ -1.00002, \tag{83}$$

and the same for SS to 1.0×10^{-4}. Two remarks make this a statement rather than a fitted factor. First, the recursion's R proved *independent of the thickness of the layer above* — identical for $d = 50, 100, 200, 400\,\mathrm{m}$ — so it is already referenced at the target layer, and the two-way phase correction that seemed obviously required in fact manufactures a spurious slowness-dependent offset. Second, what is checked is not the value -1 but that the factor is *constant*: a convention cannot depend on angle, and a physics error would not be constant. The sign itself is pinned by (80), which owes nothing to either code.

The mode-converted coefficients are covered by an identity that needs no knowledge of the convention at all. If the two differ by a diagonal rescaling $\mathrm{diag}(d_P, d_S)$ — the displacement-versus-flux factor, invisible to any per-mode ratio test — then the off-diagonal ratios are d_P/d_S and d_S/d_P, whose product is unity whatever $\mathbf{D}$ is. Measured: 1.00008, 1.00010, 1.00008.

Finally, a control isolates what this test uniquely reaches. Building $\Delta\mathscr{A}$ at $\mathbf{k} = 0$ while leaving $\boldsymbol{\Gamma}$ at the true $\mathbf{k}$ changes *nothing* at normal incidence (relative difference 0, exactly) and breaks the answer by 6.8×10^{-2} off it. The k dependence of the contrast operator is therefore load-bearing, and correct.

> **What this establishes, and what it does not.** The system matrix $\mathscr{A}$, the propagator $\boldsymbol{\Gamma}$, their composition, and the resummation reproduce an independently validated reflectivity code across contrast and across angle. That is the first external validation in this note. It does *not* validate the cube: the augmentation, the form factors and the moments of §9.2 play no part in a laterally invariant problem. Those remain checked only against other formulations.

13 The static moments A, B and C, read off

This is what Task 1 was for: the cube's propagator moment obtained from $\partial_3 \mathbf{q} = \mathscr{A}\mathbf{q} + \mathbf{d}$ rather than from the Kelvin tensor, and compared with a closed form the first-order machinery knows nothing about.

13.1 Which realisation, and why it is the easy one

B multiplies $(\delta_{ik}\delta_{jm} + \delta_{im}\delta_{jk})$ in the moment tensor, so

$$B = I_{xyxy} = \int_V \partial_y \partial_x G_{xy}\, dV, \tag{84}$$

and *that* realisation carries only lateral derivatives. In the lateral Fourier domain they are factors of ik, with no ∂_z and hence no $\mathscr{A}\boldsymbol{\Gamma}$ product to form:

$$B = \frac{1}{(2\pi)^2} \int d^2k\, (ik_x)(ik_y)\, F(\mathbf{k}) \int_{-a}^{a} G_{xy}(\mathbf{k}; -z')\, dz'. \tag{85}$$

Every other realisation of B involves ∂_z. The z integral is closed form from (76); the lateral one is quadrature.

13.2 The sign, pinned independently

The two factors of i give $(ik_x)(ik_y) = -k_x k_y$. Dropping them costs exactly a sign, which is invisible against $|B|$ and would be reported as success. So the sign is fixed by an evaluation owing nothing to the first-order machinery: by the divergence theorem,

$$\int_V \partial_y \partial_x G_{xy}\, dV = \oint_{\partial V} n_y \left(\partial_x G_{xy}\right) dA, \tag{86}$$

which is legitimate — and, more to the point, *complete* — because the divergence theorem applied to a locally integrable F returns the full *distributional* integral. Any delta at the origin is therefore included automatically, which is what makes this a safe way to evaluate a quantity whose volume form has an Eshelby delta in it. Here $\partial_x G_{xy} \sim 1/r^2$, integrable in three dimensions and non-singular on faces standing at distance a from the centre; only the $y = \pm a$ faces contribute and their integrands add, being odd in y. Evaluated,

$$\oint_{\partial V} n_y \partial_x G_{xy}\, dA = +2.6137073561 \times 10^{-12} = B \quad \text{to } 1.7 \times 10^{-13}, \tag{87}$$

so (84) is the right realisation and the sign is positive.

13.3 The result

Λ	nodes per half-period	B from (85)	relative
400	8	$2.6144253494 \times 10^{-12}$	2.7×10^{-4}
400	12	$2.6138081029 \times 10^{-12}$	3.9×10^{-5}
800	12	$2.6137960462 \times 10^{-12}$	3.4×10^{-5}
1200	14	$2.6137524215 \times 10^{-12}$	$\mathbf{1.7 \times 10^{-5}}$

Table 12. Against $B = \sqrt{3}(\lambda+\mu)/6\pi\mu(\lambda+2\mu) = 2.6137073561 \times 10^{-12}$, at $\omega = 6$ so that $(k_S a)^2 = 10^{-6}$ and the static form is the right comparison. The residual falls under refinement with no sign of a floor, so it bounds the quadrature and is not a discrepancy.

Dropping the two factors of i returns $-1.000275 \times B$: the control flips the sign exactly, as it must.

13.4 A and C, and the delta they carry

The same index map gives the other two with lateral derivatives alone:

$$I_{xxyy} = A, \qquad I_{xxxx} = A + 2B + C, \tag{88}$$

so nothing here needs ∂_z either. What they do have, and B does not, is the Eshelby delta. It lives in the δ_{ij} channels; B has $i \neq j$ and so carries none, which is why its integral converges outright while these two do not. *Convergence tracking the delta's presence is a check on the setup in itself.*

The split closes by representation, not by magnitude

Since $\widehat{g} \to C(\theta)/k^2$, the asymptotic integrand is

$$-k_x^2 F(\mathbf{k}) \, \frac{C(\theta)}{k^2} = -\cos^2\theta \; C(\theta) \, F(\mathbf{k}), \tag{89}$$

a function of *direction* only. Its k-space integral is a pure pole with no finite part, so adding it back restores exactly the divergence the subtraction was introduced to remove: **the split cannot be closed in k-space at all**, as §10 already found for the tail there. It closes by representation instead.

The remainder stays in k, and is regular at the origin — which is the point, because $-k_x^2$ times $-C/k^2$ is $+\cos^2\theta\, C$, bounded. No taper, no cutoff, no matching of one region to another: it is independent of the box half-width to seven digits across a fourfold change, where the unsubtracted integral drifted by per cent.

The local part goes to real space. Writing $h(\theta) = -\cos^2\theta\, C(\theta) = \sum_n h_n e^{in\theta}$, the $n = 0$ term is a delta contributing $h_0 \, \mathbf{1}_S(0) = h_0$. For $n \neq 0$ the inverse transform of $e^{in\theta}$ is a homogeneous distribution of degree -2,

$$\check{f}_n(r,\varphi) = \frac{(-i)^n |n|}{2\pi} \, \frac{e^{in\varphi}}{r^2}, \qquad \int_0^\infty k \, J_n(kr) \, dk = \frac{|n|}{r^2}, \tag{90}$$

whose principal-value integral over the square is elementary, the $\ln\epsilon$ of $\int_\epsilon^R dr/r$ cancelling because $e^{in\varphi}$ has zero angular mean:

$$\int_S \frac{e^{in\varphi}}{r^2}\,d^2x = \int_0^{2\pi} e^{in\varphi}\,\ln R(\varphi)\,d\varphi, \qquad R(\varphi) = \frac{a}{\max(|\cos\varphi|, |\sin\varphi|)}. \tag{91}$$

D_4 symmetry admits $n = 0, \pm4, \pm8, \dots$, and the measurement says the series stops at the first of them: $n = 4$ carries the whole correction and everything above it contributes 4×10^{-9} relative. Taking the local term as the delta alone is wrong by 5.5%, so the dipole is load-bearing and not decorative — both are gated.

The result, and the relations tested

	first order	closed form	relative
A	$-1.2205739 \times 10^{-11}$	$-1.2201107 \times 10^{-11}$	3.8×10^{-4}
B	$+2.6137524 \times 10^{-12}$	$+2.6137074 \times 10^{-12}$	1.7×10^{-5}
C	$-3.5777391 \times 10^{-12}$	$-3.5870553 \times 10^{-12}$	2.6×10^{-3}

Because all three come out of $\partial_3\mathbf{q} = \mathscr{A}\mathbf{q} + \mathbf{d}$ separately, the structural relations of §9.2 become *tests* rather than definitions. On first-order numbers only, with nothing substituted from the closed forms:

$$A \text{ vs } B - \tfrac{1}{3\mu} : \ 3.8 \times 10^{-4}, \qquad C \text{ vs } -(5 - 2\pi/\sqrt{3})\,B : \ 2.6 \times 10^{-3}. \tag{92}$$

13.5 Two things that nearly went wrong

The grid. The integrand is $4\sin(k_x a)\sin(k_y a)$ times the kernel, a product of sincs separable in the *Cartesian* components. On a polar grid it does not converge even at fixed cutoff, and that drift, read as cutoff dependence, looks exactly like a logarithmic divergence — which invites a hunt for a missing Eshelby delta. There is none. On Cartesian panels cut at the sinc half-period π/a the integral converges cleanly. The resolution study at *fixed* cutoff is what separates the two readings, and nothing else does.

A near miss. On the polar grid at $\Lambda = 50$ the value was 2.6147×10^{-12} — agreeing with B to 3.8×10^{-4}, and entirely an artefact of where the truncation happened to land.

Finally, the normalisation factors ϵ_c play no part: ϵ scales column i of $\mathbf{D}_z$ and divides row i of its inverse, so it cancels out of every projector. Dropping it removes the one step in the construction that cancels — the step of §11 that limits the useful range in k — and lets the whole assembly batch.

14 Which formulation is more accurate

§12 validates the operator and the propagator but cannot compare the two formulations, because a laterally invariant problem is one the cube moment method does not address at all. The geometry that does compare them is the one this package already uses to score itself: a periodic patch

of space-filling cubes, which for identical cubes *is* a homogeneous layer, and whose reflection coefficient the reflectivity recursion gives exactly. Both routes then build a single-site T for the same cube, both pass through the same lattice sum, the same contact correction and the same Foldy–Lax [5, 17] solve, and both are scored against the same reference. Only T_0 differs.

14.1 Substituting at the one place both routes speak the same language

$\mathbf{T} = \Delta\mathbf{C}(\mathbf{I} - \mathcal{G}\Delta\mathbf{C})^{-1}$ is not in the package's 9×9 convention — the strain block differs in sign and by the engineering doubling, and $\Delta\mathbf{C}_{\text{eff}}$ is in velocity units besides (§7.3). Mapping all of that would risk measuring a convention as though it were physics. The substitution is therefore made through the four *dimensionless* amplification factors A_u, A_θ, $A_{e,\text{diag}}$, $A_{e,\text{off}}$, which are ratios and carry no units. The package's own mapping to renormalised contrasts and its own T_0 construction are then used unchanged for both routes, so nothing distinguishes them but those four numbers. What licenses this is that the Navier route reproduces the package's own factors to 1.5×10^{-6}, re-checked rather than assumed.

14.2 The measurement

contrast	ordinary	first order	ratio
weak, $\Delta\mu/\mu \approx 0.4\%$	2.08×10^{-3}	2.09×10^{-3}	1.007
moderate, $\approx 4\%$	2.00×10^{-2}	1.94×10^{-2}	0.966
strong, $\approx 13\%$	5.59×10^{-2}	5.02×10^{-2}	0.899

Table 13. Relative error in R_{PP} against the reflectivity reference, $M = 8$, $N_z = 16$, $a = 0.5\,\text{m}$, $\omega = 60$, $p = 0$. At weak contrast the two agree, as they must — they agree at Born order — and the advantage appears where the $O(\Delta c^2)$ difference lives.

14.3 Why this is not cancellation

At weak contrast the two T_0 differ by 1.8×10^{-5} and yet both miss the reference by 2.1×10^{-3}: **most of the error is shared**, and belongs to the lattice rather than to T_0. A difference in a mostly-shared error is exactly the shape an accidental cancellation would take — the first-order $O(\Delta c^2)$ term happening to offset the lattice's own. Refining the lattice discriminates between the two readings:

M	shared error at weak contrast	ratio at strong contrast
8	2.0995×10^{-3}	0.8987
12	1.7157×10^{-3}	0.8766
16	1.4690×10^{-3}	0.8570

The shared error falls and the advantage *grows*. Cancellation would erode as the thing being cancelled against is removed; instead the advantage was being *masked* at coarse M. So 14.3% is a **lower bound** on the part attributable to T_0, not an estimate of it — it is still improving at the finest lattice tested.

A remark on how that check was posed, since the first version of it was wrong: it demanded that the ratio be *stable* under refinement, and it failed. Stability is not what should be expected. As the shared error goes to zero the ratio should approach the T_0-attributable value, and monotone movement towards a limit — not constancy — is the correct signature.

14.4 Which channel carries it

The improvement can be attributed, because the substitution runs through four separable factors. Replacing one at a time, at the strongest contrast and the finest lattice:

substituted	error	ratio
none (all ordinary)	3.7826×10^{-2}	1.0000
A_u only	3.7826×10^{-2}	1.0000
A_θ only	3.2418×10^{-2}	**0.8570**
$A_{e,\mathrm{diag}}$ only	3.7826×10^{-2}	1.0000
$A_{e,\mathrm{off}}$ only	3.7826×10^{-2}	1.0000
all four	3.2418×10^{-2}	0.8570

The whole of it is in A_θ, the trace — which is the A_{1g} channel, invariant under the full cubic group. So the advantage measured at normal incidence is the genuine O_h-invariant resummation, and the x_3- preferring part of the $O(\Delta c^2)$ difference contributes nothing. The reason is elementary once seen: a normally incident P wave is pure compression along z and never probes a shear channel at all. At $p = 0$ the measurement is blind to precisely the part of the formulation that singles out x_3.

14.5 What the x_3 preference is worth, bracketed

Two facts have to be put together. First, a structural one: `effective_stiffness_voigt` takes a *single* $\Delta\mu^*_{\mathrm{off}}$ and writes it into all three shear slots, so the package's T_0 is cubic-symmetric by construction and **cannot represent an** e_{12}/e_{13} **split**. The symmetry breaking was therefore excluded from §14 not by choice but for want of a slot. Second, *SH* incidence at $p \neq 0$ does probe the shear channels, and there:

SH at $p = 10^{-4}$	error in R_{SH}	ratio
all ordinary	4.7933×10^{-3}	1.0000
first order, e_{12} amplitude	4.7933×10^{-3}	1.0000
first order, e_{13} amplitude	1.8432×10^{-3}	**0.3845**

The e_{12} row reproducing the ordinary route *exactly* is §7 confirmed end to end: the two differ only in the traction channel, and the in-plane shear is untouched. The e_{13} row is the interesting one. An *SH* wave reflecting off a layer near normal incidence is dominated by $\partial_z u_y$ — a yz shear, the z-involving one — and the x_3-distinguished amplitude gives an error 2.6 times smaller. The first-order route separates the two by 7.2×10^{-3}; the ordinary route does not separate them at all.

That brackets the effect rather than realising it. Feeding the e_{13} amplitude through a cubic-symmetric T_0 writes it into the e_{12} slot as well, which is also wrong; it is the other *uniform* choice, not the split.

14.6 The tetragonal T_0

What the split needs is a T_0 of tetragonal rather than cubic symmetry — $e_{12} \neq e_{13} = e_{23}$, one extra shear parameter, which is the minimum that can carry it. The in-plane shear keeps its own amplitude and the two depth-involving shears share the other.

The whole of the correctness is the slot assignment, and it is measured rather than inherited. The package's Voigt ordering is (z, x, y) with $1 \to z$, $2 \to x$, $3 \to y$, so slot 3 is (x, y), slot 4 is (z, y) and slot 5 is (z, x); the first-order gate's paper indices put e_{12} in slot 3 and e_{13}, e_{23} in slots 4 and 5. The two agree — but two docstrings agreeing is not evidence, so each slot is perturbed on its own and the *SH* channel is asked which one it reads:

+10% on	moves the *SH* error by
slot 3, in-plane (x, y)	9.390×10^{-1}
slot 4, depth (z, y)	9.537×10^{0}
slot 5, depth (z, x)	0

which is exactly the physics of an *SH* wave in the (x, z) plane polarised along y: $\partial_z u_y$ dominates, $\partial_x u_y$ contributes near normal incidence, and a zx shear does not enter an *SH* problem at all. Slot 5 reading *identically* zero is the sharpest part of that.

With the slots so assigned:

SH at $p = 10^{-4}$	error in R_{SH}	ratio
ordinary	4.7933×10^{-3}	1.0000
first order, cubic T_0	4.7933×10^{-3}	1.0000
first order, tetragonal T_0	1.4905×10^{-3}	**0.3110**

The tetragonal form collects *more* than the bracket of the previous subsection predicted — 3.2× against 2.6× — and for a reason: the uniform-e_{13} substitution also wrote that amplitude into slot 3, where it is wrong and where slot 3 demonstrably contributes. Nothing is traded away for it: the normal-incidence channel is unchanged at 0.8570.

> **The answer, with its scope.** On the question as posed — same geometry, same lattice machinery, same external reference, only T_0 swapped — the first-order T is the more accurate of the two, and the margin grows with contrast. The part of that measured at normal incidence is entirely the O_h-invariant resummation; the x_3-preferring part is separately shown to be worth a further factor of 2.6 in the one channel that probes it, but cannot be carried by the present T_0. This is one geometry, one frequency, normal incidence, three contrasts, and an unconverged lattice. It scores the *single-site* T only, so an error the two share is invisible by construction. And it does not touch §7.5: the O_h symmetry breaking remains a genuine $\approx 4.2 \times 10^{-4}$ debit

against the first-order route in a homogeneous whole space, where there is no preferred direction for it to be right about.

15 The layered background

This is the payoff, and it is also where the route stops having a competitor. Everything up to here has had a closed-form whole-space $\mathbf{G}$ standing behind it, so the moment method could be scored against something. Over a stack there is no such object, and $\mathbf{\Gamma}$ must simply *be* the stratified propagator.

15.1 Nothing here is new machinery

The stratified 6×6 already exists and is validated, and §8 has already measured that it is the Green's matrix of *this* $\mathbf{A}$. What it is not is a function of depth: the block-Riccati sweep is indexed by *interface*, while the moment integral needs $\mathbf{\Gamma}(z, z')$ at continuously varying z and z'.

The tempting repair — propagate away from the nearest interface with $\exp(\mathbf{A}\,\delta)$ — is the Thomson–Haskell instability reintroduced deliberately, and an earlier attempt did exactly that: a propagator-matrix traversal that reduced correctly in the no-contrast limit, to 7.4×10^{-15}, and was unusable, $\mathrm{cond}(\mathbf{Y}_{\mathrm{up}})$ reaching 9.0×10^{15}, because forming $\exp(\mathbf{A}\,\delta)$ forms the growing evanescent branch explicitly. It was discarded rather than patched.

The stable construction does not propagate at all. The model is *cut* at the depths wanted, with zero contrast across the cut; the sweep is unchanged, is still a Riccati recursion, and still never forms the growing branch. Cutting must then be harmless, and is: between two *fixed* physical depths, two cuts and six cuts agree to 2.5×10^{-15} across the propagating and evanescent regimes alike. The bridge identity, previously verified only at the interfaces of a fixed stack, is then re-measured at depths lying on no lattice and holds to a fraction of one arithmetic floor.

15.2 The source normalisation, which the bridge could not see

§8 tests a *ratio* of two receiver depths. The source-side operator cancels out of a ratio, so that gate says nothing about how $\mathbf{\Gamma}$ is normalised at the source — and the normalisation is what makes it a Green's matrix rather than merely a solution. Measuring it needs both receivers carried back onto the source plane,

$$e^{-\mathbf{A}\delta}\,\mathbf{\Gamma}(z'+\delta, z') \;-\; e^{+\mathbf{A}\delta}\,\mathbf{\Gamma}(z'-\delta, z') \;=\; \mathbf{I}_6, \tag{93}$$

and (93) holds to 7×10^{-14}, flat in δ over a factor of eight. So the defining relation (28) holds with *no* calibration constant anywhere, and the 6×6 half of the source-convention question is closed. Reciprocity in the thesis basis, $\mathbf{\Gamma}(-\mathbf{k}; z', z)^{\mathsf{T}} = \mathbf{J}_6\mathbf{\Gamma}(\mathbf{k}; z, z')\mathbf{J}_6$ with source and receiver exchanged, holds to 1.2×10^{-15}.

15.3 The uniform limit, which cannot be what was planned

The plan called for the uniform limit of the layered assembly to reduce to the whole space. It cannot, and the reason is structural rather than numerical: layer 0 of the stratified solver is

acoustic, so a uniform *elastic* whole space is not a model the solver can be given. The ocean floor is always there and always reflects.

What replaces it is sharper, and has no free parameter. Whatever the ocean floor returns is, at *every* receiver, downgoing — it left the source upward, turned once, and is coming back down — and the half-space below radiates, so nothing else can contribute. Therefore

$$\mathbf{\Gamma}_{\text{layered}}(z,z') - \mathbf{\Gamma}_{\text{whole space}}(z-z') \;=\; \mathbf{P}_d\,[\,\cdot\,], \qquad \mathbf{P}_d = \mathbf{D}_z\,\mathrm{diag}(1,1,1,0,0,0)\,\mathbf{D}_z^{-1}, \tag{94}$$

with the split taken from the column ordering of (76) and nothing classified numerically. Measured, $\|\mathbf{P}_u\Delta\|/\|\Delta\|$ is 2.6×10^{-16} to 5.2×10^{-14} over five wavenumbers and three receiver depths, *including* receivers above the source, where the whole-space term is upgoing and the difference is still not. The difference itself is 54%–74% of $\|\mathbf{\Gamma}\|$, so the projection is not being satisfied by a vanishing numerator.

Beyond the critical wavenumber the returned field is evanescent over the whole round trip and the difference vanishes outright: at $k_x = 6, 7, 9$ against $\omega/\beta = 5.66$, $\|\Delta\|/\|\mathbf{\Gamma}\|$ is 3×10^{-15}–3×10^{-14}. There the layered $\mathbf{\Gamma}$ *is* the whole-space $\mathbf{\Gamma}$, to the arithmetic floor — which is the reduction the plan wanted, holding exactly where the boundary cannot reach and failing elsewhere only by the boundary term it should fail by.

The control matters here, because (94) is a statement about a projection and a projection can be satisfied for uninteresting reasons. Moving the contrast *below* the receiver instead sends an upgoing field back, and $\|\mathbf{P}_u\Delta\|/\|\Delta\|$ rises from 10^{-16} to 1.9×10^{-1}–4.8×10^{-1}: twelve orders, on the same geometry.

16 From the integral equation to invariant imbedding

Status. The equivalence of §16.1 is a derivation, and its pivot — that the jump is exactly $\mathbf{I}_6$ — is measured, at 7×10^{-14}. Everything from (100) to §16.12 is built and gated: the coefficient matrices and their conditioning, the impedance form and the wavenumber range it reaches, the march itself, and its score against a closed form and against Kennett, on a graded slab where no closed form exists. These report measurements, not expectations.

The laterally coupled case of §17 — where $\Delta\mathscr{A}$ varies laterally and couples $\mathbf{k}_\perp$, so that $\mathbf{Y}$ is a matrix on (3 components) × (lateral grid) — is built and gated in $2\frac{1}{2}$-D, the thesis's own geometry. What is not built *is the extension to full three-dimensional lateral structure, and the end-to-end comparison against the thesis's architecture (120) on a problem both could solve; §17 says what that leaves open.*

16.1 The kernel is one-sided, so the integral equation splits

The propagator of §11 is one-sided by construction: downgoing projectors below the source, upgoing above,

$$\mathbf{\Gamma}(z-z') = \begin{cases} +\sum_{i\,\in\,\text{down}} e^{\lambda_i (z-z')}\,\mathbf{P}_i, & z > z', \\ -\sum_{i\,\in\,\text{up}} e^{\lambda_i (z-z')}\,\mathbf{P}_i, & z < z', \end{cases} \qquad \mathbf{P}_d + \mathbf{P}_u = \mathbf{I}_6. \tag{95}$$

The Lippmann–Schwinger equation therefore splits at the *receiver* — down from above, up from below:

$$\mathbf{q}(z) = \mathbf{q}^0(z) + \int_{-\infty}^{z} \mathbf{\Gamma}_d(z-z')\,\Delta\mathscr{A}(z')\,\mathbf{q}(z')\,dz' + \int_{z}^{\infty} \mathbf{\Gamma}_u(z-z')\,\Delta\mathscr{A}(z')\,\mathbf{q}(z')\,dz'. \tag{96}$$

Both limits of integration move with z, so differentiating (96) produces two Leibniz boundary terms in addition to the interior derivative:

$$\partial_z\mathbf{q} = \underbrace{\mathbf{A}\mathbf{q}^0}_{\partial_z\mathbf{q}^0} + \underbrace{\mathbf{A}\,(\mathbf{q}-\mathbf{q}^0)}_{\partial_z\mathbf{\Gamma}=\mathbf{A}\mathbf{\Gamma}} + \underbrace{\mathbf{\Gamma}_d(0^+)\,\Delta\mathscr{A}\,\mathbf{q}}_{\text{upper limit}} - \underbrace{\mathbf{\Gamma}_u(0^-)\,\Delta\mathscr{A}\,\mathbf{q}}_{\text{lower limit}}. \tag{97}$$

By (95) the two boundary terms are $\mathbf{P}_d\,\Delta\mathscr{A}\,\mathbf{q}$ and $+\mathbf{P}_u\,\Delta\mathscr{A}\,\mathbf{q}$, and they sum to $(\mathbf{P}_d+\mathbf{P}_u)\Delta\mathscr{A}\,\mathbf{q} = \mathbf{I}_6\,\Delta\mathscr{A}\,\mathbf{q}$. Hence

$$\boxed{\partial_z\mathbf{q} = \big[\mathbf{A} + \Delta\mathscr{A}(z)\big]\mathbf{q}} \tag{98}$$

— the integral equation and the local first-order system with the *total* operator are the same statement.

Two things are worth naming about (98).

The jump is load-bearing, not bookkeeping. The coefficient of $\Delta\mathscr{A}$ in (98) is exactly one *because* $\mathbf{P}_d + \mathbf{P}_u = \mathbf{I}_6$. Any other normalisation of $\mathbf{\Gamma}$ would leave a constant standing there and the equivalence would be false. That is the quantity (93) measures on the stratified object, at 7×10^{-14} and flat in δ; §15 presents it as closing a convention question, and this is the reason it was worth closing.

What the integral form carries that the ODE does not. The ODE needs boundary conditions; (96) already has them, in the one-sidedness of its kernel — radiation at $\pm\infty$ is built into which projectors appear on which side. The integral equation is the ODE *plus* its boundary conditions, which is exactly why it is the useful form to discretise.

16.2 One-sidedness makes the discretised operator tridiagonal

Because the kernel is one-sided, a depth discretisation of (96) couples only neighbouring depths. The thesis states this directly: in Chapter 5 the array $(\mathbf{I} - \mathscr{S}_{int}\mathscr{E})$ — the discretised Lippmann–Schwinger operator — is *block tridiagonal* (`GstratRep.tex`, eq. `Amat`), and block-tridiagonal elimination *is* the Riccati recursion, given there as Algorithm 5.1. The Introduction names the connection: Riccati factorisation for discretely layered media is "implicit in the invariant embedding schemes of Kennett (1984), Tromp & Snieder (1988)" — published as [27], and tested in §24.6 — after Chin, Hedstrom & Thigpen (1984) and Ha (1985).

So the three descriptions — the split integral equation, the two-point boundary-value ODE, and the invariant imbedding — are one object seen three ways, and the sibling repository's sweep is its discretisation.

16.3 The imbedding equation

Write (98) in mode coordinates, $\mathbf{q} = \mathbf{D}_z\mathbf{a}$ with $\mathbf{a} = (\mathbf{a}_d, \mathbf{a}_u)$ split by the column ordering of (76), and let $\mathbf{V} = \mathbf{D}_z^{-1}\,\Delta\mathscr{A}\,\mathbf{D}_z$ carry the same block structure:

$$\mathbf{a}_d' = \Lambda_d\mathbf{a}_d + \mathbf{V}_{dd}\mathbf{a}_d + \mathbf{V}_{du}\mathbf{a}_u, \qquad \mathbf{a}_u' = \Lambda_u\mathbf{a}_u + \mathbf{V}_{ud}\mathbf{a}_d + \mathbf{V}_{uu}\mathbf{a}_u. \tag{99}$$

Define $\mathbf{R}(z)$ as the operator returning the upgoing amplitude produced by everything *below* z in response to the downgoing amplitude there, $\mathbf{a}_u(z) = \mathbf{R}(z)\,\mathbf{a}_d(z)$. Substituting into (99) and requiring the result to hold for every $\mathbf{a}_d$ gives the matrix Riccati equation

$$\mathbf{R}' = \mathbf{V}_{ud} + \Lambda_u\mathbf{R} - \mathbf{R}\,\Lambda_d + \mathbf{V}_{uu}\mathbf{R} - \mathbf{R}\,\mathbf{V}_{dd} - \mathbf{R}\,\mathbf{V}_{du}\,\mathbf{R}, \qquad \mathbf{R} = \mathbf{0} \text{ below the heterogeneity.} \tag{100}$$

The initial condition is the radiation condition, and it is literally the sibling repository's `Y[M+1] = 0`: nothing comes back from the radiating half-space, so the sweep starts from zero reflection and marches up.

Why this is the stable form. $\mathbf{R}$ is a reflection operator and is bounded — in an energy norm, $\|\mathbf{R}\| \le 1$ for a passive medium — so integrating (100) never forms a growing exponential. A propagator matrix does form one, which is the Thomson–Haskell instability and is what the discarded build of §15 ran into at cond = 9.0×10^{15}. The nonlinearity $-\mathbf{R}\mathbf{V}_{du}\mathbf{R}$ is not a complication to be removed: it is the resummation of the two-way multiple scattering, and it is what keeps the marched quantity bounded.

16.4 What this replaces, and what it does not

For a laterally invariant contrast, (100) supersedes the moment assembly this note has been building toward. There is no moment integral, no split-and-subtract, and — the substantive point — **no large-wavenumber treatment at all**. The delta has been absorbed *analytically*, by the Leibniz boundary terms of (97), rather than subtracted numerically. Everything in §13.4 exists to handle a singularity that, in this formulation, never has to be represented. $\boldsymbol{\Gamma}$ stops being an object to build and becomes an object to factorise.

That is a statement about the *background*, and it is worth being exact about how far it reaches. It does not retire the single-site T: a compact scatterer in a Foldy–Lax solve needs a T_0, and a T_0 is not what a depth march produces. The two decompositions answer different questions, and the moment route remains the one that answers the second.

16.5 Building the coefficient matrices

Every piece of $\mathbf{V} = \mathbf{D}_z^{-1}\,\Delta\mathscr{A}\,\mathbf{D}_z$ already exists. $\Delta\mathscr{A}$ is the difference of two (Akdef) matrices; $\mathbf{D}_z$ is analytic, from (Peigen), (SVeigen) and (SHeigen); and $\mathbf{D}_z^{-1}$ comes from (77), with the balanced numerical inverse taking over where §11 says it must. The row scaling cancels out of $\mathbf{V}$ identically — $(\mathbf{RD})^{-1}(\mathbf{R}\Delta\mathscr{A}\mathbf{R}^{-1})(\mathbf{RD}) = \mathbf{D}^{-1}\Delta\mathscr{A}\mathbf{D}$ — so the two routes must agree, and they do, to 2×10^{-16}, which checks both.

$\Delta\mathscr{A}$ is sparse for a structural reason: every entry of (Akdef) built from $\mathbf{k}$ alone is kinematic

and cancels in the difference, leaving 12 of 36. The compliance block is exactly diagonal, $\mathrm{diag}(\Delta a, \Delta b, \Delta b)$, so $\mathbf{V}_{du}$ has a three-term source; the inertia block carries $\Delta\rho\,\omega^2$ and the $\Delta\zeta, \Delta\chi, \Delta\mu$ pair; and the two remaining blocks are nilpotent and are each other's negative transpose.

That last is not a coincidence. $\Delta\mathscr{A}$ inherits (ATdef) from the two operators it differences, and (77) carries the relation into the mode basis:

$$\mathbf{J}_6\,\mathbf{V}^{\mathsf{T}}(-\mathbf{k}) + \mathbf{V}(\mathbf{k})\,\mathbf{J}_6 = 0 \iff \begin{cases} \mathbf{V}_{du}(\mathbf{k}) = \mathbf{V}_{du}^{\mathsf{T}}(-\mathbf{k}), \\ \mathbf{V}_{ud}(\mathbf{k}) = \mathbf{V}_{ud}^{\mathsf{T}}(-\mathbf{k}), \\ \mathbf{V}_{dd}(\mathbf{k}) = -\mathbf{V}_{uu}^{\mathsf{T}}(-\mathbf{k}), \end{cases} \tag{101}$$

verified to 3×10^{-14}. Only half the blocks are independent, and (100) preserves the symmetry of $\mathbf{R}$ — which is reciprocity of the reflection operator, and a gate rather than an assumption.

16.6 Is the march stiff?

Yes, and the answer is more useful than that, because of *where* the stiffness sits. The linear part of (100) is diagonal on the entries of $\mathbf{R}$: entry (c, c') sees

$$\mu_{cc'} = -\big(\lambda_{u,c} - \lambda_{d,c'}\big) = -i\,\big(k_{z,c} + k_{z,c'}\big). \tag{102}$$

Inside the propagating window k_z is real, so (102) is purely imaginary: the march is *oscillatory*, not stiff, and the step is set by resolving the phase. Beyond it $k_z = i|k_z|$ and $\mu_{cc'}$ is real and negative marching upward — decaying, at a rate up to $2|\mathbf{k}|$. That is stiffness, and the ratio grows as $|\mathbf{k}|\beta/\omega$: measured 1.0, 1.3, 3.5, 10.6, 35.5, 106.4 at $|\mathbf{k}| = 0.5, 8, 20, 60, 200, 600$ against $\omega/\beta = 5.66$.

But (102) is *diagonal*, so an integrating factor removes it exactly and for nothing: march $\tilde{\mathbf{R}} = e^{-\Lambda_u z}\,\mathbf{R}\,e^{\Lambda_d z}$ and no stiff linear term remains. No implicit solver is needed, and a general-purpose stiff integrator would be the wrong tool.

This is not a new prescription. It is what the sibling repository's layer recursion already does: in a homogeneous layer $\mathbf{V} = 0$ and the exact update is $\mathbf{R} \leftarrow e^{\Lambda_u h}\,\mathbf{R}\,e^{-\Lambda_d h}$, which is precisely the phase factor it applies. The recursion is an exponential integrator with the layer as its step, which is why it is stable at any layer thickness.

16.7 The real obstruction is non-normality

Stiffness turns out not to be the difficulty. Physics requires the spectrum of $\mathbf{V}$ to scale with $\Lambda \sim |\mathbf{k}|$ — a fixed material contrast cannot strengthen with lateral wavenumber — and it does: $\max|\mathrm{eig}\,\mathbf{V}|/2|\mathbf{k}|$ is flat at 0.0656 over four decades, $|\mathbf{k}| = 2$ to 2000. The *norm* does not. $\|\mathbf{V}\|/2|\mathbf{k}|$ runs 0.07, 1.8, 16.7, 186, 1671, 18566 over the same range, growing as $|\mathbf{k}|^2$. Nor is it a scaling choice: Parlett–Reinsch balancing of $\mathbf{V}$ returns the identity, so the non-normality is structural — (101) pins $\mathbf{V}_{du}$ and $\mathbf{V}_{ud}$ to comparable size and leaves a diagonal rescaling nothing to do.

The mechanism is measurable and is physical. As $|\mathbf{k}|$ grows, both $k_{z,P}$ and $k_{z,S}$ tend to $i|\mathbf{k}|$, so the P and S columns of $\mathbf{D}_z$ become parallel: $\cos(+P, +S)$ reaches 1.000000, and $\mathrm{cond}(\mathbf{D}_z)$ grows as $|\mathbf{k}|^{2.001}$ *after* the symplectic row scaling has been applied. This is the elastostatic degeneracy — at short lateral scale there is no distinction between P and S, both being the same decaying static

field — and unlike the unit mismatch of §11, which the row scaling does remove, it cannot be scaled away. Up- and downgoing modes are untouched: $\cos(+P, -P)$ falls to 2×10^{-6}.

The ceiling this imposes. At $|\mathbf{k}| = 600$, $\mathbf{V}_{du}$ and $\mathbf{V}_{ud}$ are each 2.0×10^{6} while the eigenvalues of their product are only 79^2 — a nine-order cancellation. The similarity $\mathrm{eig}(\mathbf{V}) = \mathrm{eig}(\Delta\mathscr{A})$, which is exact in theory, degrades accordingly: 1.5×10^{-15}, 4.2×10^{-13}, 3.5×10^{-8}, 2.0 at $|\mathbf{k}| = 2, 20, 200, 600$. **The mode basis is sound to $|\mathbf{k}| \approx 200$ here, about 35 times the S wavenumber, and not beyond.** Whether that is a real limit depends on the lateral resolution the scatterer demands, $|\mathbf{k}|_{\max} \sim \pi N/a$, and so it is recorded as a range rather than dismissed.

What it suggests. Where the modes coalesce, the mode basis is the wrong basis, and the remedy is not to condition $\mathbf{V}$ but to avoid forming it: march an impedance in the physical $(\mathbf{u}, \boldsymbol{\tau}_3)$ variables, where $\mathbf{D}_z$ never appears and the coefficient matrices are the blocks of $\mathbf{A} + \Delta\mathscr{A}$ themselves, polynomial in $\mathbf{k}$. §16.8 builds it and finds that it does clear the ceiling.

16.8 The impedance form, and the ceiling cleared

> *A correction, twice over. An earlier draft of the paragraph above asserted that marching an impedance "is what the existing Riccati solver and the classical imbedding codes do". The survey that preceded this subsection found otherwise: no matrix impedance exists in this package, in the sibling repository, or in the thesis. What exists is a* scalar *SH impedance used as an eigenvector normalisation convention, and two Riccati sweeps — both of which march a reflection matrix in* mode amplitudes. *That much stands.*
>
> *The draft then concluded "the construction below is new", and* that *is wrong. The survey covered this package, the sibling repository, the thesis and the symbolic archive; it did not cover the literature, and the literature has it. Haines, Hulme & Yu (2004) give exactly this construction — an impedance operator satisfying a Riccati equation, solved as an initial value problem, together with a one-way equation for the particle velocity — and handle lateral heterogeneity by expanding in a truncated Fourier basis, which turns their impedance operator into a finite matrix. That is the same object as $\mathbf{Y}$ below and the same means of carrying the lateral coupling inside it. Thomson (2015a,b) then established the symmetry of that operator, which is the identity used here as the main structural check. §24 sets out the correspondence term by term and says what, if anything, is left over.*

Split $\mathbf{q} = (\mathbf{u}, \boldsymbol{\tau}_3)$ and $\mathbf{A}$ into 3×3 blocks, and let $\boldsymbol{\tau}_3 = \mathbf{Y}\mathbf{u}$. Then $\mathbf{u}' = (\mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{Y})\mathbf{u}$ and $\boldsymbol{\tau}_3' = (\mathbf{A}_{21} + \mathbf{A}_{22}\mathbf{Y})\mathbf{u}$, while $\boldsymbol{\tau}_3' = \mathbf{Y}'\mathbf{u} + \mathbf{Y}\mathbf{u}'$, so

$$\mathbf{Y}' = \mathbf{A}_{21} + \mathbf{A}_{22}\mathbf{Y} - \mathbf{Y}\mathbf{A}_{11} - \mathbf{Y}\mathbf{A}_{12}\mathbf{Y}. \tag{103}$$

The coefficients are the blocks of $\mathbf{A}$ itself, polynomial in $\mathbf{k}$, with no eigenvectors anywhere. In a homogeneous medium $\mathbf{Y}' = 0$ and (103) becomes an algebraic Riccati equation whose stabilising solution is the downgoing half-space impedance; the equation is confirmed by checking that the impedance read off $\mathbf{D}_z$ satisfies it, to 10^{-14}, where $\mathbf{D}_z$ is still trustworthy.

Why it behaves differently, in one line. $\mathbf{V}$ needs the *full* inverse $\mathbf{D}_z^{-1}$, which mixes up- and downgoing and is where the cancellation lives. $\mathbf{Y} = \mathbf{T}_d\mathbf{U}_d^{-1}$ needs only the three *downgoing* columns.

Measured side by side over $|\mathbf{k}| = 2$ to 6000: the impedance residual tracks $\mathrm{cond}(\mathbf{U}_d)\,\epsilon$ linearly, reaching 7.2×10^{-10}, while $\mathbf{V}$ returns 2.000 from $|\mathbf{k}| = 600$ on. That 2.000 is $|-ik_z - ik_z|/|k_z|$: a lost branch, not a lost digit.

Refining with A alone. Newton's method on (103) — the correction solving a Sylvester equation built from $\mathbf{A}$ — holds the residual at 10^{-16} at *every* wavenumber tested. The division of labour is the natural one: $\mathbf{D}_z$ is analytic, so it names the right branch however imprecise it has become, and $\mathbf{A}$ supplies the precision. Continuation in $|\mathbf{k}|$ was tried instead of a $\mathbf{D}_z$ start and *flipped branch* at ω/α, returning the upgoing root; the arbiter caught it at exactly 2.000. The branch is then confirmed without reference to $\mathbf{D}_z$ at all, by requiring every eigenvalue of $\mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{Y}$ to have negative real part.

Y is bounded, where V was not. This is the whole difference. $\|\mathbf{Y}\|/\mu|\mathbf{k}|$ converges to 1.529022 and $\mathrm{cond}(\mathbf{Y})$ to 2.0000 across four decades — the elastostatic half-space stiffness, $O(1)$ conditioning, no growth relative to its own spectrum. The limit is also ω-independent, as a static limit must be: two frequencies a factor of four apart agree to 7.9×10^{-6} at $|\mathbf{k}| = 6000$, converging as $|\mathbf{k}|^{-2}$.

The arbiter, and the verdict. Self-consistency would not settle this, so $\mathbf{Y}$ is scored against an independently known analytic quantity: $\mathrm{eig}(\mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{Y})$ must be $\{ik_{z,P}, ik_{z,S}, ik_{z,S}\}$ with $k_{z,c}$ from (kzcDef) in closed form. The impedance holds it to 3.4×10^{-10} at $|\mathbf{k}| = 6000$, drifting as about $|\mathbf{k}|^{1.6}$ from 2×10^{-13} at $|\mathbf{k}| = 60$. It is not exact. What it does not do is fail. **The usable range goes from $|\mathbf{k}| \approx 200$ to beyond 6000 — a factor of thirty — and the failure mode changes from losing the branch to losing digits slowly.**

16.9 What Y is: the Dirichlet-to-Neumann map

Prescribe the displacement $\mathbf{u}$ on the plane $x_3 = z$ — Dirichlet data — solve the field equations in the region below it subject to the radiation condition at depth, and read off the traction $\boldsymbol{\tau}_3$ that results — Neumann data. The operator carrying the one to the other is the Dirichlet-to-Neumann map of that region, and it is $\mathbf{Y}$:

$$\boldsymbol{\tau}_3 = \mathbf{Y}(z)\,\mathbf{u}, \tag{104}$$

so $\mathbf{Y}(z)$ is the Dirichlet-to-Neumann map of the half-space below z.

This is not a gloss on (103); it is what (103) is an equation *for*. It is also the single idea the rest of this note keeps meeting under different names — the imbedding equation, the impedance form, the Möbius step, the symplectic identity, the boundedness of $\mathbf{Y}$ where $\mathbf{V}$ was unbounded, the dominance of the quadratic term — and each of those is a consequence of (104) rather than an independent finding. The purpose of this subsection is to say so once, carefully, and to record which of the consequences are measured. They are gated in `gate_first_order_lateral_impedance_3d.py`, part 14.

The scalar case, which is the whole idea. Take one dimension, an acoustic medium of density ρ and speed c, modulus $M = \rho c^2$. The first-order system is $u' = \tau/M$, $\tau' = -\rho\omega^2 u$, so

$A_{11} = A_{22} = 0$, $A_{12} = 1/M$, $A_{21} = -\rho\omega^2$. In a half-space $Y' = 0$ and (103) collapses to the algebraic statement

$$0 = -\rho\omega^2 - \frac{Y^2}{M} \qquad \Longrightarrow \qquad Y = \pm i\omega\sqrt{\rho M} = \pm i\omega\rho c, \tag{105}$$

and the radiation condition picks the sign. So Y *is* the acoustic impedance ρc, carrying the factor $i\omega$ that converts displacement to velocity; "impedance operator" is not an analogy. The elastic matrix reduces to exactly this: at normal incidence the three channels decouple and $\mathrm{diag}(\mathbf{Y}) = i\omega\rho\,(\alpha, \beta, \beta)$ to 10^{-13}, with the off-diagonal at 10^{-16} of the whole.

Two media in welded contact share both $\mathbf{u}$ and $\boldsymbol{\tau}_3$, so $\mathbf{Y}$ is *continuous* across an interface — which is the entire content of the interface condition, and is why reflection coefficients fall out of it. Writing $\mathbf{u} = \mathbf{U}_d\mathbf{a}_d + \mathbf{U}_u\mathbf{a}_u$ and imposing (104),

$$\mathbf{R} = -\big(\mathbf{T}_u - \mathbf{Y}\,\mathbf{U}_u\big)^{-1}\big(\mathbf{T}_d - \mathbf{Y}\,\mathbf{U}_d\big) = \mathbf{U}_u^{-1}\,(\mathbf{Y}_u - \mathbf{Y})^{-1}(\mathbf{Y} - \mathbf{Y}_d)\,\mathbf{U}_d, \tag{106}$$

which in the scalar case with $\mathbf{Y}_{u,d} = \mp i\omega\rho_1 c_1$ and $\mathbf{Y} = i\omega\rho_2 c_2$ is $|R| = |Z_2 - Z_1|/(Z_2 + Z_1)$, $Z = \rho c$ — the textbook coefficient, and the identity the impedance-null test of §18 rests on. *This is the first place the advantage shows.* Equation (106) says $\mathbf{R}$ is $\mathbf{Y}$ read in a chart, and reaching that chart costs the mode matrices $\mathbf{U}_u, \mathbf{U}_d$ — that is, $\mathbf{D}_z$ and its inverse. $\mathbf{Y}$ needs no chart at all.

Existence, which is the argument [9] actually make. Displacement and traction are physical fields; they exist whatever the medium does, and so does the map between them. An up/down decomposition does not: it presupposes a mode structure that, in [9]'s words, "can become poorly defined for laterally varying media". Their objection is about existence, not about conditioning, and (104) is what answers it. The conditioning argument of §16.8 is separate and is this note's own; (106) shows the two are related — the chart that may fail to exist is the same one that is ill-conditioned when it does.

The transformation law is Möbius, and the march applies it. Let $\mathbf{M} = e^{-\mathbf{A}h}$ propagate $\mathbf{q} = (\mathbf{u}, \boldsymbol{\tau}_3)$ up through a uniform step. The graph of $\mathbf{Y}_{\mathrm{bot}}$ is carried to the graph of

$$\mathbf{Y}_{\mathrm{top}} = \big(\mathbf{M}_{21} + \mathbf{M}_{22}\mathbf{Y}_{\mathrm{bot}}\big)\big(\mathbf{M}_{11} + \mathbf{M}_{12}\mathbf{Y}_{\mathrm{bot}}\big)^{-1}, \qquad \mathbf{M} = e^{-\mathbf{A}h}, \tag{107}$$

a matrix fractional-linear — Möbius — transformation. The `mobius` method of the march is literally one application of (107), which is why it is exact for a piecewise-constant medium and why it is therefore a layer stack. In a uniform medium the half-space map is the *fixed point* of this action, and that is how (107) is checked here: to 9.2×10^{-16} at $h = 5$ and 3.3×10^{-15} at $h = 80$, with the propagator that acts symplectic to 4×10^{-8}.

The Riccati equation is the infinitesimal form of (107). Put $\mathbf{M} = \mathbf{I} - \mathbf{A}h + O(h^2)$, expand, and the $O(h)$ term is (103). So "imbed the problem in the family indexed by the position of the surface and differentiate along the family" — the move of [1] and [20], and the imbedding step of §16 — and "write down the Möbius action of the propagator and take its generator" are the same derivation.

Which way it marches, and why there are two directions. The question is natural and the answer is not a convention. $\mathbf{Y}(z)$ maps what lies *below* z, so the one depth at which it is known

in advance is the **base**, where the radiation condition supplies the half-space impedance of §16.8. At the surface $\mathbf{Y}$ is the answer, not the input. The Riccati equation therefore runs **upward**, from the base to the surface, and Algorithm 1 is written as an upsweep for that reason rather than for a stylistic one.

Stability then agrees, and does so for a reason worth isolating. Perturbing the half-space impedance and integrating (103) both ways over one slab thickness, the perturbation is 0.92 of its initial size marching towards the surface and 3.2×10^{3} marching towards depth. **But that asymmetry is carried entirely by the evanescent channels.** Where every lateral mode propagates, $\mathrm{eig}(\mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{Y})$ is purely imaginary — to 10^{-17} — the fixed point is neutral, and neither direction is preferred. A direction test run on a grid too coarse to reach k_P measures nothing, which is how this one first came out inconclusive.

The second direction is real, and belongs to the other pass. Once $\mathbf{Y}$ is known, the field is recovered from the one-way equation $\mathbf{u}' = (\mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{Y})\mathbf{u}$ integrated **downward** from the surface, in the direction of energy flow — the structure [9] state explicitly and which §24 records. *The Riccati runs against the energy flow and the field runs with it*, and neither is a choice.

A third object, easily confused with the first, genuinely does march downward from the top: the Dirichlet-to-Neumann map of the region *above* z, started from $\mathbf{Y} = \mathbf{0}$ at a traction-free surface. That is the operator for a problem whose radiation condition sits on the other side — a source beneath the structure rather than above it — and it is not the one marched here.

The graph is Lagrangian, and that is reciprocity. Set $\mathbf{W} = \left(\begin{smallmatrix} \mathbf{I}_{3N} \\ \mathbf{Y} \end{smallmatrix}\right)$, whose columns span the downgoing subspace. Restricting the symplectic form to it,

$$\mathbf{W}^{\mathsf{T}}\mathbf{J}_{6}\mathbf{W} = \mathbf{Y} - \mathbf{Y}^{\mathsf{T}}, \tag{108}$$

so $\mathbf{Y}$ *symmetric* and *the downgoing subspace is Lagrangian* are one statement. Since $\mathbf{J}_{6}\mathbf{A} = (\mathbf{J}_{6}\mathbf{A})^{\mathsf{T}}$ makes the flow symplectic, and a symplectic flow carries Lagrangian subspaces to Lagrangian subspaces, symmetry is **preserved by the march rather than approached by it**. The measurement says exactly that: $\|\mathbf{Y} - \mathbf{Y}^{\mathsf{T}}\|/\|\mathbf{Y}\| \le 2.7 \times 10^{-16}$ at every contrast and every step count tried, with *no* improvement under refinement, because there is nothing to improve. Physically it is reciprocity; the symplectic identity used throughout this note as a discretisation check is this same fact one level down.

This also says what $\mathbf{Y}$ is as a geometric object: a point of the *Lagrangian Grassmannian*, the space of Lagrangian subspaces, of which symmetric matrices are one coordinate chart.

Why Y is bounded where V was not. The Grassmannian is compact; the chart is not. $\mathbf{Y}$ blows up exactly where its graph stops being transverse to $\{\mathbf{u} = 0\}$, and $\mathbf{R}$ blows up — by (106) — where $\mathbf{Y}_{u} - \mathbf{Y}$ becomes singular. **Both are chart singularities, not singularities of the field**, and they sit in different places. That is the structural reason behind the measurement of §16.8: $\mathbf{V}$ returning 2.000 from $|\mathbf{k}| = 600$ is the reflection chart failing, not the solution failing.

Whether the $\mathbf{Y}$ chart fails is then a physical question with a physical answer. $\mathbf{Y}$ becomes singular only if the region below has a resonance with a *clamped* surface, and a region radiating into a half-space at real ω has none, because energy leaks out of every candidate mode. That is

an argument rather than a proof, and it is offered as one; what supports it is the measurement already reported — $\|\mathbf{Y}\|/\mu|\mathbf{k}| \to 1.529022$ and $\mathrm{cond}(\mathbf{Y}) \to 2.0000$ across four decades, with no growth anywhere.

Energy: the anti-Hermitian part, and what its kernel counts. A Dirichlet-to-Neumann surface can lose energy downward but cannot be fed from below, so $(\mathbf{Y} - \mathbf{Y}^\dagger)/2i$ must be positive *semi*-definite. Semi, not definite: an evanescent channel carries no flux and contributes an exact zero, so the kernel is not slack in the statement but a count. One zero per P-evanescent lateral mode of the half-space below, two per S-evanescent one. At the gate's parameters ($k_P = 0.0120$, $k_S = 0.0200$, $\max|\mathbf{k}| = 0.0178$) that predicts 20 null channels of 108, and 20 is what is measured — with the most negative eigenvalue at -5.5×10^{-16} of scale and a gap of 0.24 of scale to the smallest surviving one, so the count is unambiguous.

Two things are worth drawing out. The prediction is a *count*, which no tolerance can be tuned to satisfy; and the regime it counts is precisely the one [9] single out as awkward — wavenumbers at which P has gone evanescent while S still propagates. Every one of the 20 is such a channel. The construction meets that case by not distinguishing it.

A Schur complement, which is why the quadratic term is irreducible. Discretise the region below the surface and eliminate every interior degree of freedom. With $\mathbf{K}$ the stiffness, b the boundary and i the interior,

$$\mathbf{Y} = \mathbf{K}_{bb} - \mathbf{K}_{bi}\mathbf{K}_{ii}^{-1}\mathbf{K}_{ib}, \tag{109}$$

which is static condensation of an entire half-space onto one surface, and $\mathbf{Y}\mathbf{A}_{12}\mathbf{Y}$ in (103) is $\mathbf{K}_{bi}\mathbf{K}_{ii}^{-1}\mathbf{K}_{ib}$ in differential form. This settles structurally what §17.4 measures. That term is 69% of the step and is dense in *every* basis, because the Schur complement of a connected region is dense however the region is parametrised — so no change of basis reaches it. It is also, for the same reason, the term that carries the multiple scattering: it is the only place where what is below the surface acts back on it.

The exact transparent boundary condition. A Dirichlet-to-Neumann map is a relation on a surface that replaces an infinite domain with no approximation whatever. An absorbing layer is an approximation to this object; here the object itself is in hand, which is why the slab is closed at its base by $\mathbf{Y}$ rather than by a sponge, and why a vertical absorbing layer never arises as a question. It arises laterally, where the grid is periodic and no such map is available — and an absorbing layer tried there is the one part of this construction that failed to earn its place, recorded in the gate rather than here because it produced no result to report.

$\mathbf{Y}$ is a complete state. Nothing below the surface enters except through (104), so the march carries no history. Stopping it at mid-slab and restarting from the value reached returns the same answer: to the arithmetic floor when the two stages use the same step size — it is then the same arithmetic in the same order — and to 1.4×10^{-7} when they do not, which is truncation error and not a discrepancy. The practical consequence is that regions compose by handing over one matrix, and the march exposes this as a starting value rather than hiding it. It is also, in a second guise, the reuse of §24.4: $\mathbf{Y}$ contains no reference to a source, so one march serves any number of them.

Its other names. The same operator is the *admittance* or *impedance operator* of [7, 8, 9], the *static condensation* of structural analysis, the *transparent* or non-reflecting boundary condition of numerical wave propagation, and — in one dimension — the Weyl–Titchmarsh m-function. Whether the map determines the medium is the Calderón problem, whose answer is that it does. That is worth recording because it identifies what the march computes as exactly the object an inverse problem would want to measure, although nothing in this note pursues it.

What it does not give, and the second sweep that supplies it. The map is a statement on one surface: it carries no information about the field in the interior. Reflection is therefore available from $\mathbf{Y}$ at the top alone, by (106), but *transmission is not* — transmission is the field carried through the slab, and recovering it needs the one-way equation

$$\mathbf{u}' = \big(\mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{Y}(z)\big)\mathbf{u}, \tag{110}$$

integrated **downward** from the surface while the Riccati runs upward against it. That is the double sweep of [9], and it is now built: §16.10 reports it. It costs one thing the single sweep did not need — $\mathbf{Y}$ at every depth rather than only at the top — which is why the upward sweep is run at twice the resolution and every value kept, so that the downward Runge–Kutta finds $\mathbf{Y}$ already evaluated at each of its stage depths and no interpolation enters.

Two limitations are unaffected by it. The map is frequency-by-frequency, hence non-local in time, so nothing here transfers to a time-domain march without an additional construction. And [9] reuse *fundamental solutions* as well as the operator; the second sweep is the machinery those are built from, but assembling and reusing them is not done here.

16.10 The reflection/transmission response, from both sweeps

Reflection needs one sweep and transmission needs two, so the second sweep is what turns an impedance into a scattering response. With unit downgoing amplitude above the slab the total field at the top is $\mathbf{u}(0) = (\mathbf{U}_d + \mathbf{U}_u\mathbf{R})$, and sweeping (110) down to the base – where the field is purely downgoing – gives

$$\mathbf{T} = \mathbf{U}_d^{-1}\,\mathbf{u}(2a). \tag{111}$$

The exact limit, first. At zero contrast every entry is known in closed form, which tests the sweep before any arbiter is involved: $\mathbf{R}$ vanishes to 10^{-15}, $\mathbf{T}$ is diagonal to 7×10^{-16} — a uniform slab mixes no modes — and $\mathbf{T}$ approaches $\operatorname{diag} e^{ik_{z,c}\cdot 2a}$ at *fourth* order, 3.96, 4.00, 4.00 over 4 to 32 steps. The downward sweep therefore carries the same order as the upward one, which is what the stage-matched $\mathbf{Y}$ history is for.

Every channel at once. The Mie comparison of §18 scores one column, because the plane-wave spectrum implemented alongside it is the $m = 0$ one. Energy balance scores *every entry of both matrices*: every incident channel, every outgoing channel, P, SV and SH alike, and every propagating diffraction order. Rescaling to the flux basis by $w_c(q) = \sqrt{c^2 k_{z,c}(q)/\omega}$ and restricting to the propagating channels, $\mathbf{R}^\dagger\mathbf{R} + \mathbf{T}^\dagger\mathbf{T} = \mathbf{I}$ holds to

steps	residual	order
8	2.32×10^{-3}	
16	7.56×10^{-5}	4.94
32	2.44×10^{-6}	4.95
64	8.23×10^{-8}	4.89

Table 14. Energy balance of the full response over the 44 channels of 108 that the 6×6 grid counts as propagating, with the contrast on. Only 23 of them are distinct orders (5 *P*, 9 *SV*, 9 *SH*); the other 21 sit at a zeroed Nyquist index and repeat the $k = 0$ component of a lower order, so the check includes them but they add no independent information. The rescaling w_c is not asserted — this project has twice claimed a displacement-to-flux relation from diagonal ratios and been wrong — it is written down and the identity it should produce is measured. Reaching 8×10^{-8} is the evidence that it is right.

Against exact Mie on the column its spectrum covers, the specular reflection agrees to 2.2×10^{-2} — the residual of §18 at $\varepsilon = 0.1$ — and the specular transmission to 4.6×10^{-4}. The two are not comparable with one another: transmission is dominated by the unscattered wave, so its relative error is necessarily the smaller.

Every order, in complex value, needs the sphere's lateral position. Mie places the sphere at the lateral origin; the march places it at the centre of the periodic cell, $(x_c, y_c) = \left(\frac{N_x-1}{2}\frac{L_x}{N_x}, \frac{N_y-1}{2}\frac{L_y}{N_y}\right)$, where N_x, N_y are the lateral grid sizes and L_x, L_y the periods. A lateral translation multiplies each scattered order by

$$e^{-i(k_x x_c + k_y y_c)}, \tag{112}$$

which is exactly 1 on the specular order and nowhere else. With it, every order of the *P* column agrees with Mie in complex value: over the 49 scoreable orders of an 8×8 grid the relative 2-norm residual is 0.070 for $\mathbf{R}$ and 0.038 for the scattered part of $\mathbf{T}$. Without it the same residual is 1.38 — every non-specular phase scrambled while every magnitude is right — so a comparison of specular entries, or of magnitudes, cannot see the omission. The orders indexed at a zeroed Nyquist wavenumber are excluded: they are labelled with a wavenumber they do not carry, and $(N_x/2, 0)$ reads as the specular order.

***S* incidence, scored.** The comparison above is the *P* column, because the plane-wave spectrum it uses is the $m = 0$ one — which is what a *P* wave at normal incidence excites. *S* incidence excites $m = \pm 1$, and that spectrum is now available: Section 7 of `Mathematica/MieSphericalWaves.wl`, "the channel spectra, general in m", carries it in closed form for all three families with the Sommerfeld path on both branches, and it is ported and checked against that file's 72-case exact-field reference at 1.5×10^{-8}.

Its normalisation is not assumed. Stationary phase relates a spectrum to a far field by $f = -2\pi i k_z c$, which for $m = 0$ reproduces $\sum_n a_n (-i)^n P_n$ exactly; requiring the same of $m = 1$ fixes the renormalisation $\rho(n) = -1/[n(n+1)]$, and the result then reproduces an independently written *SV* far field to 4×10^{-6} across a spread of angles. With that, the *SV* column of the response scores against Mie as

column	channel	orders	median march/Mie	
R	*P*	20	0.954	(range 0.935–0.966)
T	*P*	20	1.058	(range 0.992–1.112)
R	*S*	41	0.933	

Table 15. *SV* incidence at the Γ order, $k_S a = 2.4$, on a 6×6 grid of period 600 m: the median over orders of $|\mathbf{R}_{\text{march}}|/|\mathbf{R}_{\text{Mie}}|$, and likewise for **T**. The scatter is the residual of §18; the zeroed-Nyquist orders are excluded. A magnitude ratio is blind to phase; the complex residual is in the next table.

column	*P* channel	*S* channels	*S*, toroidal omitted
R	0.066	**0.098**	0.414
T	0.104	**0.075**	0.198

Table 16. The same column in complex value: the relative 2-norm residual $\|\mathbf{R}_{\text{march}} - \mathbf{R}_{\text{Mie}}\|/\|\mathbf{R}_{\text{Mie}}\|$ over the scoreable orders of each channel, with the translation phase (112) applied and, for **T**, the unscattered wave removed from both sides. The last column drops the toroidal term (113) from the prediction; including it lowers the *S* residual in both **R** and **T**.

One entry is a symmetry rather than a measurement and is worth keeping: the $m = 1$ angular function carries a factor $\sin\theta$, so an *S* wave *cannot* backscatter into *P* exactly along the axis. Mie gives identically zero there, and so does the march, to 8×10^{-15} of the specular $SV \to SV$: the square array is not axisymmetric, but it keeps the mirror $x \to -x$, under which the specular *P* amplitude of an *x*-polarised *S* wave is odd. The same mirror requires march/Mie to be equal at $\pm k_x$ in every channel, and it is, to 6.5×10^{-14} — a check on the cross-section of §18 that needs no arbiter.

The toroidal term. An *x*-polarised *S* wave excites the *toroidal* (*M*-type) family, $\mathbf{u} = \nabla \times (\mathbf{r}\chi)$, as well as the spheroidal one. Its potential is odd in azimuth, $\chi = h_n(k_S r)\, P_n^1(\cos\theta)\sin\varphi$, and its coefficient is

$$t_n = i k_S\, c_n\, \rho(n), \tag{113}$$

where c_n is the *SH* Mie coefficient, the scattered response per unit of its own incident coefficient $(2n+1)i^n/(ik_S)$. The factor ik_S is the ratio of the plane wave's own *M*-type and *N*-type expansion coefficients: with $\alpha_n = -(2n+1)i^n/[ik_S\, n(n+1)]$ on $\nabla \times \nabla \times (\mathbf{r}\, j_n P_n^1 \cos\varphi)$,

$$\hat{\mathbf{x}}\, e^{ik_S z} = \sum_n \alpha_n\, \nabla \times \nabla \times (\mathbf{r}\, j_n P_n^1 \cos\varphi) + ik_S\, \alpha_n\, \nabla \times (\mathbf{r}\, j_n P_n^1 \sin\varphi),$$

one curl fewer costing one power of k_S. This is determined, not fitted, and three ways independent of the march: the identity is reproduced from the exact Cartesian families in `Mathematica/ToroidalIncidentExpansion.wl` to 10^{-26}, where the opposite sign fails at $O(1)$; a least-squares fit of the expansion by finite-difference curls in Python recovers both coefficients to 10^{-8}; and the optical theorem for *S* incidence on the lossless sphere, $\sigma_{\text{sc}} = \sigma_{\text{ext}}$ with $\sigma_{\text{sc}} = \oint[(\alpha/\beta)|\mathbf{f}_P|^2 + |\mathbf{f}_S|^2]\, d\Omega$ and $\sigma_{\text{ext}} = (4\pi/k_S)\,\text{Im}\,\hat{\mathbf{x}} \cdot \mathbf{f}_S(0)$, holds to 10^{-8} with (113) and fails for $-ik_S$ (1.145), k_S (0.516) and 1 (3.13). The toroidal family conserves energy on its own, so the theorem alone cannot distinguish (113) from omitting the term; the expansion fixes the magnitude and the theorem the sign. The package's *SH* far field is not an arbiter here: it drops the toroidal part as "$O((ka)^4)$", which fails

at $k_S a = 2.4$, where $|c_n|$ runs 1.5 to 2.5 times the spheroidal ones, $|b_n \rho(n)|$ with b_n the $SV \to SV$ Mie coefficient.

16.11 The symplectic picture, and what it guarantees

The identity $\mathbf{J}_6\mathbf{A} = (\mathbf{J}_6\mathbf{A})^{\mathsf{T}}$ has been used three times already in this note — as a discretisation check in §3.3, as the reason the lateral operator must be derived rather than read off in §17.2, and as the source of $\mathbf{Y} = \mathbf{Y}^{\mathsf{T}}$ in §16.9. They are one structure, and it is worth setting out on its own, because it is what makes several of this note's results exact rather than approximate.

The algebra and the group. Written as $\mathbf{A}^{\mathsf{T}}\mathbf{J}_6 = -\mathbf{J}_6\mathbf{A}$, the identity is exactly the defining condition of the symplectic Lie algebra $\mathfrak{sp}(6N)$. Its exponential therefore lies in the symplectic group:

$$\mathbf{M} = e^{-\mathbf{A}h} \in \mathrm{Sp}(6N), \qquad \mathbf{M}^{\mathsf{T}}\mathbf{J}_6\mathbf{M} = \mathbf{J}_6, \tag{114}$$

measured here to 4×10^{-8} at $h = 5$ and 5×10^{-7} at $h = 80$ — a looser figure than the algebraic identity behind it, which holds to 8.7×10^{-17}, because it is the accuracy of the matrix exponential rather than of the structure. Physically the conserved form is the vertical power flux: $\mathbf{J}_6$ pairs $-\boldsymbol{\tau}_3$ with $\mathbf{v}$, which §1.3 chose for exactly that reason, so (114) is energy flux conservation written as a group property.

Lagrangian subspaces. A $3N$-dimensional subspace L of the $6N$-dimensional phase space is *Lagrangian* when the symplectic form vanishes identically on it — the largest dimension at which that is possible. The downgoing solutions span such a subspace, and (108) is the statement in coordinates: for the graph $\mathbf{W} = (\mathbf{I}, \mathbf{Y})^{\mathsf{T}}$, $\mathbf{W}^{\mathsf{T}}\mathbf{J}_6\mathbf{W} = \mathbf{Y} - \mathbf{Y}^{\mathsf{T}}$, so

$$\mathbf{Y} \text{ symmetric} \iff \text{the downgoing subspace is Lagrangian} \iff \text{reciprocity.} \tag{115}$$

The three are the same fact in three languages, and the middle one is the reason the outer two are exact.

The Lagrangian Grassmannian. Collect all such subspaces and the result is a compact manifold $\Lambda(3N)$ of dimension $3N(3N+1)/2$, on which $\mathrm{Sp}(6N)$ acts. Symmetric matrices are *one coordinate chart* on it, and (107) is that action written in the chart — which is why the transformation law is fractional-linear and why its generator is a Riccati equation. Every object this note marches is a curve in $\Lambda(3N)$; $\mathbf{Y}$ and $\mathbf{R}$ are two ways of writing down where that curve is.

What this buys: reciprocity is exact, and needs no symplectic integrator. This is the part worth stating carefully, because the obvious inference is wrong. One might expect that preserving $\mathbf{Y} = \mathbf{Y}^{\mathsf{T}}$ requires a structure-preserving integrator — and it does not. Write the right-hand side of (103) as $\mathscr{R}(\mathbf{Y})$. The symplectic identity makes $\mathbf{A}_{21}$ and $\mathbf{A}_{12}$ symmetric and $\mathbf{A}_{22} = -\mathbf{A}_{11}^{\mathsf{T}}$, so

$$\mathscr{R}(\mathbf{Y})^{\mathsf{T}} = \mathbf{A}_{21} + \mathbf{Y}^{\mathsf{T}}\mathbf{A}_{22}^{\mathsf{T}} - \mathbf{A}_{11}^{\mathsf{T}}\mathbf{Y}^{\mathsf{T}} - \mathbf{Y}^{\mathsf{T}}\mathbf{A}_{12}\mathbf{Y}^{\mathsf{T}} = \mathscr{R}(\mathbf{Y}^{\mathsf{T}}), \tag{116}$$

so $\mathscr{R}$ carries symmetric matrices to symmetric matrices. A Runge–Kutta step is a linear combination of evaluations of $\mathscr{R}$ added to $\mathbf{Y}$, and linear combinations of symmetric matrices are symmetric. **Symmetry is therefore preserved exactly by any explicit Runge–Kutta method, not merely to the order of the method**, and the measurement shows the signature: $\|\mathbf{Y} - \mathbf{Y}^{\mathsf{T}}\|/\|\mathbf{Y}\| \le 2.7 \times 10^{-16}$, unchanged between 8 and 32 steps and between zero and six per cent contrast. A quantity that is being approximated improves under refinement; this one does not move. It is also why the Möbius step and RK4 agree on reciprocity although only the former is symplectic.

What this buys: a discretisation test with a control. The identity belongs to the continuum operator, and a discretisation inherits it only if it is built correctly — which makes it the sharpest available check on the lateral operator. §17.2 derives the traction rows with each derivative *outside* its modulus, and that is what the identity is testing: the derived ordering gives 8.7×10^{-17}, the plausible wrong ordering with the modulus pulled outside gives 1.3×10^{-1}, and the two are indistinguishable to 1.3×10^{-17} in a laterally uniform medium. A check that cannot fail proves nothing, so the control is the point: this identity is the only test in the note that distinguishes the two orderings at all.

Where the charts fail. The subspaces that are *not* transverse to the reference one form the Maslov cycle of $\Lambda(3N)$, and that is exactly the set where the $\mathbf{Y}$ chart blows up; $\mathbf{R}$ blows up on a different cycle, which is §16.9's point that the two fail in different places. Since $\Lambda(3N)$ is compact and the charts are not, some such failure is unavoidable for any coordinate — the question is only whether the physical problem drives the curve into that particular cycle, and §16.9 argues on radiation grounds that for $\mathbf{Y}$ it does not. Counting crossings is the Maslov index, which in one dimension is the oscillation theory of Sturm and Liouville; nothing here uses it, and it is named only so that the structure is recognisable.

16.12 Marching it through a slab, and scoring it

§16.8 only ever solved the *algebraic* Riccati equation, in a homogeneous medium. This is the march: integrate (103) upward from the base of a slab, starting from the downgoing impedance of the background half-space — the radiation condition, and the continuum form of $\mathbf{Y}_{M+1} = 0$ — then read a reflection matrix off the result using continuity of $\mathbf{u}$ and $\boldsymbol{\tau}_3$ and nothing else — equation (106), which §16.9 reads as the statement that $\mathbf{R}$ is $\mathbf{Y}$ expressed in a chart that costs a mode decomposition.

Two integrators are carried so that each checks the other. For a piecewise-constant medium the step is the *linear-fractional transform* (107) and is exact, so one step settles a uniform slab outright; RK4 applied to (103) directly never forms $e^{-\mathbf{A}h}$ at all, and is the control.

Absolute, before anything relative. At normal incidence one step of (107) reproduces the closed form to 1.4×10^{-16}, 4.5×10^{-16} and 3.8×10^{-15} at contrasts of +2%, +10% and +40% — with no discretisation in it anywhere. RK4 then converges on the same answer at ratios 11.4, 15.3, 15.9, 15.9: fourth order, confirming that (103) is what is being integrated.

Against Kennett. The thesis and Kennett amplitude conventions differ, and §16.13 determines exactly how; for the moment what is compared is the three *diagonal* ratios, which must be constant because a convention cannot depend on angle. Across $\sin i = 0$ to 0.75 they are constant to 4.6×10^{-14}, and equal $(-1, -1, +1)$ to 4.8×10^{-14}. **The *SH* sign does not flip where *P–SV* does**; the split is real, and lumping the three together is what made this gate's first run report a spurious 2.000. The collocation route of §12 reaches the same three constants, $(-1.000042,\ -1.000166,\ +1.000166)$ at 120 cells, by an entirely different path — so this is a cross-check between two independent routes rather than a fitted offset, and the march reaches it ten orders more sharply. The constants survive past both critical slownesses, where the march has to carry evanescence.

The graded slab, which is the point. A uniform slab never needed a march. Grade the contrast with depth and no closed form exists — while Kennett, cut at the *same* depths, is solving the identical piecewise-constant problem. The two must therefore agree to the arithmetic floor at finite step count, not merely converge together, and they do: 3.6×10^{-15}, 2.2×10^{-15} and 1.1×10^{-14} at 4, 16 and 64 steps, at normal and oblique incidence alike. Reversing the depth profile — same layers, same thicknesses, reordered — changes $\mathbf{R}$ by 50%, so the agreement is not the insensitivity of a march that cannot see where the contrast sits.

16.13 What the convention actually is

> *A correction, and a instructive one. Both this section and §12 previously said the two conventions "differ by a fixed matrix". That was asserted on the strength of the three* diagonal *ratios, which cannot establish it — and it is wrong twice: the relation is two-sided rather than a single matrix, and it is not fixed.*

A convention is a renormalisation of mode amplitudes, and the incident and reflected sets need not be renormalised the same way. If $\mathbf{a}_d^{\mathrm{th}} = \mathbf{S}_d\,\mathbf{a}_d^{\mathrm{ken}}$ and $\mathbf{a}_u^{\mathrm{th}} = \mathbf{S}_u\,\mathbf{a}_u^{\mathrm{ken}}$ then

$$\mathbf{R}_{\mathrm{th}} = \mathbf{S}_u\,\mathbf{R}_{\mathrm{ken}}\,\mathbf{S}_d^{-1}, \tag{117}$$

which is a *two-sided* transform. If both are diagonal, the elementwise ratio $Q_{ij} = (\mathbf{R}_{\mathrm{th}})_{ij}/(\mathbf{R}_{\mathrm{ken}})_{ij}$ satisfies $Q_{ij} = (\mathbf{S}_u)_i/(\mathbf{S}_d)_j$ — rank one in log space, so

$$Q_{PP}\,Q_{SS} = Q_{PS}\,Q_{SP} \tag{118}$$

exactly. That is a sharp test of diagonality, and the diagonal-only measurement never performed it. Measured, (118) holds to 1.9×10^{-14}: both renormalisations *are* diagonal.

Writing $Q_{ij} = -s_i/s_j$ with $s = (1,\ ic)$, the off-diagonals give c, and

$$c = \sqrt{k_{z,S}/k_{z,P}}\,, \tag{119}$$

to 5.6×10^{-15} across the sweep. This is the displacement-versus-flux difference: Kennett normalises modes to unit vertical energy flux, which carries $\sqrt{k_z}$ per mode, while the columns here are normalised by their own displacement component. **c runs from 1.293 to 2.466 over $p = 2 \times 10^{-5}$ to 1.95×10^{-4} and diverges as $k_{z,P} \to 0$ at the P critical slowness, so the convention**

is not fixed. What is fixed is its diagonal — $\mathbf{S}_u = -\mathbf{S}_d$ on *P–SV* and $\mathbf{S}_u = +\mathbf{S}_d$ on *SH*, giving $(-1,-1,+1)$ — which is all §16.12 and §12 ever saw.

Two further things are worth stating precisely. The overall scale is not observable at all: (117) is unchanged by $\mathbf{S}_{u,d} \to \lambda\mathbf{S}_{u,d}$ on both sides, so only $\mathbf{S}_u\mathbf{S}_d^{-1}$ and the P/S ratio within each set are determined — reporting a single convention matrix would be overclaiming a third time. And the standing signature of a displacement-versus-flux difference is that it is *invisible to any per-mode ratio test*, which is precisely the test that was being run.

16.14 What this is, against the thesis's own development

It is worth being exact about which part of this is new, because most of it is not, and the parts that are not are the load-bearing ones.

The problem the thesis set. The target is laterally varying propagation in which *the dominant material variation is in depth.* That is not a simplification of convenience: it is the structure of the medium. Depth-average properties generally increase with depth, so the reference carries a strong gradient, while the lateral departure from that depth-average is superimposed and comparatively weak. The thesis's two-potential split is built on exactly that asymmetry — the depth-dependent profile is solved *exactly and never expanded*, and only the lateral contrast is carried as a series — and, because the contrast is a deviation from the depth-average at each depth, it has zero lateral mean by construction, so the reference is the optimal expansion point and not merely a convenient one.

The thesis already says this is the regime for a Riccati sweep. Its Introduction states the criterion directly: the Riccati sweep "factorizes the reference medium Green's tensor, but does not include any of the scattered field in the reference solution. Hence, a multiple scattering series based on this scheme converges rapidly, *even when there are strong gradients in the reference model*, provided that the superimposed heterogeneity is not too strong." Strong gradients in the reference, weak superimposed heterogeneity: that is the stated regime, and it is the regime above. The complementary case — smooth reference, strong heterogeneity — the thesis assigns to its cumulative forward-scattering factorisation instead, so the choice between them is already made and is not being re-opened here.

What is therefore borrowed. The formulation, the regime, the criterion, the vertical operator and the economy that makes it affordable are all the thesis's [29]: the block-tridiagonal structure (eq. `Amat`), the Riccati factorisation and Algorithm 5.1, the $\mathbf{P}^z$ vertical sweep, and the observation that $Q^{\partial} = (\mathbf{I} - \mathscr{S}_{int}\mathscr{E})^{-1}\mathscr{S}_{int}$ depends on the reference alone and so is built *once* per frequency, after which each Krylov iterate is a cheap forward–backward sweep. Nothing here displaces any of that.

What is new, and it is two things. First, the marched variable. The thesis factorises in reflection/transmission operators on mode amplitudes; §16.7 measures why that has a ceiling — P and S coalesce at large lateral wavenumber, $\mathrm{cond}(\mathbf{D}_z)$ grows as $|\mathbf{k}|^2$, and the mode-basis

coefficients lose the *branch* by $|\mathbf{k}| = 600$ — and §16.8 shows the impedance is free of it, because $\mathbf{Y}$ needs only the downgoing columns and never the full inverse. This is not academic for the present problem: resolving lateral structure of scale a on N points demands $|\mathbf{k}|_{\max} \sim \pi N/a$, so the ceiling is reached by refining the lateral mesh, which is precisely what a laterally varying calculation does.

Second, the march is a *continuum* one. The thesis factorises a stack of interfaces; (103) is an ODE in z. For a stack the two agree exactly — §16.12 measures agreement with Kennett to 10^{-14} on a graded stack cut at the same depths — but for a *gradient* they do not, and that is the case this problem is made of.

16.15 What the continuum march buys, measured

A layer stack approximates a gradient by staircasing it. That caps the accuracy at second order in the step, however the layer response is computed, because the error is then in the medium and not in the solver. Integrating (103) through the true profile has no such cap. Measured on a smooth gradient, against a converged reference:

steps	staircase	order	impedance ODE	order
8	3.35×10^{-2}	—	1.42×10^{-2}	—
16	8.27×10^{-3}	2.02	8.74×10^{-4}	4.02
32	2.06×10^{-3}	2.00	5.42×10^{-5}	4.01
64	5.15×10^{-4}	2.00	3.38×10^{-6}	4.00
128	1.29×10^{-4}	2.00	2.11×10^{-7}	4.00

Table 17. Second order against fourth, on the same profile. To reach 10^{-8} the staircase needs 16 384 steps and the ODE 512 — a factor of 32 in the depth direction, which multiplies every lateral wavenumber and every frequency.

The fourth order has to be earned and was not, at first: sampling the medium at each sublayer's midpoint and running RK4 inside that sublayer integrates the staircase *exactly* rather than the profile *accurately*, and returns order 1.96 — the medium approximation, not the integrator, being the error. Evaluating $\mathbf{A}$ at the Runge–Kutta stage depths is what makes the order real. The distinction is recorded because it is invisible in any test against a layered arbiter: §16.12 agrees with Kennett to 10^{-14} either way, since there both are solving the same piecewise-constant problem.

What it does not buy. The economy above is in the depth direction only. It does not change the lateral cost, it does not change the number of Krylov iterations, and it says nothing about strong scattering — for which the thesis's own criterion, weak superimposed heterogeneity, still governs whether the series converges at all.

17 The lateral coupling, carried inside the march

17.1 Where the thesis puts it, and where this puts it

The thesis already solves the laterally varying problem, and it makes a definite architectural choice. Its eq. `(CBPdef0)` assembles

$$\mathbf{P}_{r,s} \;=\; \mathbf{P}^{z}_{r,s} + \mathbf{P}^{x}_{r,s} \;=\; -i\,\Theta^{z}_{in}\,Q^{\partial}\,\Theta^{z}_{ou} \;+\; \mathrm{diag}\big[\mathbf{P}^{x}_{r,s,1},\;\ldots,\;\mathbf{P}^{x}_{r,s,L}\big], \tag{120}$$

acting on an $8LN_x$ vector — eight augmented components, L depth rows, N_x lateral sites. Its contrast array `(dCBC)`, `(dCBCrho)` is doubly block diagonal: over the L rows, and within each row over the N_x sites, each site carrying its own 8×8 $\Delta\mathbf{C}_{\mathrm{eff}}(\mathbf{R}_j;k_y)$ weighted by the cell area. So the vertical operator is *diagonal in lateral wavenumber* — Algorithm 5.3 is a mode-domain reflection/transmission recursion at fixed k_x — and every lateral coupling lives in the site-diagonal contrast together with the sideways sweeps $\mathbf{P}^{x}$ of Algorithm 5.2, the two being reconciled by an outer iteration on $(\mathbf{I}-\mathbf{P}\,\Delta\mathbf{C}_{\mathrm{eff}})$.

What follows does the opposite, and that is the only part of it that is new. The lateral grid is carried *inside* the marched variable: $\mathbf{Y}$ becomes a matrix on (3 components) $\times$ (N_x lateral points) and the lateral coupling is integrated along with depth, with no outer iteration for it at all. The geometry is the thesis's own $2\frac{1}{2}$-D — heterogeneity in (x,z), the third direction carried by the parameter k_y — so that the thesis's solver remains available as an arbiter.

17.2 The operator has to be derived, not read off

A laterally varying $\mathscr{A}$ cannot be transcribed from `(Akdef)`. That equation is a *symbol* in k_x; it says nothing about where a spatially varying modulus sits relative to a derivative, and the two orderings that differ are indistinguishable in any laterally uniform medium. Momentum balance does say. With $\zeta = 4\mu(\lambda+\mu)/\kappa$, $\chi = 2\mu\lambda/\kappa$ and $\gamma = \lambda/\kappa$, eliminating T_{xx} and T_{xy} gives

$$\partial_3 T_{xz} = -\rho\omega^2 u_x - \partial_x\big[\zeta\,\partial_x u_x + \chi\,\partial_y u_y + \gamma\,T_{zz}\big] - \partial_y\big[\mu\,(\partial_x u_y + \partial_y u_x)\big], \tag{121}$$

with the derivative *outside* the modulus. That ordering is forced by $\partial_j T_{ij}$, not chosen, and it is what leaves the discretised operator symplectic. The lateral derivative is spectral on a periodic grid, hence exactly antisymmetric, which turns the symplectic relation into an available test rather than an approximation:

$$\mathbf{J}_6\,\mathscr{A}(k_y) \;=\; \big[\mathbf{J}_6\,\mathscr{A}(-k_y)\big]^{\mathsf{T}_x}. \tag{122}$$

Here T_x denotes transposition in the *lateral grid index alone*, the six component indices being left as they are — it is not the ^{T} used elsewhere in this note. The parameter k_y reverses with it because in $2\frac{1}{2}$-D it is a parameter and not a grid direction, and the formal transpose of ∂_x is $-\partial_x$ while a modulus is its own transpose. (122) reduces to (74) when the medium is laterally uniform. It holds to 1.9×10^{-16} for a laterally varying medium, and the plausible wrong ordering — $\mu\,\partial_x^2$ in place of $\partial_x\mu\,\partial_x$ — breaks it by 1.2×10^{-1} while agreeing with (121) *identically* when the medium does not vary. No laterally invariant test could have separated them.

Nothing is trusted on the strength of that alone. Transformed to the lateral wavenumber domain in a uniform medium the operator must reproduce `(Akdef)` block by block, and does, to

10^{-15} at every grid wavenumber, for two media and for $k_y = 0$ and $k_y \neq 0$.

17.3 What survives, measured

The reduction is exact. With the lateral contrast switched off but the depth gradient left on, the coupled march must return a $\mathbf{Y}$ that is block diagonal in k_x whose nth block is the 3×3 march of §16.12. It is, to 3×10^{-15}, with no spurious coupling generated by the discretisation (6×10^{-16}). Two structural identities come with it: at $k_y = 0$, *SH* decouples from *P–SV* exactly however the medium varies laterally; and shifting the medium one grid point conjugates the marched impedance by the shift, to 10^{-15}, which is what catches an index slip between the medium array and the derivative matrix.

The fourth order survives the coupling. On a profile smooth in *both* depth and lateral position, against a converged reference:

steps	staircase	order	impedance ODE	order
12	2.73×10^{-3}	—	3.75×10^{-4}	—
24	6.82×10^{-4}	2.00	2.30×10^{-5}	4.02
48	1.70×10^{-4}	2.00	1.43×10^{-6}	4.01
96	4.26×10^{-5}	2.00	8.88×10^{-8}	4.01

Table 18. The orders of §16.15 are unchanged by the lateral coupling.

An independent algorithm on the same operator agrees. Rather than marching the quadratic Riccati for a $3N_x \times 3N_x$ impedance, one may march the *linear* system for a $6N_x \times 3N_x$ fundamental block started at $[\mathbf{I}; \mathbf{Y}_0]$ and form $\mathbf{Y} = \mathbf{T}\mathbf{U}^{-1}$ only at the top. The two disagree by their own truncation, and the discriminating test is that the disagreement falls at fourth order — 2.2×10^{-5}, 1.4×10^{-6}, 9.2×10^{-8}, 5.8×10^{-9} at $24, 48, 96, 192$ steps, orders 3.91, 3.97, 3.99. A genuine error in the Riccati reduction would plateau. The linear route is also where the reason for marching an impedance becomes visible directly. At fixed step size, as the slab thickens cond($\mathbf{U}$) runs 1.3×10^{4}, 3.4×10^{11}, 3.7×10^{22}, 5.2×10^{28} at 200, 600, 1200, 2000 m and the route fails outright — the two answers disagree by 99% at the last two — while $\|\mathbf{Y}\|$ stays finite throughout and within a factor 1.27 of itself.

The explicit step is limited, and the lateral mesh tightens it. This was found by the gate failing, and it matters for the economy claimed in §16.15. The right-hand side of (103) has decay rates set by the vertical wavenumbers, so the *largest lateral* wavenumber on the grid sets the largest rate, and explicit RK4 is stable only while $2|k_x|_{\max} h$ is of order one — measured, the march returns `nan` at $2|k_x|_{\max} h = 1.31$ and is stable from 0.98 down. The Möbius step (107), being exact per sublayer, has no such limit and is stable at every step tried.

The consequence is that the lateral mesh and the depth step are coupled: refining laterally raises $|k_x|_{\max}$ and therefore forces a smaller depth step for the explicit integrator. That offsets part of the fourth-order economy of §16.15, and how much of it is offset in any particular calculation has not been measured. An implicit or exponential integrator would not pay it, and neither would the Möbius step at the cost of returning to second order.

17.4 Where the cost actually is

An earlier draft of this section asserted that $\mathbf{R}\,\mathbf{V}_{du}\,\mathbf{R}$ — "two dense products per step" — is the cost. Building it says otherwise on both counts, and the reason is structural. $\mathbf{A}_{12}$ is the inverse vertical stiffness, $\mathrm{diag}(1/\kappa, 1/\mu, 1/\mu)$: a pure multiplication operator carrying no derivative, hence *exactly* diagonal on the lateral grid. So $\mathbf{Y}\mathbf{A}_{12}\mathbf{Y}$ needs one dense product, not two — measured, 0.189 ms naive against 0.093 ms exploiting the diagonal. The other blocks are block sparse at the component level, occupying 2, 3 and 2 of nine $N_x \times N_x$ sub-blocks, each occupied sub-block that carries a derivative being dense laterally. Counting one right-hand side in units of N_x^3:

term	flops	structure
$\mathbf{Y}(\mathbf{A}_{12}\mathbf{Y})$	27	one dense $(3N_x)^3$ product; $\mathbf{A}_{12}$ diagonal
$\mathbf{A}_{22}\mathbf{Y}$	6	2 of 9 sub-blocks occupied
$\mathbf{Y}\mathbf{A}_{11}$	6	2 of 9 sub-blocks occupied

Table 19. The quadratic term is 69% of the step — the dominant single cost, but because the linear terms are block sparse, not because it is two products.

So the conclusion the earlier draft reached survives, with its stated reason replaced. The quadratic term is where a near-field preconditioner would act, and that remains a claim about physics — it is the term carrying the back-scattering — rather than about arithmetic.

17.5 What this does not establish

The march is built, reduces exactly, keeps its order and its symplectic structure, and stays on the downgoing branch with $\|\mathbf{Y}\|$ constant to six figures across a sixteen-fold change in step count. What is *not* measured is any end-to-end advantage over the thesis's own architecture (120) on a problem either could solve. That comparison needs a laterally coupled arbiter, and the natural one — the thesis's $2\frac{1}{2}$-D solver — has not been run against it. Until it has, the case for putting the coupling inside the march rests on the structural results above, on the exact-sphere verification of §18, and not on a measured speed advantage against the thesis's own solver.

18 Three-dimensional lateral structure, and the exact-sphere verification

18.1 Both lateral directions carried as grid directions

Section 17.2 carries the lateral coupling inside the marched variable along one direction, with the third held as the parameter k_y. A medium invariant in y is then represented exactly and a medium that is not is not represented at all: no single k_y describes a sphere, however fine the lateral grid.

Both directions become grid directions by the same construction. The impedance $\mathbf{Y}$ acts on (3 components) $\times$ $(N_x N_y)$ lateral points, the lateral index running x-slowest so that

$$\mathbf{D}_x = D_x \otimes \mathbf{I}_{N_y}, \qquad \mathbf{D}_y = \mathbf{I}_{N_x} \otimes D_y, \tag{123}$$

each factor the spectral derivative of its own direction. Every ik_y of (121) becomes $\mathbf{D}_y$, and the modulus/derivative ordering that momentum balance forces is the same ordering applied twice: the two horizontal traction rows read

$$\partial_3 T_{xz} = -\rho\omega^2 u_x - \partial_x\big[\zeta\,\partial_x u_x + \chi\,\partial_y u_y + \gamma\,T_{zz}\big] - \partial_y\big[\mu(\partial_x u_y + \partial_y u_x)\big], \tag{124}$$

$$\partial_3 T_{yz} = -\rho\omega^2 u_y - \partial_y\big[\chi\,\partial_x u_x + \zeta\,\partial_y u_y + \gamma\,T_{zz}\big] - \partial_x\big[\mu(\partial_x u_y + \partial_y u_x)\big], \tag{125}$$

with every derivative outside its modulus. The constitutive rows keep the modulus inside, because they are not divergences.

The symplectic statement becomes *simpler*, not more complicated. In $2\frac{1}{2}$-D it had to be written $\mathbf{J}_6\mathbf{A}(k_y) = [\mathbf{J}_6\mathbf{A}(-k_y)]^{\mathsf{T}}$ — a transpose in the lateral index together with an explicit sign flip, because k_y is a parameter and transposition does nothing to a scalar. Here $\mathbf{D}_y$ is a real antisymmetric matrix, so it reverses its own sign under transposition and the identity closes outright,

$$\mathbf{J}_6\mathbf{A} = (\mathbf{J}_6\mathbf{A})^{\mathsf{T}}, \tag{126}$$

with no parameter to flip. It holds at 8.7×10^{-17}, and the ordering with the modulus outside every derivative pair — indistinguishable in a laterally uniform medium — breaks it at 1.3×10^{-1}.

The operator reduces to objects established earlier: to $\mathbf{A}$ of (4) at each of the N_xN_y wavenumber pairs in a uniform medium (2.1×10^{-15}), to the $2\frac{1}{2}$-D operator block by block in k_y when the medium is invariant in y (3.6×10^{-16}), to the $2\frac{1}{2}$-D *march* at every k_y (3.9×10^{-15}), and to the scalar march of §16.8 at zero lateral contrast (3.8×10^{-15}). Shift covariance holds in each direction separately, which is what distinguishes an index slip between the two Kronecker factors from a consistent relabelling.

18.2 The cross-section enters through its area, not its radius

A sphere of radius a centred at depth z_c cuts the plane at depth z in a disc of radius $R(z) = \sqrt{a^2-\zeta^2}$, $\zeta = z - z_c$. Painting that disc onto the lateral grid cell by cell makes the marched coefficient piecewise constant in depth: it changes in jumps as $R(z)$ crosses successive grid radii, the integrator reproduces the staircase exactly, and the order of §16.15 is lost.

The disc has an exact two-dimensional transform,

$$F(q,R) \;=\; \frac{2\pi R\,J_1(qR)}{q} \;=\; 2\pi R^2\,\frac{J_1(qR)}{qR}, \qquad F(0) = \pi R^2, \tag{127}$$

and $J_1(x)/x$ is even and entire. So F is a power series in $(qR)^2$, hence in $R^2 = a^2-\zeta^2$ — a polynomial in depth. *The contrast taken in wavenumber is therefore an analytic function of z across the whole of the sphere's bounding slab, poles included*, because the cross-sectional area enters and not the radius, and the area vanishes linearly in R^2 where the radius has a square root. Marching over exactly $[z_c-a,\,z_c+a]$ places the only non-smoothness — the junction to the identically zero exterior — at the endpoints, never inside a step.

Measured on the sphere's own bounding slab, with the cross-section built from (127) and sampled at the stage depths:

	depth steps			
	32	64	128	256
exact transform	3.03×10^{-5}	1.72×10^{-6}	1.02×10^{-7}	6.16×10^{-9}
observed order		4.14	4.08	4.05
voxelised control	6.29×10^{-3}	2.06×10^{-3}	1.77×10^{-3}	3.60×10^{-4}
observed order		1.61	0.22	2.30

Fourth order is attained on a sphere with a sharp boundary. The control does not merely converge more slowly: it has no order at all, because a staircase's error depends on where the step boundaries happen to fall relative to the radius crossings, so refinement moves the error rather than reducing it.

18.3 The array relation

The march is laterally periodic, so it solves an array of spheres, one per cell of area $A = L_x L_y$. At normal incidence every sphere sees the same incident phase, so the array field is $\sum_{\mathbf{R}} \mathbf{u}_{\mathrm{sc}}(\mathbf{r} - \mathbf{R})$, and Poisson summation gives

$$\mathbf{u}_{\mathrm{array}} = \frac{(2\pi)^2}{A} \sum_{\mathbf{g}} \hat{\mathbf{u}}(\mathbf{g})\, e^{i\mathbf{g}\cdot\boldsymbol{\rho} + ik_z(\mathbf{g})|z|}, \tag{128}$$

so each diffraction order carries the isolated sphere's angular spectrum times $(2\pi)^2/A$. Referring the incident wave from the top of the bounding slab to the sphere centre and the scattered order back again supplies the two phases, and the reflection matrix extracted from $\mathbf{Y}$ at the top of the slab satisfies

$$\mathbf{R}[\mathbf{g} \leftarrow \mathbf{0}] = \frac{(2\pi)^2}{A}\, c_{\mathrm{mode}}(\mathbf{g})\, e^{i(k_P + k_z(\mathbf{g}))a}. \tag{129}$$

The mode amplitudes c_{mode} are those of the elastic Mie solution, assembled as plane-wave amplitudes rather than as a field: at one lateral wavenumber P and S carry different k_z, so they cannot be summed into a single vector before a depth is chosen.

Equation (129) is exact but for scattering *between* array elements, which is second order in the contrast where the signal is first. It is written for a sphere at the lateral origin; the march places the sphere at the centre of the periodic cell, and each order then carries the translation phase (112) as well.

18.4 What the three refinement axes measure

The comparison has three independent resolutions — the depth step, the lateral grid at fixed period, and the period at fixed pitch — and they do not measure the same kind of quantity.

Write ε for the fractional perturbation of α, β and ρ inside the sphere. Because the single-sphere signal is $O(\varepsilon)$ and the array coupling $O(\varepsilon^2)$, the *relative* departure of the march from (129) is linear in ε. At $k_P a = 1.44$, with a period of 2.5 diameters, on an even (8×8) and an odd (9×9) grid:

ε	0.001	0.003	0.010	0.030	0.100
departure/ε, $N = 8$	0.157	0.158	0.163	0.175	0.218
departure/ε, $N = 9$	0.190	0.190	0.193	0.202	0.232

constant over two decades, the drift at the largest contrast being the next order appearing.

Linearity does not by itself identify the departure as coupling: any defect that reaches the specular order only at second order in the contrast scales the same way. One such defect is easy to make. On an even grid the Nyquist wavenumber of the lateral derivative is set to zero, and a cross-section built from (127) with that wavenumber inherits a component $F(0)\,(-1)^j$ at the Nyquist modes, where the true content is exactly zero — a disc centred between grid points is symmetric about its centre, and $e^{i\pi j}$ is antisymmetric about it. That component overshoots the disc by 75% at every N, breaks its mirror symmetry at order one, and produces a departure of $1.45\,\varepsilon$ that is just as linear. The odd grid, which has no Nyquist mode, is the control; with the Nyquist modes zeroed the even grid agrees with it to within the lateral resolution, the disc's overshoot is the Gibbs constant 1.0895, and it is mirror-symmetric to round-off.

The depth ladder converges to the residual rather than through it. At $\varepsilon = 10^{-3}$ on the 8×8 grid, differenced against the finest march so that the part no depth step removes cancels:

depth steps	16	32	64	128
departure	4.569×10^{-4}	1.664×10^{-4}	1.572×10^{-4}	1.567×10^{-4}
less the finest	4.031×10^{-4}	2.528×10^{-5}	1.576×10^{-6}	9.28×10^{-8}
observed order		4.00	4.00	4.09

Fourth order in the depth step, measured against an exact solution rather than against the march itself.

The lateral grid at fixed period does move the residual, because the band-limited disc converges to the sharp one only algebraically at its edge. At $\varepsilon = 10^{-3}$ and $L = 600\,\mathrm{m}$ the departure is 0.157, 0.209, 0.231, 0.244 and 0.252 times ε at $N = 8$, 12, 16, 20 and 24, each refinement moving it less than the last — roughly as $1/N$ — so the converged residual at this spacing is about $0.3\,\varepsilon$.

The period at fixed pitch moves it too — 1.57, 1.56 and 2.45×10^{-4} at 2.5, 3.8 and 5.0 diameters and a 75 m pitch — without falling monotonically, which is what an interference does and a $1/A$ falloff would not.

The residual, predicted. Because the array and the isolated sphere share their first-order response exactly, their difference begins with double scattering, and second-order Born computes it without the march. Per lateral wavenumber $\mathbf{q}$ the double-scattering amplitude is

$$I(\mathbf{q}) = \iint dz\,dz'\; e^{ik_P(z+z')}\,F(q,z)\,F(q,z')\,K(\mathbf{q};z-z'), \tag{130}$$

with F the transform (127) of the cross-section at each depth and K the two-vertex kernel: the weak-form Born vertex at each end and the spectral Green's tensor $\hat{\mathbf{G}} = [k_S^2 \hat{g}_S \mathbf{I} + \mathbf{D}\mathbf{D}(\hat{g}_S - \hat{g}_P)]/\rho\omega^2$, $\mathbf{D} = (iq_x, iq_y, \partial_\zeta)$, $\hat{g}_K = (i/2\kappa_K)\, e^{i\kappa_K|\zeta|}$, between them. Up to four ζ-derivatives act on $\hat{g}_S - \hat{g}_P$; their δ' and δ'' parts cancel between S and P identically in $\mathbf{q}$ and the contrasts, and the one contact term that survives, $(\delta\lambda + 2\delta\mu)^2 k_P^4\, \delta(\zeta)/\rho\omega^2$, is independent of $\mathbf{q}$ — the spectral form of the Eshelby self-term. The kernel is derived twice, in `Mathematica/ElasticBornSecondOrderKernel.wl` and independently in `sympy`, agreeing to 1.7×10^{-15}.

It is also tested against the exact sphere, at every $\mathbf{q}$ and with its contact term, which the array's difference from one sphere cannot see. The isolated sphere's integral converges slowly — its large-q tail is the local self-interaction, falling only as $q^{-1.4}$ — so the kernel built on the static (Kelvin) tensor, which has the same contact term and approaches the dynamic one as $(k/q)^2$, is subtracted: the remainder falls as q^{-5} and is integrated numerically, and the static self-term is added back in real space, exactly. For a ball the static potentials of a polynomial source are polynomials inside it, so with the plane waves expanded to 36 Taylor terms every step is elementary (`Mathematica/BornSecondOrderStatic.wl`, checked against Poisson's equation, the biharmonic identity and the static Navier equation). First order with the ε^2 part of the contrast, -95.01999, plus the remainder, $60.32414 + 136.34050i$, plus the static self-term, -0.52061, gives $-35.21646 + 136.34050i$ for the ε^2 coefficient of the backscatter; Mie's own is $-35.21616 + 136.34011i$, agreement to 3.3×10^{-6} of the double scattering.

The array takes $(1/A)\sum_{\mathbf{g}}$ over diffraction orders and the isolated sphere $\int d^2q/(2\pi)^2$, so the relative departure per unit contrast is

$$\frac{1}{4\pi\rho\alpha^2 f_1}\Big[\frac{1}{A}\sum_{\mathbf{g}} I(\mathbf{g}) - \int \frac{d^2q}{(2\pi)^2}\, I(\mathbf{q})\Big], \tag{131}$$

f_1 the first-order backscatter amplitude, which agrees with Mie's linear coefficient to 10^{-6}. By Poisson summation the bracket couples a sphere only to images at least $L - 2a$ away, so the same smooth window $e^{-(q/Q)^8}$ may be applied to both terms; the prediction does not move, to six figures, over $Q = 0.08$ to 0.16, while each term moves by three per cent. It is $0.2880 - 0.0556i$ at 2.5 diameters. The march, converged in depth, gives an imaginary part of -0.0558 at every lateral resolution, and a real part that rises under lateral refinement — 0.146, 0.201, 0.224, 0.237, 0.245 at $N = 8$ to 24 — and extrapolates in $1/N$ to between 0.280 and 0.290.

The array, solved exactly. Second-order Born settles the residual only as $\varepsilon \to 0$. At finite contrast the array has an exact solution of its own, by layer multiple scattering: each sphere's outgoing multipole coefficients are $\mathbf{b} = \mathbf{T}(\mathbf{I} - \mathbf{G}_0\mathbf{T})^{-1}\mathbf{a}_{\text{inc}}$, with $\mathbf{T}$ the Mie T-matrix of each order and $\mathbf{G}_0$ the lattice coupling of the L, M, N multipoles — the scalar translation contracted with the planar structure constants, and the toroidal family through $\mathbf{M}_{nm} = -i\mathbf{L}\psi_{nm}$ with $\mathbf{L}$ the angular momentum about the multipole's own centre, which reduces its lattice sum to scalar ones. The structure constants are the lattice field projected on a sphere of radius $0.7\,L$, where their product with j_q behaves like 0.7^q and nothing is lost to order $q = 36$. With the coupling switched off it reproduces the isolated sphere's predictions at every order to 3×10^{-12}; it converges in the multipole order to 1.2×10^{-8} at $N_{\max} = 10$; and as $\varepsilon \to 0$ its departure from the isolated sphere is $0.28805 - 0.05570i$ per unit contrast, against the second-order Born $0.28801 - 0.05560i$. At

$\varepsilon = 0.1$ the march's complex residual against it falls under lateral refinement — 4.3×10^{-2}, 2.0×10^{-2}, 1.2×10^{-2} at $N = 6, 9, 12$ — while its residual against the isolated sphere stays at 0.07: at finite contrast the march converges to the periodic problem it solves.

Energy, over every open order. The array is super-wavelength: at $L = 600\,\mathrm{m}$, $k_S L = 12$, and 5 *P*, 9 *SV* and 9 *SH* diffraction orders propagate on each side — 46 channels in all. Each has a lateral wavevector $\mathbf{G}$ in the reciprocal lattice, and $e^{i\mathbf{G}\cdot\mathbf{R}} = 1$ on every site, so every incidence lies in the Bloch sector of normal incidence and one coupling $\mathbf{G}_0$ serves them all. A far field $\mathbf{f}\,e^{ikr}/r$ radiated by every site is, in each order,

$$\sum_{\mathbf{R}} \mathbf{f}\,\frac{e^{ik|\mathbf{r}-\mathbf{R}|}}{|\mathbf{r}-\mathbf{R}|} = \sum_{\mathbf{G}} \frac{2\pi i}{A\,k_z(\mathbf{G})}\,\mathbf{f}(\hat{\mathbf{k}}_{\mathbf{G}})\,e^{i(\mathbf{G}\cdot\boldsymbol{\rho}+k_z(\mathbf{G})|z|)}, \tag{132}$$

with A the cell area and $\hat{\mathbf{k}}_{\mathbf{G}}$ the direction of order $\mathbf{G}$. With the reference plane through the sphere centres and the flux rescaling w_c of §16.10, the 46×46 scattering matrix $\mathbf{S}$ — incidence from above and below, every mode, every open order — satisfies $\mathbf{S}^\dagger\mathbf{S} = \mathbf{I}$ to between 6×10^{-16} and 2×10^{-14} at $N_{\max} = 2$, 4, 6 and 10, to 2×10^{-14} at $\varepsilon = 0.3$, and to 1×10^{-14} at $L = 900\,\mathrm{m}$, where 118 channels are open. That the residual does not fall with $N_{\max}$ is the expected result, not a coincidence: each order's Mie T is unitary, so the truncated problem conserves energy exactly whenever the anti-Hermitian part of $\mathbf{G}_0$ matches the sum (132) over the open orders. The check therefore measures the reciprocal-space lattice sums — the only place the open orders enter — and not the truncation. It can fail: with the coupling switched off the residual is 7.2×10^{-3}, and restricted to the specular channels it is 3.7×10^{-3}, the flux of 0.29 to 0.36 per cent that each specular incidence sends into diffracted orders.

A second implementation shares no code with the first: the Mie operator from the symplectic bracket, its own Ewald sums projected on a different sphere with a different Ewald parameter, and the vector translations extracted numerically rather than through $\mathbf{L}$. At $N_{\max} = 3$ it is unitary to 7.7×10^{-13}, and it agrees with the first *entry by entry, in complex value*, to 4.1×10^{-10} of the scattered part. It also fixes the two conventions that the balance is sensitive to. The prefactor $i/(2Ak_z)$ in place of $2\pi i/(Ak_z)$ breaks it to 1.1×10^{-3}; an incident expansion that gives an *SV* wave only $\mathbf{N}$ multipoles, and an *SH* wave only $\mathbf{M}$, breaks it to 4.0×10^{-2} — a transverse plane wave carries both families, whatever its polarisation.

18.5 What this establishes

The continuum impedance march, with the lateral coupling carried inside the marched variable on a full three-dimensional lateral grid, reproduces the exact elastic response of a sphere at fourth order in the depth step. The sphere has a sharp boundary, and the order survives it because the cross-section enters through (127). The residual after the depth error is exhausted is linear in the contrast, converges under lateral refinement to about $0.29\,\varepsilon$ at 2.5 diameters, and varies with the period as an interference — and second-order Born of the periodic medium, less that of the isolated sphere, predicts it in real and imaginary part without any of the march's machinery. It is the array's own double scattering, and the march converges to it — and, at finite contrast, to the array's exact layer-multiple-scattering solution.

19 The algorithm

Everything above is stated as identities and measurements; what follows collects it as a procedure, in the form Chapter 5 of the thesis uses for Algorithms 5.1–5.3. One algorithm covers every case treated here. With $N_x = N_y = 1$ and the lateral derivatives replaced by ik_x, ik_y it is the laterally invariant march of §16.8; with $N_y = 1$ and $\mathbf{D}_y = ik_y\mathbf{I}$ it is the $2\frac{1}{2}$-D march of §17.2; with both directions gridded it is the three-dimensional march of §18. Nothing in the body of the march distinguishes them, which is the point of carrying the coupling inside the marched variable.

Three things in it are not free choices, and each is marked where it occurs: the modulus/derivative ordering in $\mathbf{A}_{21}$ and the lower rows of $\mathbf{A}_{22}$, which momentum balance forces; the sampling of $\mathbf{A}$ at the *stage* depths rather than once per sublayer, without which the order falls to two; and the starting impedance, which must solve the algebraic Riccati equation of the half-space rather than be read off $\mathbf{D}_z$ alone.

Algorithm 1 A CONTINUUM IMPEDANCE MARCH FOR THE REFLECTION OF A LATERALLY VARYING SLAB

- given a slab $0 \le z \le H$ over a laterally uniform half-space, a lateral grid of $N = N_x N_y$ points with spectral derivatives $\mathbf{D}_x$, $\mathbf{D}_y$, and a depth partition $0 = z_0 < z_1 < \cdots < z_M = H$ numbered *downward* as in Chapter 5, with sublayer thicknesses $h_m = z_m - z_{m-1} > 0$, compute the impedance $\mathbf{Y}$ at the top of the slab and the reflection matrix $\mathbf{R}$
  - (**the operator at one depth — every modulus is a multiplication operator on the lateral grid**)
- **Procedure** $\mathbf{A}(z)$:
  - $\kappa = \lambda + 2\mu, \quad \gamma = \lambda/\kappa, \quad \zeta = 4\mu(\lambda+\mu)/\kappa, \quad \chi = 2\mu\lambda/\kappa,$ all evaluated from (α, β, ρ) at depth z
  - $\mathbf{A}_{11} = \begin{bmatrix} 0 & -\gamma\mathbf{D}_x & -\gamma\mathbf{D}_y \\ -\mathbf{D}_x & 0 & 0 \\ -\mathbf{D}_y & 0 & 0 \end{bmatrix}, \qquad \mathbf{A}_{12} = \operatorname{diag}\left(\kappa^{-1}, \mu^{-1}, \mu^{-1}\right)$
  - $\mathbf{A}_{21} = -\begin{bmatrix} \rho\omega^2 & 0 & 0 \\ 0 & \rho\omega^2 + \mathscr{X} & \mathscr{C} \\ 0 & \mathscr{C}^* & \rho\omega^2 + \mathscr{Y} \end{bmatrix}, \qquad \mathbf{A}_{22} = -\begin{bmatrix} 0 & \mathbf{D}_x & \mathbf{D}_y \\ \mathbf{D}_x\gamma & 0 & 0 \\ \mathbf{D}_y\gamma & 0 & 0 \end{bmatrix}$
  - with, **the derivative outside the modulus**,

$$\mathscr{X} = \mathbf{D}_x\zeta\mathbf{D}_x + \mathbf{D}_y\mu\mathbf{D}_y, \qquad \mathscr{C} = \mathbf{D}_x\chi\mathbf{D}_y + \mathbf{D}_y\mu\mathbf{D}_x,$$
$$\mathscr{Y} = \mathbf{D}_y\zeta\mathbf{D}_y + \mathbf{D}_x\mu\mathbf{D}_x, \qquad \mathscr{C}^* = \mathbf{D}_y\chi\mathbf{D}_x + \mathbf{D}_x\mu\mathbf{D}_y$$

  - (*$\zeta\mathbf{D}_x\mathbf{D}_x$ *in place of* $\mathbf{D}_x\zeta\mathbf{D}_x$ *agrees exactly in a laterally uniform medium and breaks* (126)*)
  - **Return** $(\mathbf{A}_{11}, \mathbf{A}_{12}, \mathbf{A}_{21}, \mathbf{A}_{22})$
  - (**bottom of the slab,* $z_M = H$*: the half-space impedance, one* 3×3 *block per lateral wavenumber**)
- **For** each lateral wavenumber (k_x, k_y) of the grid
  - $\mathbf{Y}^{(0)} = \mathbf{T}_d\mathbf{U}_d^{-1}$ from the three *downgoing* columns of $\mathbf{D}_z$
  - **Repeat** until the residual of (103) stops falling
    - $\mathbf{F} = \mathbf{A}_{21} + \mathbf{A}_{22}\mathbf{Y} - \mathbf{Y}\mathbf{A}_{11} - \mathbf{Y}\mathbf{A}_{12}\mathbf{Y}$
    - solve $(\mathbf{A}_{22} - \mathbf{Y}\mathbf{A}_{12})\,\mathbf{E} - \mathbf{E}\,(\mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{Y}) = -\mathbf{F}$ for $\mathbf{E}$
    - $\mathbf{Y} \leftarrow \mathbf{Y} + \mathbf{E}$
  - (*$\mathbf{D}_z$ *names the branch;* $\mathbf{A}$ *supplies the precision**)
- assemble the blocks block-diagonally and transform to the lateral grid

Algorithm 1 *continued*

- (**upsweep**)
- **For** $m = M, \cdots, 1$ — $\mathbf{Y}$ enters each step at z_m and leaves it at z_{m-1}, a distance h_m *upward*
  - $\mathbf{K}_1 = \mathscr{F}(\mathbf{Y},\ \mathbf{A}(z_m))$
  - $\mathbf{K}_2 = \mathscr{F}\left(\mathbf{Y} - \frac{h_m}{2}\mathbf{K}_1,\ \mathbf{A}(z_m - \frac{h_m}{2})\right)$
  - $\mathbf{K}_3 = \mathscr{F}\left(\mathbf{Y} - \frac{h_m}{2}\mathbf{K}_2,\ \mathbf{A}(z_m - \frac{h_m}{2})\right)$
  - $\mathbf{K}_4 = \mathscr{F}(\mathbf{Y} - h_m\mathbf{K}_3,\ \mathbf{A}(z_{m-1}))$
  - $\mathbf{Y} \leftarrow \mathbf{Y} - \frac{h_m}{6}(\mathbf{K}_1 + 2\mathbf{K}_2 + 2\mathbf{K}_3 + \mathbf{K}_4)$
  - (**the minus signs are the upsweep:* $h_m > 0$ *is a thickness, and the march runs against increasing* z*)
  - (*$\mathbf{A}$ *is rebuilt at each stage depth. Freezing it per sublayer integrates a staircase exactly and returns second order**)
  - (**or, unconditionally stable and second order, with* $\mathbf{A}$ *frozen at the midpoint — one solve, no exponential**)
- $\mathbf{A}_{\text{mid}} = \mathbf{A}\left(z_m - \frac{h_m}{2}\right)$, solve $\left(\mathbf{I} + \frac{h_m}{2}\mathbf{A}_{\text{mid}}\right)\mathbf{W} = \left(\mathbf{I} - \frac{h_m}{2}\mathbf{A}_{\text{mid}}\right)\begin{bmatrix}\mathbf{I}\\ \mathbf{Y}\end{bmatrix}$ for $\mathbf{W}$
- $\mathbf{Y} \leftarrow \mathbf{W}_{\text{lower}}\,\mathbf{W}_{\text{upper}}^{-1}$
- (**no matrix exponential is formed. The signs are the upsweep: the step advances from* z_m *to* z_{m-1}*, and follows from* $\mathbf{q}' = \mathbf{A}\mathbf{q}$*)
  - (**the reflection matrix at the top**)
- transform $\mathbf{Y}$ to the lateral wavenumber domain, giving $\hat{\mathbf{Y}}$
- build $\mathbf{U}_d, \mathbf{T}_d, \mathbf{U}_u, \mathbf{T}_u$ block-diagonally, each column a unit-displacement mode of the medium above
- $\mathbf{R} = \left(\mathbf{T}_u - \hat{\mathbf{Y}}\mathbf{U}_u\right)^{-1}\left(\hat{\mathbf{Y}}\mathbf{U}_d - \mathbf{T}_d\right)$

where $\mathscr{F}(\mathbf{Y}, \mathbf{A}) = \mathbf{A}_{21} + \mathbf{A}_{22}\mathbf{Y} - \mathbf{Y}\mathbf{A}_{11} - \mathbf{Y}\mathbf{A}_{12}\mathbf{Y}$ is the right-hand side of (103).

Which integrator, and when. The two marching steps are alternatives, not successive stages. The explicit one attains fourth order but is conditionally stable: the right-hand side of (103) carries decay rates set by the largest vertical wavenumber, so the largest lateral wavenumber on the grid sets the limit, and the step becomes unstable once $2\,|k|_{\max} h \gtrsim 1.3$. *Refining the lateral grid therefore forces a smaller depth step*, which offsets part of the advantage measured in §16.15 by an amount that has not been quantified. The stable step freezes the medium at the midpoint and is second order — confirmed, with errors falling by 4.00, 4.00, 4.00 over a fourfold refinement

— and it is one solve of the $6N$ system rather than an exponential of it. The next paragraph says why, because the imbedding literature's step is the exponential and the difference is worth stating.

Why the stable step is a solve and not an exponential. The Möbius step of the imbedding literature freezes the medium in each sublayer and propagates with $\exp(\mathbf{A}h)$, which is exact for that frozen medium. It is also an exponential of the *full* $6N \times 6N$ operator, and the impedance form exists precisely to avoid handling that object. Measured against one explicit step, at $h = -H/32$:

grid	$3N$	explicit step	$\exp(\mathbf{A}h)$	ratio	one solve
6×6	108	3.48 ms	10.45 ms	3.0	2.51 ms
8×8	192	9.46 ms	33.43 ms	3.5	7.93 ms
10×10	300	25.13 ms	71.06 ms	2.8	20.93 ms

about three times the explicit step, the ratio flat in N because both scale as N^3: $(6N)^3 = 8(3N)^3$ against four right-hand sides of roughly four $(3N)^3$ products each.

The exactness buys nothing. Freezing the medium at the midpoint already makes the step second order, so reproducing the frozen sublayer exactly is accuracy spent where the method cannot use it. Replacing the exponential by its $(1,1)$ Padé, $\mathbf{M} = (\mathbf{I} - \frac{h}{2}\mathbf{A})^{-1}(\mathbf{I} + \frac{h}{2}\mathbf{A})$, keeps the order and the A-stability and costs one solve:

depth steps	4	8	16	32
$2\lvert k\rvert_{\max}h$	1.26	0.63	0.31	0.16
$\exp(\mathbf{A}h)$	2.80×10^{-2}	6.99×10^{-3}	1.74×10^{-3}	4.30×10^{-4}
one solve	4.17×10^{-2}	1.08×10^{-2}	2.72×10^{-3}	6.81×10^{-4}

observed orders $2.00, 2.00, 2.02$ against $1.95, 1.99, 2.00$, and a constant factor of about 1.6 in the error for a factor of 3.3 in cost. It holds where the explicit step does not: at $2\lvert k\rvert_{\max}h = 2.51$ the explicit march returns 4.33 — it has diverged — and the solve returns 1.41×10^{-1}. That is what the stable branch is for, and it is why the algorithm above forms no exponential.

Two remarks, one of them a dead end worth recording. The exponential is never needed even when it is used, because the update takes only $\mathbf{M}_{21} + \mathbf{M}_{22}\mathbf{Y}$ and $\mathbf{M}_{11} + \mathbf{M}_{12}\mathbf{Y}$, the two halves of $\mathbf{M}[\mathbf{I}; \mathbf{Y}]$; but computing that *action* by a Krylov method, while it agrees to 2.5×10^{-15}, runs about twice as slowly as forming the exponential outright, because at these dense sizes scaling-and-squaring is the faster algorithm. And the Padé is not the exponential: for $\lvert\lambda h\rvert$ large it tends to $-\mathbf{I}$ where the exponential tends to $\mathbf{0}$, so a strongly evanescent sublayer mode is held rather than decayed. That did not bite over the range above, where the largest lateral wavenumber on the grid was carried at $2\lvert k\rvert_{\max}h$ up to 2.51; a slab stiff enough to expose it would want the exponential back.

Cost. $\mathbf{A}_{12}$ is diagonal — it carries no derivative — so the quadratic term of $\mathscr{F}$ is a single dense product of $3N \times 3N$ matrices. It nevertheless dominates, because the linear terms are block sparse: two of the nine $N \times N$ sub-blocks of $\mathbf{A}_{11}$ and $\mathbf{A}_{22}$ are non-zero. Per right-hand side, in units of N^3: the quadratic term 27, $\mathbf{A}_{22}\mathbf{Y}$ and $\mathbf{Y}\mathbf{A}_{11}$ six apiece, so the quadratic term is 69% of the work and the preconditioner, when one is built, belongs there.

19.1 The plane-wave R/T arrays of a sphere

Algorithm 1 stops at $\mathbf{R}$, which is all one sweep can give. The procedure below is the whole response — both matrices, every incident and outgoing channel — and it is stated separately because three of its steps are not refinements of Algorithm 1 but additions to it: the second sweep, the impedance *history* that sweep consumes, and the cross-section that puts a sphere on a periodic grid without a staircase.

Four things in it are not free choices. The cross-section enters through its *area* and not its radius, so the disc is band-limited rather than voxelised (§18). The march domain is exactly $[-a, +a]$ about the centre, so the only non-smoothness in the coefficient — the junction to the homogeneous exterior — sits at the endpoints and never inside a step. The upward sweep is run at *twice* the depth resolution of the downward one and every value kept, so that the downward Runge–Kutta finds $\mathbf{Y}$ already evaluated at each of its stage depths and no interpolation enters; without that the second sweep is second order and the first is wasted. And the azimuthal convention on the array's specular order is shared between the mode matrices and the multipole spectrum, because on axis neither is defined alone.

Algorithm 2 The plane-wave reflection and transmission arrays of a sphere

- given a sphere of radius a and contrast $(\Delta\lambda, \Delta\mu, \Delta\rho)$ centred in a cell of area $A = L_x L_y$ repeated periodically, a lateral grid of $N = N_x N_y$ points, and M depth steps, compute $\mathbf{R}$ and $\mathbf{T}$ for every incident and outgoing plane-wave channel
  - (**the medium at one depth: a disc of the right AREA, band-limited to the grid**)
- **Procedure** slice(ζ):
  - $R^2 = a^2 - \zeta^2$ (and the slice is the exterior medium where $R^2 \le 0$)
  - $\widehat{\Pi}(\mathbf{q}) = 2\pi R^2 \, \dfrac{J_1(|\mathbf{q}|R)}{|\mathbf{q}|R}$ on the grid's lateral wavenumbers, with the value πR^2 at $\mathbf{q} = 0$
  - (**$J_1(x)/x$ is even and entire, so $\widehat{\Pi}$ is analytic in $R^2 = a^2 - \zeta^2$ and therefore smooth across the pole. A voxelised disc is not, and the fourth order does not survive it**)
  - $\Pi(\mathbf{x})$ = the inverse lateral transform of $\widehat{\Pi}/A$, centred on the cell by the shift phase
  - **Return** $(\lambda, \mu, \rho)(\mathbf{x}) = (\lambda_0, \mu_0, \rho_0) + \Pi(\mathbf{x})\,(\Delta\lambda, \Delta\mu, \Delta\rho)$
  - (**sweep one: upward, at DOUBLE resolution, keeping every value**)
- partition $[-a, +a]$ into $2M$ steps $\zeta_0 = -a < \cdots < \zeta_{2M} = +a$, where $\zeta = z - a$ is measured from the sphere's centre and increases *downward*: ζ_{2M} is the base and ζ_0 the top
- $\mathbf{Y} \leftarrow$ the half-space impedance of the exterior medium, by the Newton loop of Algorithm 1
- store $\mathbf{Y}_{2M} \leftarrow \mathbf{Y}$
- **For** $j = 2M, \cdots, 1$: advance $\mathbf{Y}$ from ζ_j to ζ_{j-1} by one step of Algorithm 1, then store $\mathbf{Y}_{j-1} \leftarrow \mathbf{Y}$
- (**the stored values at even j are the downward sweep's step edges and those at odd j its midpoints — exactly its stage depths**)

Algorithm 2 *continued*

- (**the mode matrices, and the one convention that is shared**)
- **For** each lateral wavenumber $\mathbf{q} = (k_x, k_y)$ of the grid, build the 3×3 blocks $\mathbf{U}_d, \mathbf{T}_d, \mathbf{U}_u, \mathbf{T}_u$ of the exterior medium, columns ordered P, SV, SH and each of unit displacement
- on the specular order $\mathbf{q} = \mathbf{0}$ take the azimuth $\psi = 0$, so that $\hat{\mathbf{e}}_{SV} = \hat{\mathbf{x}}$ and $\hat{\mathbf{e}}_{SH} = \hat{\mathbf{y}}$
- (**the azimuth has no limit there, but the field does: the ψ-dependence of these polarisations cancels against that of the $m = \pm 1$ multipole spectrum they are compared with, so the SAME branch must be taken in both. Taking $0/0 = 0$ instead zeroes the entire specular column**)
  - (**reflection, from the top of the first sweep alone**)
- $\hat{\mathbf{Y}} \leftarrow \mathbf{Y}$ transformed to the lateral wavenumber domain
- $\mathbf{R} = \left(\mathbf{T}_u - \hat{\mathbf{Y}}\mathbf{U}_u\right)^{-1}\left(\hat{\mathbf{Y}}\mathbf{U}_d - \mathbf{T}_d\right)$
  - (**sweep two: downward, WITH the energy flow, while the Riccati ran against it**)
- $\mathbf{u} \leftarrow \mathbf{U}_d + \mathbf{U}_u\mathbf{R}$ — the total displacement at the top for unit incidence in each channel, transformed to the lateral grid
- **For** $m = 1, \cdots, M$ — with $j = 2(m-1)$ and step $h = \zeta_{j+2} - \zeta_j > 0$
  - $\mathbf{K}_1 = \mathscr{O}_j\,\mathbf{u}, \quad \mathbf{K}_2 = \mathscr{O}_{j+1}\left(\mathbf{u} + \frac{h}{2}\mathbf{K}_1\right)$
  - $\mathbf{K}_3 = \mathscr{O}_{j+1}\left(\mathbf{u} + \frac{h}{2}\mathbf{K}_2\right), \quad \mathbf{K}_4 = \mathscr{O}_{j+2}(\mathbf{u} + h\mathbf{K}_3)$
  - $\mathbf{u} \leftarrow \mathbf{u} + \frac{h}{6}\left(\mathbf{K}_1 + 2\mathbf{K}_2 + 2\mathbf{K}_3 + \mathbf{K}_4\right)$
- (**the signs are the downsweep, and it is the stable direction here for the same reason the upsweep is stable for* $\mathbf{Y}$*)
  - (**transmission: below the slab the field is purely downgoing**)
- $\mathbf{T} = \mathbf{U}_d^{-1}\,\hat{\mathbf{u}}$, with $\hat{\mathbf{u}}$ the field at ζ_{2M} returned to the wavenumber domain

where $\mathscr{O}_j = \mathbf{A}_{11}(\zeta_j) + \mathbf{A}_{12}(\zeta_j)\,\mathbf{Y}_j$ is the one-way operator of (110), built from the stored impedance at that depth and never re-marched.

What it costs, against Algorithm 1. Measured at $N = 36$, $M = 16$: 2.49× the reflection alone. Most of that is not the second sweep. The *upward* sweep is what doubles — it runs at $2M$ steps so that the downward one finds $\mathbf{Y}$ at its stage depths — and the downward sweep itself is M steps of four matrix–vector products against $\mathscr{O}$, cheaper per step than the Riccati by the whole quadratic term. So the price of transmission is paid mostly in the sweep one already had.

An earlier draft of this paragraph said 1.5×*, reasoning from the cheapness of the second sweep and forgetting that the first one had been doubled.* The memory cost is the history, $2M + 1$ matrices

of $3N \times 3N$, which is the only new storage in the procedure and the reason the doubling is stated as part of the algorithm rather than left to the caller.

Checking it without an arbiter. Two conditions are available before any comparison with Mie, and both are stated in §16.10. At zero contrast every entry is known in closed form, $\mathbf{R} = \mathbf{0}$ and $\mathbf{T} = \operatorname{diag} e^{ik_{z,c}\cdot 2a}$, which tests the sweeps and the mode matrices together. At finite contrast, rescaling to the flux basis by $w_c(\mathbf{q}) = \sqrt{c^2 k_{z,c}/\omega}$ and restricting to propagating channels, $\mathbf{R}^\dagger\mathbf{R} + \mathbf{T}^\dagger\mathbf{T} = \mathbf{I}$ — which scores every entry of both matrices at once, and needs nothing external.

Against the isolated sphere. To compare with Mie the array must be undone. One sphere per cell of area A, all seeing the same incident phase at normal incidence, gives by Poisson summation

$$\mathbf{R}[\mathbf{G} \leftarrow \Gamma] = \frac{(2\pi)^2}{A}\, c_{\mathrm{mode}}(\mathbf{G})\, e^{i(k_{\mathrm{inc}} + k_z(\mathbf{G}))a}, \tag{133}$$

with the same expression for $\mathbf{T}$ plus the unscattered wave $e^{2ik_{\mathrm{inc}}a}$ on the specular entry of the incident channel. The two phases are the incident wave referred from the top of the slab to the sphere's centre, where Mie normalises it, and the scattered order referred back from the centre to the plane. The spectrum c_{mode} is the $m = 0$ one for P incidence and the $m = \pm 1$ one for S; (133) is exact but for scattering *between* spheres, which is what growing A removes and what the residual of §18, about $0.3\,\varepsilon$ at 2.5 diameters, bounds.

20 Summary of checks

Table 20. All checks pass. The symbolic scripts report 25/25 and 44/44; the T-matrix assembly 16/16, the Schwinger settlement 17/17, the layer comparison 7/7, the thesis spectral representation 13/13, the slab comparison 7/7, the static moments 11/11, the Riccati blocks 11/11, the impedance 9/9, its march 14/14, the layered $\mathbf{\Gamma}$ 8/8 and the lateral impedance march 33/33 and the same march in three dimensions 56/56; against multiple scattering 21/21; and the propagator gate exits clean. These tallies are machine-checked against the gates by `scripts/check_paper_tallies.py`, because a hand-written count of someone else's output is exactly the thing that goes stale.

object	what is established	checks
$\mathscr{A}$	built from c_{ijkl}; matches the published isotropic blocks	4
	matches the thesis $\mathbf{A}(k_x, k_y)$ under one similarity transform	3
	symmetry relations, spectrum, normal incidence	6
	stiffness matrices, $\mathbf{U}_{\alpha l}$, ν_1, ν_2	9
	source vector, negative control	3
$\Delta\mathscr{A}$	exact contrast operator; Navier body force on a generic field	4
	nonlinearity in the contrast, quantified	3
	augmentation: constant, reproduces the form, divergence exhibited	4
	O_h/trial-space structure, density channel	6

object	what is established	checks
$\Delta\mathbf{C}_{\text{eff}}$	$\mathbf{J}_6\Delta\mathscr{A}$ symmetric; test space derived; controls	5
	9×9 symmetric, rank 9, rigid block, basis trap	7
	Born agreement, $O(\Delta c^2)$ difference, localisation	9
	O_h decomposition, positive and negative controls	6
$\boldsymbol{\Gamma}$	propagates by $\exp(\mathbf{A}\,dz)$ with all multiples present	3
moments	face integrals evaluated; j_1 also by solid angle	4
	$\int dV/r$ in two independent closed forms	1
	B, and $A = B - 1/3\mu$, $C = -(5 - 2\pi/\sqrt{3})B$, G	4
	Kelvin coefficients against the h_c form	1
	the trace identity (45), derived without the recursion	1
	the tabulated route reproduces all four	1
$\mathbf{T}$	coupling rebuilt independently, matches the symbolic derivation	2
	calibration: reproduces the existing amplification factors	1
	the finite-size density–strain term, isolated and sized	2
	density channel identical; O_h split on the e_{12}/e_{13} pair	6
	Born limits agree	1
Schwinger	$\mathscr{F}[\partial_\alpha(\mathbf{1}_V c)] = ik_\alpha\mathscr{F}[\mathbf{1}_V c]$, with a sign control	3
	$\mathscr{A}$ in the paper basis $= S^{-1}\mathscr{A}_{\text{thesis}}S$; spectrum;	4
	$\mathscr{A}(-\mathbf{k})^{\mathsf{T}}\mathbf{J}_6 = -\mathbf{J}_6\mathscr{A}(\mathbf{k})$, with a control	
	$\boldsymbol{\Gamma}$ inverted laterally *is* the Kelvin tensor	1
	the radiation reaction (75), no static counterpart	1
	reciprocity (74), pointwise and z-integrated	2
	the two-$\Delta\mathscr{A}$ assembly is the Galerkin double-volume moment	1
	the gap to the calibrated $\mathscr{G}$, factorised	1
	the 9×9 matrix is symmetric, with a sign control and a not-vacuous control	3
thesis	$\mathbf{D}_z^{-1}$ from (77) is diagonal on the thesis eigenvectors, and exact after normalisation	2
	$\mathbf{D}_z\boldsymbol{\Lambda}\mathbf{D}_z^{-1}$ reproduces (Akdef)	1
	the quasi-Hamiltonian relation as the thesis states it	1
	quasi-SV and quasi-SH independent at $k = 0$	1
	$\boldsymbol{\Gamma}$ reciprocal in the sense (74), and the Kelvin tensor with the radiation reaction, from the analytic modes alone	3
	the valid range in k, recorded	1
	the branch point is real, and attenuation removes it; complex ω is the complex slowness	4
A, B, C	the batched eigenvectors are the scalar ones	1
	(86) pins the realisation and the sign	1
	B from the first-order system matches the closed form	1

object	what is established	checks
	the two factors of i are load-bearing	1
	the subtracted remainder is cutoff-independent	1
	the local term stops at $n = 4$; the dipole is load-bearing	2
	A and C match their closed forms	2
	the two structural relations, tested on first-order numbers	2
slab	the Navier route reproduces the package's amplification factors	1
	the advantage grows as the shared lattice error falls	1
	the first-order T_0 is the more accurate of the two	1
	the improvement survives either choice of the split channel	1
	SH incidence does probe the shear channels	1
	the tetragonal T_0 collects what the cubic one cannot, without spoiling the channel the cubic one gets right	2
Riccati blocks	$\Delta\mathscr{A}$ is 12/36 sparse, the compliance block diagonal	2
	$\mathbf{V}$ built two ways, the row scaling cancelling	1
	$\mathbf{V}$ is quasi-Hamiltonian (101)	1
	the linear part is oscillatory inside the window, stiff outside	2
	$\mathrm{eig}(\mathbf{V}) \propto \mathbf{\Lambda}$ but $\|\mathbf{V}\|$ does not, and balancing cannot fix it	2
	$\mathrm{cond}(\mathbf{D}_z) \sim \|\mathbf{k}\|^2$ from P/S coalescence, with the ceiling recorded	3
impedance	(103) is satisfied by the $\mathbf{D}_z$ impedance	1
	$\mathbf{Y}$ degrades linearly where $\mathbf{V}$ fails outright	2
	Newton on $\mathbf{A}$ alone holds the residual at the floor	1
	and lands on the downgoing root, tested without $\mathbf{D}_z$	1
	$\mathrm{cond}(\mathbf{Y}) = O(1)$; the static limit is ω-independent	2
	the arbiter: $\mathbf{Y}$ reproduces $k_{z,P}$ and $k_{z,S}$ to $\|\mathbf{k}\| = 6000$	2
impedance march	one exact step (107) reproduces the closed form	1
	RK4 on (103) converges at fourth order	2
	the conventions are $(-1,-1,+1)$, constant across slowness, and agree with the collocation route	2
	they survive past both critical slownesses	1
	the GRADED slab matches Kennett to the floor at every mesh	1
	reversing the profile changes $\mathbf{R}$ by 50%	1
	the convention is DIAGONAL, by the rank-one test (118)	1
	its P/S part is $\sqrt{k_{z,S}/k_{z,P}}$, and so not fixed	1
	$\mathbf{S}_u = -\mathbf{S}_d$ on P–SV and $+\mathbf{S}_d$ on SH	1
	a layer stack is second order on a smooth gradient, the ODE fourth, and the step count to fixed accuracy differs by 32×	3
layered	the source jump (93) is exactly $\mathbf{I}_6$	1

object	what is established	checks
$\boldsymbol{\Gamma}$	zero-contrast cuts are harmless, so any depth is reachable	1
	the bridge identity at depths on no lattice	1
	reciprocity (74), source and receiver exchanged	1
	(94), with a not-vacuous control	2
	the evanescent reduction to the whole space	1
	a reflector below returns an upgoing part, by twelve orders	1
layer	normal incidence vs (80), three contrasts	3
	second-order convergence in the cell count	1
	the *PP*, *SS* and *SH* conventions are constant across slowness — see §16.13 for what they are	1
	the k dependence of $\Delta\mathscr{A}$ is load-bearing, two controls	2
lateral	the operator reproduces `(Akdef)` at every grid wavenumber, for two media and two k_y	3
impedance	the spectral lateral derivative is exactly antisymmetric	1
	the symplectic relation (122) under lateral variation	2
	the wrong modulus ordering breaks it, and is indistinguishable from (121) when the medium is uniform	2
	SH decouples at $k_y = 0$; the march is lateral-shift covariant	2
	zero lateral contrast reduces to the 3×3 march of §16.12, with no coupling made by the discretisation	4
	the linear route agrees at fourth order, and cond($\mathbf{U}$) diverges with thickness where $\|\mathbf{Y}\|$ stays finite	3
	the explicit step limit is set by the largest lateral wavenumber, and the Möbius step has no such limit	2
	second order for the staircase and fourth for the ODE, with the lateral coupling on	2
	a single contrast mode couples $k_x \pm m$, and dominantly so: the remaining coupling, which multiple scattering must generate, is 6% of it	2
	$\mathbf{A}_{12}$ is exactly diagonal; block occupancy as §17.4 tabulates it; the quadratic term is one dense product and 69% of the step	5
	the measured cost exponent is below the dense cubic estimate	1
	the coupled algebraic Riccati equation has a stabilising downgoing solution, and the march holds that branch to the top	3
	$\|\mathbf{Y}\|$ constant to six figures across a sixteen-fold change in the step count	1

21 What is not established

In keeping with the project's standard that an unvalidated formula be labelled as such, the following are *open*:

1. **The *iterative* multiple-scattering solver has never been compared on equal terms.** §22 scores the march against the voxel route, but that route solves a sphere by a *direct* dense solve of the $9N_c$ system. The GMRES solver in this project works on a uniform periodic slab, whose geometry is a fully occupied lattice with no per-cell occupancy and which therefore cannot hold a sphere — so the one configuration with an exact external arbiter is closed to it. What is needed is a Bloch-periodic Foldy–Lax over a *voxelised sphere*, which would put the iterative solver on the same problem as the march and, unlike the present comparison, on the same periodic array. Until then nothing here says anything about iterative cost, and in particular the iteration count's growth with contrast — the axis on which the sweep architecture is known to do nothing — remains unmeasured against the march, which has no iteration count at all.

2. **The T-matrix uses the collocation propagator, not the consistent one.** $\mathbf{T}$ is assembled in §9 and calibrated exactly, but $\mathscr{G}$ there is the *single* cube moment while the $\mathbf{J}_6$ pairing makes the scheme Galerkin. This is now measured rather than suspected: (73) separates the gap into a geometric factor $\int_V 1/r \big/ D_0 = 1.2644$ and a factor $i\omega$, and the double moment D_0 that the consistent pairing needs is extracted in §10.4. The $i\omega$ in (73) is *not* open — see §7.3 — so what remains is the **geometric** factor alone. Until that is resolved, everything in §9 is the first-order coupling seen through a collocation propagator.

3. **The external arbiter reaches the operator, not the cube.** §12 closes this item for $\mathscr{A}$, $\mathbf{\Gamma}$, their composition and the resummation: a layer in a whole space reproduces an independently validated reflectivity recursion across contrast and angle. But a laterally invariant problem does not touch the augmentation, the form factors or the moments, so the quantities in §7 and §9 still compare two *formulations* with each other rather than either against truth. The O_h split remains the only partial exception there, and it is one-sided: it lower-bounds the first-order error at $\approx 4.2 \times 10^{-4}$ and bounds the ordinary route not at all. §14 then answers the accuracy question for the cube by the one geometry in which both formulations apply and an external reference exists — a space-filling patch — and the first-order T wins there by a margin that grows with contrast. What remains genuinely open is the *scope* of that: one frequency, normal incidence, an unconverged lattice, and no measurement at all away from space-filling.

4. **Whether the resummation improves accuracy is unknown.** It is exact as a local inversion; that is not the same claim.

5. **O_h invariance was tested on a subgroup of order** 3, not the full group.

6. **The derivative count for a collocation scheme differs from the Galerkin one.** In the Galerkin pairing the transferred derivative lands on polynomials, so the third-derivative moment cannot arise; under collocation it transfers onto $\mathbf{\Gamma}$ and can. Do not quote the Galerkin count for a collocation build. (69) is how that transfer is carried out in practice, and it is verified; what is not yet done is the count itself.

7. **The derivative channels are verified only by symmetry.** That is a strong test — ten orders of separation from its control, and the derivative terms carry 80% of the matrix — but symmetry is a *structural* property. A bookkeeping error that is symmetric would pass it. The value of those channels has not been compared with an independent arbiter, which would need the Galerkin double-volume moments of $\partial_i\partial_j G_{kl}$; those follow from the same autocorrelation and Duffy construction as D_0, with $\partial_i\partial_j$ moved onto the weight, but they have not been computed.

8. **The tetragonal T_0 is measured but not shipped.** §14.6 builds it and shows it collects the x_3 benefit, 3.2× in the channel that probes it. But it is built in the gate, not in the package: `effective_stiffness_voigt` still takes one shear parameter and writes it into all three slots. Widening it is a small backwards-compatible change, and the right moment for it is when the first-order route is wired into the package proper rather than now, when nothing upstream in the package produces two shear amplitudes.

9. **The laterally coupled march has no external arbiter, and no measured advantage.** §17 establishes that the construction is sound — it reduces exactly to the validated 3×3 march, keeps fourth order, keeps the symplectic structure (122), and agrees with an independent algorithm on the same operator. Every one of those is internal. Nothing yet compares it with an object built outside this development, and in particular nothing compares it with the thesis's own architecture (120), which solves the same problem by keeping the vertical operator diagonal in k_x. The natural arbiter is the thesis's $2\frac{1}{2}$-D solver; until it has been run, "carrying the coupling inside the march" is a structural result and not a demonstrated improvement. The cost trade is likewise only half measured: the quadratic term is 69% of a step, but the explicit integrator's step limit tightens as the lateral mesh is refined, and how much of the fourth-order economy that consumes has not been quantified.

22 Against multiple scattering, on the one validated problem

The sphere is the only problem in this note with an exact external solution and a laterally varying medium at once, so it is the only place the imbedding route can be scored against the alternative the rest of this project is built on: a body voxelised into cubes, each carrying an analytic T-matrix, coupled and solved. Both compute the same physical quantity — the backscattered P amplitude of a sphere at $k_S a = 2.4$, contrast $\varepsilon = 0.1$ — and both are scored against exact Mie. *The voxel numbers of this section were measured again on 31 October 2026*, after the fix of 29 September to the voxel route's far field, which had counted the incident wave's linear variation twice, and with the uniform-strain closure exact at every frequency as the voxels' single site (`scripts/gate_imbedding_vs_foldy_lax.py`, its data in `data_imbedding_vs_voxel.json` and `data_angle_resolved.json` beside this note). The march's numbers are unchanged.

Two corrections to the framing, before the numbers. The voxel route *as applied to a sphere* is not iterative: it ends in a direct dense solve of the $9N_c$ system. The iterative (GMRES) multiple-scattering solver in this project works on a uniform periodic slab whose geometry is a fully occupied lattice with no per-cell occupancy, so it cannot hold a sphere — and therefore has

no externally validated problem to be compared on at all. Closing that gap needs a Bloch-periodic Foldy–Lax over a voxelised sphere, which is not built.

And on the periodization, an earlier draft of this section argued that the comparison was *biased against the march*: exact Mie is the isolated sphere, the march solves an array, and the inter-sphere coupling it is charged for is real physics the arbiter does not contain. **That argument is wrong and is withdrawn.** The question being asked is what one sphere scatters. The march cannot answer it directly — it requires a laterally periodic grid — so the array is not a different problem that happened to be posed, it is *machinery this method needs*, and the accuracy that machinery costs is the method's to carry. A route that must periodize is charged for periodizing, exactly as a route that must voxelise is charged for the staircase. Both errors below are method errors and neither is discounted.

What "error" means in every table below. The two routes return different objects, so the two columns are not the same functional and it is worth saying exactly what each is before any number is quoted.

For the **march**, the error at a diffraction order $\mathbf{G}$ is the relative error of the plane-wave coefficient itself,

$$e_{\mathrm{march}}(\mathbf{G}) = \frac{\left|\mathbf{R}_{\mathrm{march}}(\mathbf{G}) - \mathbf{R}_{\mathrm{Mie}}(\mathbf{G})\right|}{\left|\mathbf{R}_{\mathrm{Mie}}(\mathbf{G})\right|}, \tag{134}$$

with $\mathbf{R}_{\mathrm{Mie}}$ the isolated-sphere amplitude carried to that order by the Poisson relation (133), and the same for $\mathbf{T}$. It is a scalar, one mode coefficient against another.

For the **voxel route**, which returns a field rather than a coefficient, it is the volume-corrected relative error of the far-field *displacement* in the same direction, in the max norm,

$$e_{\mathrm{voxel}}(\hat{\mathbf{r}}) = \frac{\left\|\mathbf{u}_{\mathrm{FL}} - c\,\mathbf{u}_{\mathrm{Mie}}\right\|_\infty}{\left\|c\,\mathbf{u}_{\mathrm{Mie}}\right\|_\infty}, \qquad c = \frac{V_{\mathrm{sphere}}}{V_{\mathrm{voxel}}}. \tag{135}$$

The factor c divides out the staircase dilution, and is applied to this route only because only this route has a staircase; without it the comparison measures the fact that cubes do not fill a sphere (§18).

These are comparable in spirit and not identical: one is the modulus of a scalar, the other a max norm over a displacement vector, and the denominators differ correspondingly. For a P wave the displacement is nearly along $\hat{\mathbf{r}}$, so one component dominates and the two measures nearly coincide, but the difference is real and is not calibrated away anywhere.

	unknowns	error (134)/(135)	time (s)
voxel Foldy–Lax, $n_{\mathrm{sub}} = 6$	1224	0.107	0.17
impedance march, $N = 81$, $L = 900$	**243**	**0.0112**	0.84

Table 21. At matched cost *and at this frequency*: 9.6× the accuracy on a fifth of the unknowns. *Error* is $|\mathbf{R}_{\mathrm{march}} - \mathbf{R}_{\mathrm{Mie}}|/|\mathbf{R}_{\mathrm{Mie}}|$ for the march — a plane-wave coefficient (134) — and $\|\mathbf{u}_{\mathrm{FL}} - c\,\mathbf{u}_{\mathrm{Mie}}\|_\infty / \|c\,\mathbf{u}_{\mathrm{Mie}}\|_\infty$, $c = V_{\mathrm{sphere}}/V_{\mathrm{voxel}}$, for the voxel route — a volume-corrected displacement (135). The two are not the same functional. Both are backscatter, at $k_S a = 2.4$, $\varepsilon = 0.1$. The measured time exponents are 2.05 and 2.14 over the ranges tested, so the advantage is accuracy *per unknown* and not a difference in scaling. The ratio is strongly frequency dependent and falls to 1.4 at $k_S a = 0.5$ — see below.

Neither ladder converges monotonically. This is the part worth stating carefully, because the obvious reading of the table — that one method is converging faster — is wrong. Refining either knob does *not* reduce either error monotonically:

voxel route, n_{sub}	3	4	6	8
error	0.191	0.103	0.107	**0.023**
march, period (diameters)	2.5	3.8	5.0	6.7
error	0.021	**0.011**	0.023	0.016

Table 22. Backscatter error against the exact sphere: *Error* is $|\mathbf{R}_{\text{march}} - \mathbf{R}_{\text{Mie}}|/|\mathbf{R}_{\text{Mie}}|$ for the march — a plane-wave coefficient (134) — and $\|\mathbf{u}_{\text{FL}} - c\,\mathbf{u}_{\text{Mie}}\|_\infty / \|c\,\mathbf{u}_{\text{Mie}}\|_\infty$, $c = V_{\text{sphere}}/V_{\text{voxel}}$, for the voxel route — a volume-corrected displacement (135). The two are not the same functional. Both are non-monotone, for different reasons. The voxel route's error is the *staircase*, and a cubic lattice's coverage of a sphere is not monotone in n_{sub} — the voxel volume is 26%, 4.7%, 17% and 4.3% wrong at these four. The march's, at a fixed 100 m pitch, combines the array coupling, a wave interference that oscillates with spacing rather than decaying, with the lateral discretisation of §18, which does converge — roughly as $1/N$ at fixed period. Quoting a single convergence rate for either on this problem would be wrong.

The difference that survives is in the levels, not the rates. The voxel route's error moves with the staircase rather than with the cell size: 0.107 at $n_{\text{sub}} = 6$, 0.023 at 8 and 0.034 at 10. Its best is twice the march's, 0.011.

The level is geometric, and it is not the sub-cell validity limit. That limit is the obvious suspect, because at $k_S a = 2.4$ with $n_{\text{sub}} = 6$ the sub-cells sit at $ka = 0.40$. Refining to $n_{\text{sub}} = 8$ and 10 brings them to 0.30 and 0.24, and the error is 0.023 and 0.034: it does not fall steadily as the cells come inside that range, but moves with the staircase, whose volume is 4.3% wrong at $n_{\text{sub}} = 8$. What limits the voxel route is the representation of a curved boundary, which is §18's conclusion reached from the other direction and the reason the cross-section of Algorithm 2 is band-limited rather than voxelised.

So the ratio is not a constant — it is a frequency dependence. $k_S a = 2.4$ is where the march's own gate runs, which is not a neutral choice, so the comparison is repeated across the band:

$k_S a$	voxel (135)	march (134)	ratio
0.10	2.35×10^{-3}	4.66×10^{-3}	**0.50**
0.15	2.19×10^{-3}	4.77×10^{-3}	**0.46**
0.20	2.55×10^{-3}	4.91×10^{-3}	**0.52**
0.30	2.44×10^{-3}	5.27×10^{-3}	**0.46**
0.50	8.38×10^{-3}	6.00×10^{-3}	1.40
0.70	1.07×10^{-2}	6.49×10^{-3}	1.65
1.00	2.12×10^{-2}	4.24×10^{-3}	5.01
1.50	4.59×10^{-2}	1.31×10^{-2}	3.50
2.00	7.97×10^{-2}	3.41×10^{-2}	2.34
2.40	1.07×10^{-1}	1.12×10^{-2}	9.60

Table 23. Backscatter error against the exact sphere, at $\varepsilon = 0.1$, $n_{\text{sub}} = 6$ and period 900 m. *Error* is $|\mathbf{R}_{\text{march}} - \mathbf{R}_{\text{Mie}}|/|\mathbf{R}_{\text{Mie}}|$ for the march — a plane-wave coefficient (134) — and $\|\mathbf{u}_{\text{FL}} - c\,\mathbf{u}_{\text{Mie}}\|_\infty / \|c\,\mathbf{u}_{\text{Mie}}\|_\infty$, $c = V_{\text{sphere}}/V_{\text{voxel}}$, for the voxel route — a volume-corrected displacement (135). The two are not the same functional. The voxel error is nearly flat below $k_S a = 0.3$ and grows as about $(k_S a)^{1.3}$ over the band; the march's stays within a factor of eight across more than a decade, and within a factor of 1.4 below $k_S a = 1$. The two errors are different *kinds* of thing, which is why the gap opens: the march's is the array's coupling, a property of the arrangement, while the voxel route's is the staircase, whose phase error scales with the cell size measured in wavelengths. Bold marks the rows where the *voxel* route is the better of the two.

The crossover is measured and not extrapolated: it is at $k_S a = 0.43$. Below it the voxel route wins — by about 2× from $k_S a = 0.1$ to 0.3, where the staircase costs less than the array does. Above it the imbedding route wins, by a margin that grows with frequency. So the honest statement is not that one method is an order of magnitude better than the other; it is that *one degrades with frequency and the other does not*, and which to use follows from where the problem sits.

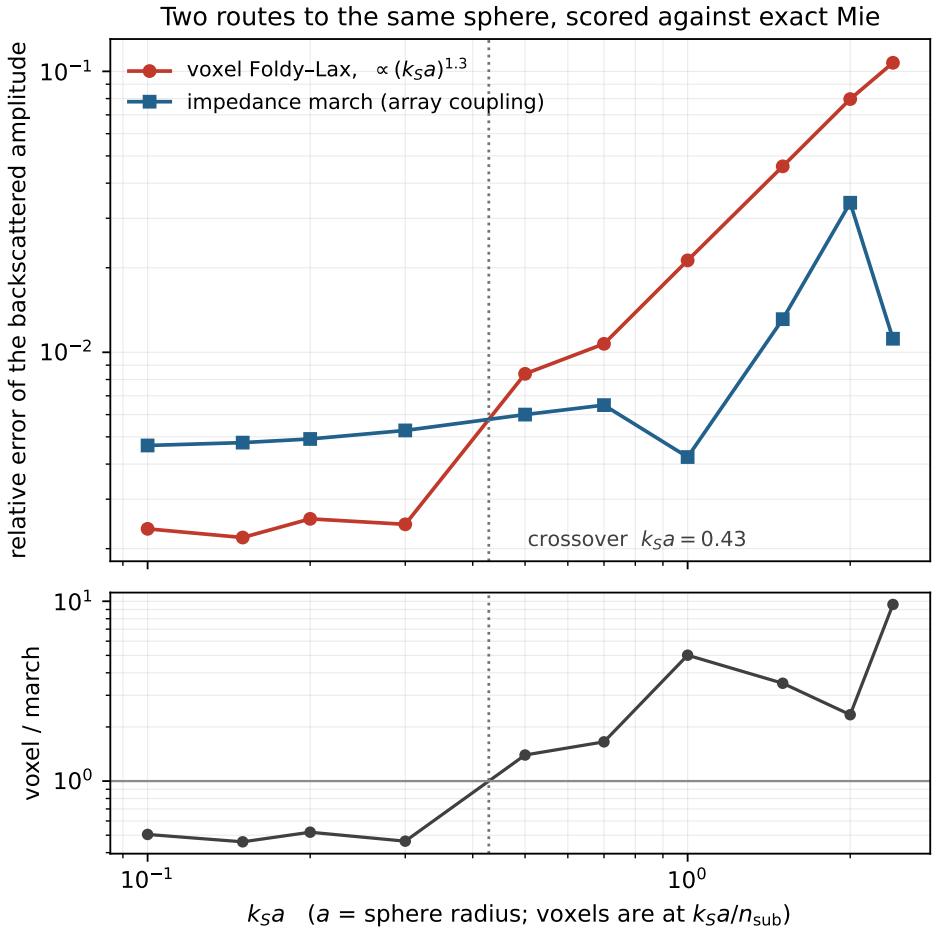


Figure 1. The same measurements. *Error* is $|\mathbf{R}_{\text{march}} - \mathbf{R}_{\text{Mie}}|/|\mathbf{R}_{\text{Mie}}|$ for the march — a plane-wave coefficient (134) — and $\|\mathbf{u}_{\text{FL}} - c\,\mathbf{u}_{\text{Mie}}\|_\infty / \|c\,\mathbf{u}_{\text{Mie}}\|_\infty$, $c = V_{\text{sphere}}/V_{\text{voxel}}$, for the voxel route — a volume-corrected displacement (135). The two are not the same functional. a is the *sphere's* radius; the voxels sit at $k_S a/n_{\text{sub}}$, a sixth of it here. The upper panel is the error of each route; the lower is their ratio, crossing one at $k_S a = 0.43$. The march's non-monotonicity above $k_S a = 1$ is the array coupling behaving as an interference rather than a decay; the voxel route's error is nearly flat below $k_S a = 0.3$, where the staircase sets it, and grows above.

Every direction, not only backscatter. Everything above scores one direction. Repeating it over every propagating order of the array — reflected and transmitted, at $k_S a = 1$ and 2.4, on the 900 m period — scores the rest:

$k_S a$	directions	march	voxel	ratio
1.0	specular only	1.39×10^{-2}	$\mathbf{1.31 \times 10^{-2}}$	0.95
2.4	specular	$\mathbf{1.67 \times 10^{-2}}$	5.83×10^{-2}	3.50
2.4	wide angle	$\mathbf{2.18 \times 10^{-2}}$	1.22×10^{-1}	5.59

Table 24. Medians, over the propagating orders in each group, of the per-order relative errors: *Error* is $|\mathbf{R}_{\text{march}} - \mathbf{R}_{\text{Mie}}|/|\mathbf{R}_{\text{Mie}}|$ for the march — a plane-wave coefficient (134) — and $\|\mathbf{u}_{\text{FL}} - c\,\mathbf{u}_{\text{Mie}}\|_\infty / \|c\,\mathbf{u}_{\text{Mie}}\|_\infty$, $c = V_{\text{sphere}}/V_{\text{voxel}}$, for the voxel route — a volume-corrected displacement (135). The two are not the same functional. The median is taken across orders spanning several scattering angles, so it summarises and does not replace the figure below. The march's scores carry the translation phase (112), exclude the zeroed-Nyquist orders, and, for the forward specular order, remove the unscattered wave from both sides, so that both routes are scored on the scattered field. Bold is the better of the pair. At $k_S a = 2.4$ the march is the better route at every order, by 5.6× at the wide angles; at $k_S a = 1$, where this period supports only the specular orders, the voxel route is marginally the better.

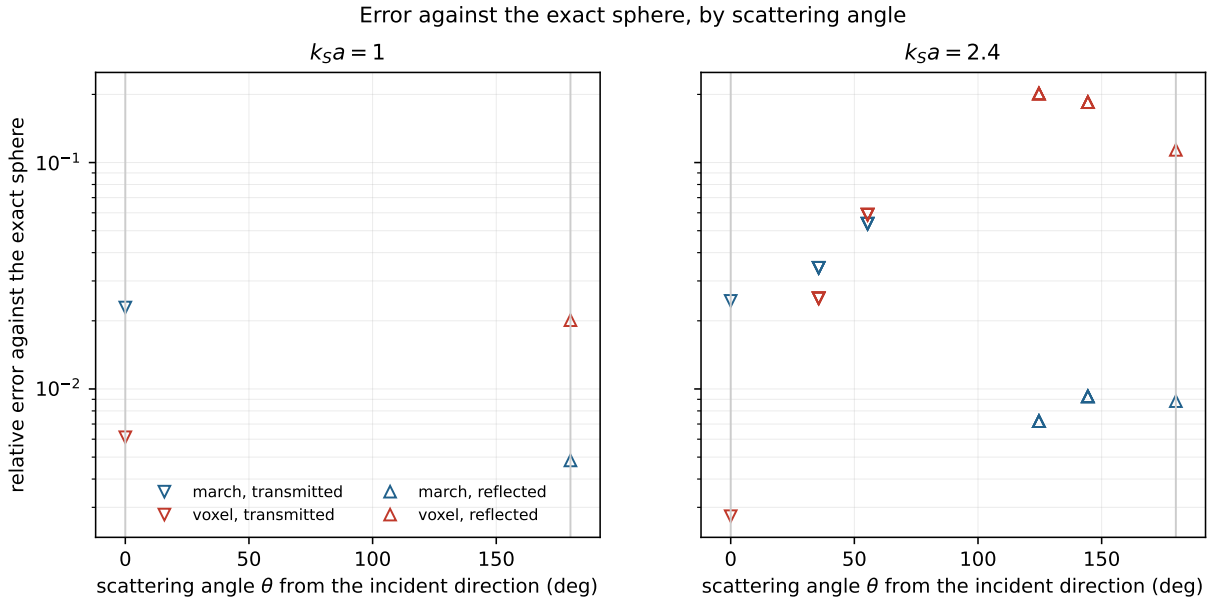


Figure 2. Every propagating order, by scattering angle, θ measured from the incident direction so that 180° is backscatter. *Error* is $|\mathbf{R}_{\text{march}} - \mathbf{R}_{\text{Mie}}|/|\mathbf{R}_{\text{Mie}}|$ for the march — a plane-wave coefficient (134) — and $\|\mathbf{u}_{\text{FL}} - c\,\mathbf{u}_{\text{Mie}}\|_\infty / \|c\,\mathbf{u}_{\text{Mie}}\|_\infty$, $c = V_{\text{sphere}}/V_{\text{voxel}}$, for the voxel route — a volume-corrected displacement (135). The two are not the same functional. At $k_S a = 1$ the period supports only the specular orders; at 2.4 it supports wide-angle ones, and the march is the more accurate route at every one of them.

Orders at one scattering angle. The exact sphere is *axisymmetric*: its far field depends on θ alone. A square array is not, but it keeps the four-fold rotation and the mirrors of the square, and for normal P incidence those relate every order at $\theta = 144.4°$ to every other. The march's errors there must therefore agree, and they do, at 0.00927 to twelve figures, against the voxel route's 0.185. These orders see two defects that the specular orders cannot. A missing translation phase (112) scores them at 1.94 to 1.98 — $|1 - e^{i\phi}|$ for ϕ near π, an order-unity error alongside correct

magnitudes — and a cross-section that breaks the mirror, as a zeroed-Nyquist disc does (§18), scores symmetry-related orders differently.

Wide angles at lower frequency, where the confound is removed. The table above has wide-angle orders at $k_S a = 2.4$ only, and that is an artefact of the period rather than a choice: an order propagates when $L > \lambda_P = 1257\,\mathrm{m}/(k_S a)$, so at $L = 900$ none exists below about $k_S a = 1.5$. The one frequency the comparison could reach was therefore the one least favourable to the voxel route. Holding the lateral pitch fixed and growing L until wide-angle orders exist removes that:

		specular		wide angle		
$k_S a$	L (m)	march	voxel	march	voxel	voxel/march
2.4	900	$\mathbf{1.67 \times 10^{-2}}$	5.83×10^{-2}	$\mathbf{2.18 \times 10^{-2}}$	1.22×10^{-1}	5.6
1.5	1800	$\mathbf{1.27 \times 10^{-2}}$	2.56×10^{-2}	$\mathbf{1.21 \times 10^{-2}}$	3.74×10^{-2}	3.1
1.0	2700	$\mathbf{9.42 \times 10^{-3}}$	1.31×10^{-2}	$\mathbf{1.02 \times 10^{-2}}$	2.39×10^{-2}	2.3

Table 25. Medians of the per-order relative errors (134) and (135), scored as in the previous table, at $\varepsilon = 0.1$ and $n_{\mathrm{sub}} = 6$, with the lateral pitch held at 112.5 m so only the period changes. Bold is the better of the pair; the last column is the ratio at wide angles. **The march is the better route at every frequency tested, at wide angles and at the specular orders**, by 2.3 to 5.6 times at wide angles and 1.4 to 3.5 times at the specular orders.

The march's errors stay between 0.9 and 2.2 per cent across the three rows, while the voxel route's grow with frequency as the staircase does, so the margin is set by the voxel route's error rather than by the march's.

Against the array the march solves. The isolated sphere scores the march on the one-sphere question, periodization included. The array's exact solution (§18) scores it on the periodic problem it actually solves, which isolates its own error. Medians over the non-specular propagating orders:

$k_S a$	L (m)	pitch (m)	march vs array	march vs sphere	array vs sphere
2.4	900	112.5	1.84%	2.18%	1.52%
		75.0	0.88%	1.78%	
		56.2	0.64%	1.72%	
1.5	1800	112.5	1.23%	1.21%	1.61%
		75.0	0.72%	1.26%	
1.0	2700	112.5	1.29%	1.02%	0.70%

Table 26. P incidence, P channel of $\mathbf{R}$ and $\mathbf{T}$, $\varepsilon = 0.1$. The array's answer is the layer-multiple-scattering solution of §18; the sphere's is the same solve with the lattice coupling switched off.

At the 112.5 m pitch of the table above, the march's own error on its periodic problem is as large as the array's coupling and partly cancels it, so its error against the sphere understates both. Refining the pitch at fixed period drives the error against the array down — 1.84, 0.88, 0.64 per cent — while the error against the sphere settles at the array's coupling. The march's wide-angle residual against one sphere is therefore periodization only once the lateral grid is resolved; at the

coarse pitch it is periodization and lateral discretisation in comparable parts. Neither approaches the voxel route's 7.5 to 31 per cent.

And the fast iterative route exists. The comparison above uses the dense direct solve that `compute_sphere_foldy_lax` ships with, and an earlier draft of this section said the block-Toeplitz FFT version — the one worth racing against the march — was not built. It is: `cubic_scattering/sphere_scattering_fft.py` maps the sub-cells to a 3-D grid, embeds the propagator in a $(2n-1)^3$ circulant block, transforms its 81 components and solves with GMRES. It reproduces the dense answer to 10^{-9}.

Measured, it does not yet win end to end, and the reason is a setup cost rather than the algorithm: the kernel construction is 99.9% of the call — 77.2 of 77.3 s at $n_{\mathrm{sub}} = 6$, 360.0 of 360.4 s at 10 — while the FFT matvec and the GMRES solve together are 0.09 and 0.43 s, against a dense solve of 1.14 and (at $n_{\mathrm{sub}} = 8$) 6.96 s that grows as $O(n^3)$. The kernel depends only on the frequency, the lattice and the contrast, so it amortises over incident directions and repeated solves exactly as $\mathbf{Y}$ does for the march. *Comparing the two properly means amortising both, and that has not been done.*

The march's floor is also not a floor of the method: it is the array, and it moves when the array does.

What this does and does not settle. It settles that on this problem, at this frequency and contrast, the imbedding route is the more accurate of the two per unknown by a wide margin, while being scored against an arbiter that does not contain the physics it is charged for. It does not settle anything about the *iterative* solver, which was never in the comparison. It also does not settle the asymptotic cost of either: four points at these sizes are descriptive, and the dense products are not in their cubic regime. The comparison that would settle the remaining question — iterative multiple scattering against imbedding, on a problem with an external arbiter — needs the periodic voxel sphere that §21 now lists.

23 Where the route belongs

The x_3 preference of §7.5 is an artefact only because the problem posed there — an isotropic cube in an isotropic whole space — has no preferred direction. A layered background does. There the formulation's directional structure matches the physics rather than contradicting it, the O_h argument does not apply, and the propagator is already available and already bridged to $\mathscr{A}$ by (29).

That was written as an expectation and is now a measurement. In the stratified geometry of §14, the x_3-distinguished shear amplitude is the one the physics wants, by a factor of 2.6 in the channel that probes it (§14.4). The same feature that is demonstrably wrong in a whole space — $\geq 4.2 \times 10^{-4}$ of error, by §7.5 — is demonstrably right once there is a stack of interfaces for it to be right about. Both statements are measured, and they are not in tension: they are about different problems.

That is also the case in which the ordinary route has no competitor: the cube moment method rests on volume moments of the whole-space Green's tensor, and in a stratified background that

object does not exist. The first-order form supplies what the moment method cannot.

24 Against the literature

The construction marched in §16.8 and extended in §17.2 and §18 is not new. It is the impedance operator approach of Haines, Hulme & Yu [9], and the symmetry relied on throughout as a structural check is the impedance-operator symmetry of Thomson [25, 26]. Neither is cited in the thesis or anywhere in the codebase this document accompanies, which is how the omission survived: the survey that precedes work here covers five places, and all five are local.

The lineage is also shared rather than parallel. Haines *et al.* build on an invariant imbedding formulation [1, 20], which is the same tradition Chapter 5 of [29] works in, and their impedance treatment begins with [7] for *SH* and [8] for general hyperbolic systems. None of those is cited in the thesis either.

24.1 The correspondence

Haines *et al.* [9] define an impedance operator $\mathbf{Z}$ by $\sigma_i^n = -(\mathbf{Z}\cdot\mathbf{v})_i$, relating stretched traction to particle velocity, and show it satisfies a first-order pseudo-differential Riccati equation

$$\partial_{\xi_3}\mathbf{Z} + \mathbf{Z}A_1\mathbf{Z} + \mathbf{Z}A_2 + A_3\mathbf{Z} + A_4 = 0, \tag{136}$$

independent of the boundary conditions and solved as an initial value problem, after which the particle velocity follows from a one-way equation $\partial_{\xi_3}\mathbf{V} = (A_1 - A_2\mathbf{Z})\mathbf{V}$.

Set against (103), $\mathbf{Y}' = \mathbf{A}_{21} + \mathbf{A}_{22}\mathbf{Y} - \mathbf{Y}\mathbf{A}_{11} - \mathbf{Y}\mathbf{A}_{12}\mathbf{Y}$: the two are the same equation. The quadratic term is $\mathbf{Z}A_1\mathbf{Z}$ against $\mathbf{Y}\mathbf{A}_{12}\mathbf{Y}$, the two linear terms correspond one for one, and the inhomogeneous term is A_4 against $\mathbf{A}_{21}$. The marched variable differs only in that $\mathbf{Y}$ maps displacement to traction where $\mathbf{Z}$ maps particle velocity to traction, a factor $-i\omega$; the sign conventions differ correspondingly.

The lateral treatment corresponds too, and this is the part worth being plain about. Haines *et al.* represent horizontal variation in a complete orthonormal basis, typically Fourier, so that the partial differential equation becomes a system of ordinary differential equations in ξ_3 for the coefficients and the impedance operator becomes a finite matrix under truncation. That is what §17.2 does with a pseudospectral grid and §18 with two of them. Carrying the lateral coupling inside the marched variable, rather than by an outer iteration on $(\mathbf{I}-\mathbf{P}\,\Delta\mathbf{C}_{\text{eff}})$, is *their* construction, not a departure from the literature; it is a departure only from the thesis's own architecture (120), which is what §17.2 actually compares against.

The symplectic identity is theirs too, and older than either of us. Haines *et al.* note in passing, at their eq. (44), that the system matrix satisfies $\mathbf{A}^T\mathbf{J} = -\mathbf{J}\mathbf{A}$ in both elastic and anelastic media — which is (122) and (126), since $(\mathbf{J}\mathbf{A})^T = \mathbf{A}^T\mathbf{J}^T = -\mathbf{A}^T\mathbf{J} = \mathbf{J}\mathbf{A}$ — and attribute it to [14], [2] and [8], remarking that each such symmetry reflects a conservation law. It is a 1978 result used here as a discretisation check, not a new identity. Thomson [26] later develops the same operator for laterally heterogeneous anisotropic media — there called a displacement-to-traction operator and identified as the elastic analogue of a Dirichlet-to-Neumann map — and proves that its kernel

equals its transpose with the arguments reversed, together with the symplectic orthogonality of the lateral modes that stands behind the $\mathbf{J}_6$ pairing used throughout.

24.2 What is left

Setting the formulation aside, three things in this document are not in the papers above, and they are smaller claims than the one they replace.

- **The contrast operator.** $\Delta\mathscr{A}$ and $\Delta\mathbf{C}_{\text{eff}}$ are derived from the thesis's own `dCBC`/`dCBCrho` and carry its O_h structure; the march is driven by that operator rather than by a general anisotropic medium. The comparison in §17.2 is against the thesis architecture (120), which the literature above does not treat.
- **Verification against an exact scatterer.** The march is scored against the elastic Mie sphere at fourth order in the depth step (§18), on a sharp boundary, through the Poisson array relation. The papers above establish the formulation and its accuracy in their own terms; this is an independent external check of the same construction.
- **Why the impedance is better *conditioned*.** Haines *et al.* do argue for the impedance over an up–down split, but on different grounds: the decomposition into upgoing and downgoing parts "can become poorly defined for laterally varying media", whereas decomposing by the direction of energy *flow* gives something that "exists and is well-defined for all wavefields". That is a statement about well-definedness, and the difficulty they name is the lateral coupling — a two-dimensional medium scatters one wavenumber into all of them, and P at a given wavenumber may be evanescent while S at the same wavenumber still propagates. They do not discuss eigenvector degeneracy, conditioning or loss of precision. The argument of §16.8 is therefore a different one and is measured: $\mathbf{V}$ needs the *full* inverse $\mathbf{D}_z^{-1}$, which mixes the two families and is where the cancellation lives, while $\mathbf{Y}$ needs only the three downgoing columns, so its residual tracks $\operatorname{cond}(\mathbf{U}_d)\,\epsilon$ linearly to 7.2×10^{-10} over $|\mathbf{k}| = 2$ to 6000 where $\mathbf{V}$ returns 2.000 — a lost branch, not a lost digit. The two arguments are complementary: theirs is that the decomposition may fail to exist, this one is what computing it anyway costs.
- **The residual as a measurement.** The residual after the depth error is exhausted is linear in the contrast and converges under lateral refinement to about $0.3\,\varepsilon$ at 2.5 diameters — a bound on the inter-sphere multiple scattering of the periodic problem the march actually solves, though not yet separated from the march's own error.

24.3 The integrator is theirs as well, and better

The fourth-order result of §16.15 is *not* a contrast with this literature. Haines *et al.* solve both the Riccati and the one-way equation with "multiply linked applications of the explicit fourth-order Runge–Kutta scheme", and state the stage requirement outright: "for a step size of h, the Runge–Kutta scheme requires function values at ξ_3, $\xi_3 + \frac{1}{2}h$ and $\xi_3 + h$: in other words at half-step intervals". That is the scheme of Algorithm 1 and the sampling discipline of §16.15. The 32× of §16.15 is measured against a *layer stack*, which is a different comparison and not one with them.

On three further counts their implementation was ahead of the one described here. All three have since been adopted, and §24.4 reports what each was worth when measured rather than estimated — which in one case is a good deal less than the estimate below, left standing so the correction is visible:

- **Adaptive step control.** They derive a step size from the leading Runge–Kutta error, $h = \theta/(\|\mathbf{D}\| + \sqrt{\theta\,\|\partial_{\xi_3}\mathbf{D}\|})$ with $\mathbf{D} = -i\omega(\mathbf{A}^{VV} - \mathbf{A}^{VT}\mathbf{Z})$, reporting relative precisions of about 1 part in 4×10^3 and 10^5 at $\theta = \frac{1}{2}$ and $\frac{1}{4}$. The march here uses a uniform step and meets the explicit method's stability wall at $2|k|_{\max}h \simeq 1.3$ with no adaptive rule at all.
- **Assembling A by FFT.** "The elements can be computed by fast Fourier transform when using Fourier series as the basis functions." Here $\mathbf{D}_x$ and $\mathbf{D}_y$ are built as dense Kronecker matrices. The saving is bounded — the quadratic term $\mathbf{Y}\mathbf{A}_{12}\mathbf{Y}$ is dense in any basis and is 69% of the work by §17.4, so the ceiling is about 1.45× — but it is a saving not taken.
- **Reuse.** "Once the impedance operators and fundamental solutions have been calculated for a given region of heterogeneity they can be used in solving any number of problems (allowing, for example, varying source and receiver geometries) involving that region." Nothing here exploits that.

They also make explicit a stability argument used only implicitly here: the one-way equation is integrated *in* the direction of energy flow and the Riccati equation *against* it, and that choice is what keeps both stable.

The Möbius step and its replacement in §19 are therefore not an improvement on this literature either. Haines *et al.* form no matrix exponential anywhere; the exponential is inherited from the layer-stack tradition, and removing it recovers a position they already held.

24.4 The three counts, adopted and measured

Each of the three was built into the march and scored against the version without it. The gate is `gate_first_order_lateral_impedance_3d.py`, now `56/56`.

Adaptive step control, and what it really controls. Their rule is implemented as written, with $\mathbf{D} = \mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{Y}$ — the operator of the one-way equation $\mathbf{u}' = (\mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{Y})\mathbf{u}$, which is their $-i\omega(\mathbf{A}^{VV} - \mathbf{A}^{VT}\mathbf{Z})$ in the conventions of this note — and $\partial_z\mathbf{D}$ by one finite difference at the start of each step. Tightening θ tightens the answer as it should:

θ	steps	relative error	mean $2\|k\|_{\max}h$
1	21	4.18×10^{-5}	0.24
$\frac{1}{2}$	39	3.06×10^{-6}	0.13
$\frac{1}{4}$	75	2.18×10^{-7}	0.07

Table 27. Adaptive stepping on the 3-D lateral grid, scored against a converged uniform march. Halving θ buys about 14×, consistent with the fourth-order error the rule is derived from.

Against a uniform step at equal accuracy the saving is 75 steps rather than 128 — worth having, and smaller than the headline. The more useful observation is structural, and it is not one this note made before implementing the rule: **the step control is also the stability control.** Since $h \le \theta/\|\mathbf{D}\|$ and $\|\mathbf{D}\|$ bounds the largest vertical wavenumber, $2|k|_{\max}h \le 2\theta$; at $\theta = \frac{1}{2}$ that is 1.0, inside the explicit method's wall at $\simeq 1.3$ measured in §16.15. This is why [9] can use an explicit Runge–Kutta scheme throughout and meet no stability wall, where the uniform march here did. The wall was never an argument for an implicit step; it was an argument for a step-size rule.

One honest qualification. $\|\mathbf{D}\|$ is taken as the maximum absolute row sum, which bounds the spectral radius at $O(n^2)$ rather than the cost of a decomposition — and measured on the operator at the base of the slab it over-bounds it by 4.4×. That factor is paid in steps. A tighter estimate would recover it; an eigendecomposition would cost more than it saves.

Fourier assembly of A, which is worth much less than estimated. Every factor of $\mathbf{D}_x$ or $\mathbf{D}_y$ is applied to the columns of the identity by transform instead of multiplied in, so a term such as $\mathbf{D}_x\zeta\mathbf{D}_x$ costs $O(N^2 \log N)$ rather than $O(N^3)$. It reproduces the dense assembly to 4.4×10^{-16}. It is also *slower* than the dense product below $N \approx 100$, where the transform's constant dominates, and reaches 1.8–2.5× at $N = 144$.

The estimate above — a ceiling of about 1.45× — was too generous, and for a reason worth stating. It compared assembly against assembly. Assembly is not the step: one Runge–Kutta step is four evaluations of the right-hand side at 13.7 ms each against 6.2 ms of assembly, so assembly is 10% of the step and a 2× saving on it is 5% of the march. The reason is the one §16.9 gives structurally: $\mathbf{Y}\mathbf{A}_{12}\mathbf{Y}$ is a Schur complement and is dense in every basis, so no transform reaches the term that dominates. The trick is real, it is now taken, and it is worth a few per cent rather than half.

Reuse, which this formulation gets for nothing. $\mathbf{Y}$ is the Dirichlet-to-Neumann map of the medium and contains no reference to a source, so one march serves any number of them: 425 ms for the march, then eight sources in 2.0 ms in total, against 3400 ms to treat each as a fresh problem. Nothing had to be added for this beyond noticing it, and the same property in a second guise — $\mathbf{Y}$ as a complete state, so that regions compose by handing over one matrix — is what §16.9 gates as the imbedding restart.

The one-way equation those fundamental solutions would be assembled from is now built — §16.10 — so what remains theirs and not taken is the assembly and reuse of the solutions themselves, and their treatment of varying receiver geometry.

24.5 What has not been checked

The correspondence above is drawn from the full text of [9], and from the abstract and stated results of [26] only. The companion paper [13], which carries the worked examples and the numerical validation, has not been consulted, so no comparison of accuracy or run time against their implementation is offered here — and on the evidence of their §2.5 there is no reason to expect one to favour the march described above.

One genuine difference of construction remains unresolved. Haines *et al.* handle material

discontinuities with a curvilinear transformation, ξ_3 constant on major interfaces, so that tractions and velocities stay continuous and are never differentiated across a jump. §18 instead takes the cross-section through its exact transform (127) and marches the sphere's own bounding slab, which earns fourth order across a sharp boundary with no transform at all — but only because the disc's *area* is polynomial in depth. That is a device for this problem, not a method for interfaces, and whether the two coincide in any useful sense has not been established.

24.6 Tromp & Snieder: the same imbedding, with the contrast linearised

The thesis Introduction names Tromp & Snieder among the invariant imbedding schemes behind the Riccati sweep (§16.2). Their paper [27] is closer to (100) than that citation suggests, and it differs from it in exactly one place — the place §4.3 identified.

Their construction. They take the first Born response of a thin homogeneous layer of width Δz in a homogeneous background, per unit width (their eqs 2.12 and 2.15), add the layer to the base of a band by the reflection/transmission addition rule, and let $\Delta z \to 0$. The result, their eqs 3.6a–d, is the Riccati system for the band's $\mathbf{R}$ and $\mathbf{T}$ from both sides, with $\mathbf{T} = \mathbf{I}$ and $\mathbf{R} = \mathbf{0}$ at zero thickness. Structurally it is (100), marched in the other direction and in the interaction picture (the background phases carried in the coefficients rather than in $\boldsymbol{\Lambda}$). They call the equations *exact*, on the ground that the nth Born term of the thin layer is $O(\Delta z^n)$, so that first Born is exact at $O(\Delta z)$.

Where that fails. Their coefficients are linear in the density and stiffness perturbations. $\Delta\mathscr{A}$ is not linear in the stiffness, because the elimination inverted $\mathbf{C}_{33}$, (16). The Born argument misses this because their Born series carries ∂_z and ∂_{z_0} of the plane-wave Green's tensor, and $\partial_z\partial_{z_0}G$ contains $\delta(z - z_0)$: for a stiffness perturbation the second Born term of the thin layer is $O(\Delta z)$, not $O(\Delta z^2)$. Physically, normal traction is continuous across the layer, so the strain inside it is rescaled by $\mu/(\mu + \Delta\mu)$ or $(\lambda + 2\mu)/(\lambda + 2\mu + \Delta\lambda + 2\Delta\mu)$ at $O(1)$, whatever the width. That is the same local field that the elimination of §4.3 sums exactly and that the thin layer's first Born response omits.

The statement proved. Let $\Delta\mathscr{A}^{(1)} = \partial_\varepsilon\, \Delta\mathscr{A}(\varepsilon\,\Delta c, \varepsilon\,\Delta\rho)\big|_{\varepsilon=0}$ be the linearisation of the contrast operator. Then:

1. their thin-layer coefficients are *exactly* the interaction-picture blocks of $\mathbf{D}_z^{-1}\Delta\mathscr{A}^{(1)}\mathbf{D}_z$, with $\mathbf{t}_D = \mathbf{V}_{dd}$, $\mathbf{r}_U = -\mathbf{V}_{ud}$, $\mathbf{t}_U = -\mathbf{V}_{uu}$, $\mathbf{r}_D = \mathbf{V}_{du}$. This holds at every slowness and for a general anisotropic perturbation;
2. consequently their eqs 3.6 return, to integration precision, the *exact* response of a fictitious medium with system matrix $\mathscr{A} + \Delta\mathscr{A}^{(1)}$;
3. $\partial_\varepsilon^2\Delta\mathscr{A} = 0$ in the density direction and $\neq 0$ in any stiffness direction, so the equations are exact for density contrast and wrong at $O(\Delta c^2)$ otherwise.

At normal incidence the fictitious medium is an ordinary one: the band with its moduli $M \in \{\lambda + 2\mu,\ \mu\}$ replaced by the linearised compliance $M_{\text{eff}} = M^2/(M - \Delta M)$. Their eq. 3.9, the transmitted phase of a smooth band, is linear in the contrast for the same reason.

The measurement. A uniform band 100 m thick in the project's background at 10 Hz, three slownesses up to 1.5×10^{-4} s/m. The Python gate `gate_tromp_snieder_uniform_band.py` (21/21) integrates their system against the Kennett recursion; `TrompSniederBornGenerator.wl` (45/45) transcribes their coefficients term by term from the printed sign patterns, integrates eqs 3.6 at 32 digits, and scores them against exact propagators $\exp(\mathscr{A}^{\text{tot}}H)$, so the two share neither the transcription nor the reference. Their result (their reflection matrices are displacement-normalised and Kennett's are flux-normalised) is compared through similarity invariants:

band	R_{PP}	R_{SS}	R_{SH}	$R_{PS}R_{SP}$
density +40%	$\le 10^{-20}$ (Mathematica), $\le 3 \times 10^{-13}$ (Python)			
project moderate	4.18×10^{-2}	2.45×10^{-2}	2.51×10^{-2}	6.24×10^{-2}
moduli +20%	0.197	0.299	0.254	0.553
moduli −20%	0.186	0.195	0.167	0.347

Table 28. Relative error of eqs 3.6 of [27] against the exact band, at $p = 8 \times 10^{-5}$ s/m; "project moderate" is $\Delta\lambda = 2$, $\Delta\mu = 1$ GPa, $\Delta\rho = 100$ kg/m^3. Over the three slownesses the moderate contrast gives 2.3–5.2% in the diagonal channels; at ±20% the $R_{PS}R_{SP}$ error reaches 1.63 at $p = 1.5 \times 10^{-4}$ s/m. The two implementations agree to ≤ 0.27% of each entry, the rounding of the printed values.

Stiffness-only bands as $\varepsilon \to 0$ give a relative error proportional to ε (log–log slope 1.03–1.07): correct at Born order, as they should be. Against $\mathscr{A} + \Delta\mathscr{A}^{(1)}$ their result agrees to $\le 10^{-20}$ in every case, so the error is entirely the linearisation and none of it is the imbedding.

What this means here. The route is sound: a thin-layer response composed by the addition rule and taken to the limit is the Riccati equation, and (100) is that equation. What the thin layer must supply is its *exact* $O(\Delta z)$ response, which is $\Delta\mathscr{A}$ itself and not its Born approximation. (100) is driven by $\Delta\mathscr{A}$, so it is untouched. The ratios in the table of §4.3, 0.957 and 0.940 at the moderate contrast, are precisely the per-unit-depth coefficients that [27] get wrong. The same point constrains any scheme that builds the depth generator from a thin slab of scatterers — a single layer of cube voxels, say — rather than from $\Delta\mathscr{A}$: the slab's in-plane local field is $O(1)$ as $\Delta z \to 0$ and must be carried, not dropped.

A What the contrast parameter scales

Every sphere measurement in this note is indexed by a single number ε, and several of the conclusions are statements about how something depends on it — the array coupling is linear in ε (§18), the Krylov iteration count grows with it (§22). Those are only readable if it is clear what ε moves. It is defined in §18 as the fractional perturbation of α, β and ρ inside the sphere; this appendix records what that forces everything else to do, because the answer is not ε and in one place is nearly five times it.

The family. Put $s = 1 + \varepsilon$ and scale all three primary quantities by it, $\alpha' = s\alpha$, $\beta' = s\beta$, $\rho' = s\rho$. The moduli are then fixed, not free:

$$\rho' = s\rho, \qquad \mu' = \rho'\beta'^2 = s^3\mu, \qquad \lambda' = \rho'\alpha'^2 - 2\mu' = s^3\lambda. \tag{137}$$

This is the construction the sphere gate already carries, and it is derived there rather than typed, because the march and the Mie arbiter must describe the same sphere and a hand conversion between (α, β, ρ) and (λ, μ, ρ) is a silent factor waiting to happen.

What follows, and why one exponent is easy to miss. Everything else in the problem is a power of s:

ε	0.02	0.10	0.20	0.40	0.80
$\Delta\alpha/\alpha = \Delta\beta/\beta = \Delta\rho/\rho = s - 1$	0.020	0.100	0.200	0.400	0.800
$\Delta\lambda/\lambda = \Delta\mu/\mu = s^3 - 1$	0.061	0.331	0.728	1.744	**4.832**
$\Delta Z/Z = s^2 - 1, \quad Z = \rho c$	0.040	0.210	0.440	0.960	2.240

Table 29. The moduli move as the *cube*. At $\varepsilon = 0.8$ — the strongest point of the Krylov sweep in §22 — the shear modulus inside the sphere is nearly six times the background, not 1.8 times, and that is what the iteration count is responding to.

Two consequences are worth naming. **Poisson's ratio is unchanged**: α/β is s-independent by construction, so the family moves the *scale* of the medium and not its elastic character, and a sweep in ε is a clean one-parameter sweep rather than a path through a two-parameter space. And **the impedance moves as s^2**, which is the quantity reflection actually responds to — so the single-interface coefficient $(Z' - Z)/(Z' + Z)$ is $(s^2 - 1)/(s^2 + 1)$, exactly ε to leading order and only that to leading order.

A symbol warning. ε is overloaded in this note: bare and without indices it is the contrast of (137); bold ($\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon}_{i3}$) or carrying indices ($\varepsilon_{33}$) it is strain. The usage is consistent throughout and the two never appear in the same expression, but they are not the same object and the distinction is carried entirely by the typography.

References

Standing rule for this note: a technique that is borrowed *rather than derived here carries an author–date reference at the point it is used. The converse matters too — a technique that is* not *borrowed must not be dressed as though it were, and a name coined in this note is a label, not a citation. Provenance, since it differs: [28] was read from the document itself; [27] was read in full and checked against Crossref; [11], [12], [19], [21], [9], [13], [25] and [26] were located by literature search; [1], [2], [7], [8], [14] and [20] were transcribed from the reference list of [9] and are not otherwise verified; the rest are from memory.* None has been verified against a library. *Authors, years and titles should be reliable and volume and page numbers less so; all must be checked before this note is circulated, and nothing in this list should be quoted from here without that check.*

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